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Department of Physics and Astronomy "A. Righi"

Theoretical Physics

Quantum Field Theory II

Lectures held by

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Contents

Introduction

In these notes, we will explore the fascinating world of Quantum Field Theory (QFT). QFT is a fundamental framework in physics that combines classical field theory, special relativity, and quantum mechanics. It provides a powerful language for describing the behavior of particles and fields at the most fundamental level. We assume are going to assume that the reader has a basic understanding of quantum mechanics and special relativity, as well as some familiarity with classical field theory and quantum field theory of free fields (Which will aniway be reviewed in the first chapter).

There exist different motivations for studying QFT: it provides a framework for describing the interactions between particles and fields, and it has been successful in explaining a wide range of phenomena, from the behavior of subatomic particles to the properties of condensed matter systems. Additionally:

- QFT is the language of the Standard Model of particle physics, which describes the fundamental particles and their interactions (except for gravity); it is used for the quantization of systems with an **infinite number of degrees of freedom**, such as fields; it is essential for understanding the behavior of particles at high energies and small distances, where quantum effects become significant; and it provides a framework for describing the behavior of particles in curved spacetime, which is important for understanding phenomena such as black holes and the early universe.

- It is a successful framework for describing the behavior of particles and fields at the most fundamental level, incorporating both quantum mechanics and special relativity:

  - At high energies we can witness **particles creation and annihilation**, which cannot be described by non-relativistic quantum mechanics; QFT provides a framework for describing these processes, in an infinite-dimensional system, where the number of particles is not fixed.

  - QFT is a relativistic theory, which means that it is consistent with the principles of special relativity, implmenting **causality** and **Lorentz invariance**; this is essential for describing the behavior of particles at high energies and small distances, where relativistic effects become significant. So two events that are spacelike separated cannot influence each other

$$(x-y)^2 = (x^0-y^0) - \sum_{i=1}^{3}(x^i-y^i)^2 < 0 \implies [\mathcal{O}(x), \mathcal{O}(y)] = 0,$$

    thus the events are causally disconnected, ensuring causality.

- Particles are described as excitations of underlying fields, which can be created and annihilated; this allows for a more complete description of particle interactions and the behavior of particles at high energies. Examples of particles and underlying fields include:
  - **Scalar field**: $\phi(x) \implies$ spin-0 particles (e.g., Higgs boson).
  - **Spinor field**: $\psi(x) \implies$ spin-1/2 particles (e.g., electrons, quarks).
  - **Vector field**: $A_\mu(x) \implies$ spin-1 particles (e.g., photons, W and Z bosons).
- The **Standard Model** of particle physics is a quantum field theory that describes the fundamental particles and their interactions (except for gravity); it has been successful in explaining a wide range of phenomena, from the behavior of subatomic particles to the properties of condensed matter systems. It is a quantum field theory that describes the electromagnetic, weak, and strong interactions, through the exchange of gauge bosons among fields of spin 0, 1/2, and 1 particles.
- The inclusion of gravity in a quantum field theory framework is an open problem in theoretical physics, but can be approached by inserting the metric tensor of general relativity

$$g^{\mu\nu} = \eta^{\mu\nu} + h^{\mu\nu}.$$

  While there are approaches to quantizing gravity, such as string theory and loop quantum gravity, a complete and consistent theory of quantum gravity has not yet been developed. Quanta of the gravitational field are called gravitons, which are hypothetical spin-2 particles that mediate the gravitational interaction, but they have not been observed experimentally. These notes will not cover quantum gravity, but it is an important area of research in theoretical physics which sometimes will be mentioned throughout the notes.

The main goal of these notes is to provide a comprehensive understanding of interacting quantum field theories, which are essential for describing the behavior of particles and fields at the most fundamental level, as well as mathematical methods for making theoretical predictions on interactions among particles. We will cover a wide range of topics, including:

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- The quantization of free fields, which provides the foundation for understanding the behavior of particles and fields in quantum field theory.
- The construction of interacting quantum field theories, which are essential for describing the behavior of particles and fields at the most fundamental level.
- The use of perturbation theory and Feynman diagrams to calculate scattering amplitudes and other physical observables in quantum field theory.
- The renormalization of quantum field theories, which is necessary to make sense of the infinities that arise in perturbation theory and to extract meaningful physical predictions from the theory.
- The application of quantum field theory to the Standard Model of particle physics, which describes the fundamental particles and their interactions (except for gravity).

QFT is one of the most successful and powerful frameworks in physics, and it has been instrumental in our understanding of the fundamental nature of the universe. This theory has been tested and confirmed by a wide range of experiments, some of which tested the predictions of QFT to an extraordinary degree of precision. For example, the anomalous magnetic moment of the electron has been measured to an accuracy of one part in a trillion ($\sim 10^{-13}$), and the predictions of QFT have been confirmed to this level of precision. Additionally, the discovery of the Higgs boson at the Large Hadron Collider in 2012 provided further confirmation of the predictions of QFT and the Standard Model.

Another essential aspect of these notes is to provide a comprehensive understanding of the importance of symmetries in quantum field theory, which play a crucial role in determining the behavior of particles and fields. We will explore the concept of gauge symmetries, which are local symmetries that arise in quantum field theories and are responsible for the interactions between particles and fields. We will also discuss the role of global symmetries, which are symmetries that do not depend on spacetime coordinates, and their implications for the behavior of particles and fields. In the end we will argue that *a QFT is essentially defined by its fields and its symmetries* (statement that will be made more precise in the end of the notes), and we will see how symmetries can be used to constrain the behavior of particles and fields, and to make predictions about their interactions.

For a more comprehensive introduction to quantum field theory, we recommend the following textbooks:

- **An Introduction to Quantum Field Theory** by Michael E. Peskin and Daniel V. Schroeder: This is a widely used textbook that provides a comprehensive introduction to quantum field theory, covering both the theoretical foundations and practical applications of the subject. **TODO:** check and change in the end
- **Quantum Field Theory** by Mark Srednicki: This textbook offers a clear and concise introduction to quantum field theory, with an emphasis on the physical intuition behind the mathematical formalism.
- **Quantum Field Theory in a Nutshell** by A. Zee: This book provides a more informal and intuitive introduction to quantum field theory, making it accessible to a wider audience while still covering essential topics.
- **The Quantum Theory of Fields** by Steven Weinberg: This three-volume series is a comprehensive and rigorous treatment of quantum field theory, written by one of the pioneers in the field. It is suitable for advanced students and researchers who want a deep understanding of the subject.

For more information about the notation and conventions used in these notes, please refer to the appendix, where we provide a detailed explanation of the symbols and mathematical conventions that we will use throughout the notes (Appendix A).

Prerequisites: *Familiarity with the Lagrangian and Hamiltonian formalisms of Classical Mechanics, standard Quantum Mechanics, Special Relativity and the basics of Quantum Field Theory (free fields) is required.*

1 Free QFT Review

A free quantum field theory does not account for interactions between particles. It describes particles that do not interact with each other, and the fields that represent them evolve independently. Free QFTs are often used as a starting point for understanding more complex interacting theories, as they provide a simpler framework to study the properties of quantum fields and particles.

The lagrangian for a free quantum field theory typically consists of kinetic terms and mass terms for the fields, without any interaction terms: it can only contain **quadratic terms** in the fields, without any higher-order interactions. Quanta of the free fields correspond to particles that do not interact with each other, and their dynamics are governed by the **free propagators** of the theory, which describe how particles propagate through spacetime without any interactions: propagators are the Green's functions of the free theory, and they determine how particles propagate from one point to another in spacetime with determined probability amplitudes.

We will review the free quantum field theories for different types of particles, including scalar fields (spin 0), fermionic fields (spin 1/2), and gauge fields (spin 1). We will also discuss the symmetries of these theories and how they lead to conservation laws through Noether's theorem. This review will provide a foundation for understanding more complex interacting quantum field theories, and it will also highlight the differences between free and interacting theories, as well as the role of symmetries in quantum field theory.

1.1 | Spin 0

For a real scalar field $\phi(x)$, the free Lagrangian is given by

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2\phi^2, \tag{1.1.1}$$

where m is the mass of the scalar particle. The equation of motion derived from this Lagrangian is the **Klein-Gordon equation**:

$$(\Box + m^2)\phi = 0, \tag{1.1.2}$$

where $\Box = \partial_\mu \partial^\mu$ is the d'Alembertian operator. The solutions to this equation describe free scalar particles with mass m. This is a classical equation of motion, having the same role of Maxwell's equations in classical electromagnetism, but for a scalar field $\phi(x)$ instead of a vector field $A^\mu(x)$.

The idea of quantum field theory is to promote the classical field $\phi(x)$ to a quantum operator $\hat{\phi}(x)$ that acts on a Hilbert space of states. The quantization procedure involves imposing commutation relations on the field operators, which leads to the creation and annihilation of particles. The free scalar field can be expanded in terms of creation and annihilation operators, which allows us to describe the particle content of the theory. Indeed, we can see how decomposing the field into its Fourier modes leads to the interpretation of the quanta of the field as particles with specific momenta and energies:

$$\phi(x) = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\, e^{i\mathbf{k}\cdot\mathbf{x}}\tilde{\phi}(t,\mathbf{k}),$$

where each mode can be computed by using this ansatz in the Klein-Gordon equation in momentum space, leading to the dispersion relation of an harmonic oscillator represented by on-shell particles with energy $E_\mathbf{k} = \omega_\mathbf{k} = \sqrt{\mathbf{k}^2 + m^2}$:

$$\left(\partial_t^2 + \omega_\mathbf{k}^2\right)\tilde{\phi}(t,\mathbf{k}) = 0,$$

showing how the free scalar field can be interpreted as a collection of independent harmonic oscillators, each corresponding to a different momentum mode.

In order to quantize the free scalar field, we have to promote the fields themselves to operators and impose the canonical commutation relations, which leads to the creation and annihilation of particles. Before showing how to do that, it is crucial to note that $\hat{\phi}(x)$ depends on both space and time, enforcing the Heisenberg picture of quantum mechanics, where operators evolve in time while states remain fixed. The canonical commutation relations for the scalar field are given by

$$[\hat{\phi}(t,\mathbf{x}), \hat{\pi}(t,\mathbf{y})] = i\delta^3(\mathbf{x}-\mathbf{y}),$$

where $\hat{\pi}(x) = \frac{\partial\mathcal{L}}{\partial(\partial_0\hat{\phi}(x))} = \partial_0\hat{\phi}(x)$ is the conjugate momentum operator. This constraint along with reality of the field $\hat{\phi}(x) = \hat{\phi}^\dagger(x)$ allows us to express the field operator in terms of **creation and annihilation operators**:

$$\hat{\phi}(x) = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\frac{1}{\sqrt{2\omega_\mathbf{k}}}\left(\hat{a}_\mathbf{k}e^{-ik\cdot x} + \hat{a}_\mathbf{k}^\dagger e^{ik\cdot x}\right),$$

where $k \cdot x = \omega_\mathbf{k}t - \mathbf{k}\cdot\mathbf{x} = k^\mu x_\mu$, and $\omega_\mathbf{k} = k^0$. The creation and annihilation (ladder) operators satisfy the following commutation relations

$$\begin{cases} [\hat{a}_\mathbf{k}, \hat{a}_{\mathbf{k}'}^\dagger] = (2\pi)^3\delta^3(\mathbf{k}-\mathbf{k}'), \\ [\hat{a}_\mathbf{k}, \hat{a}_{\mathbf{k}'}] = [\hat{a}_\mathbf{k}^\dagger, \hat{a}_{\mathbf{k}'}^\dagger] = 0. \end{cases}$$

Note that while the field operator $\hat{\phi}(x)$ is Hermitian and time dependent, the creation and annihilation operators are not, and they are responsible for creating and destroying particles in the quantum field theory. The **vacuum state** $|0\rangle$ is defined as the state that is annihilated by all annihilation operators:

$$\hat{a}_{\mathbf{k}} |0\rangle = 0 \quad \forall \mathbf{k},$$

while we can create one-particle states by acting with the creation operators on the vacuum, which adds one quanta of the field with momentum $\mathbf{k}$:

$$|\mathbf{k}\rangle = \sqrt{2\omega_{\mathbf{k}}}\hat{a}^\dagger_{\mathbf{k}} |0\rangle ,$$

and more generally, we can create multi-particle states by acting with multiple creation operators on the vacuum, which leads to the **Fock space** of the theory. With this definition of single particle states, we can compute the inner product between two single particle states, which is given by

$$\langle \mathbf{k} | \mathbf{k}' \rangle = 2\omega_{\mathbf{k}} \delta^3(\mathbf{k} - \mathbf{k}'),$$

showing that the single particle states are orthogonal and normalized with respect to the relativistic normalization convention. The normalization $\sqrt{2\omega_{\mathbf{k}}}$ is arbitrary, but important for ensuring that the probabilities computed in the theory are consistent with relativistic invariance.

Remark. *To find a Lorent invariant measure, we can start from the Minkowski space measure* $\mathrm{d}^4 k$ *and impose the on-shell condition* $k^2 = m^2$ *using a delta function along with a step function to ensure that we only consider positive energy solutions:*

$$\int \mathrm{d}^4 k \, \delta(k^2 - m^2)\theta(k^0) = \int \frac{\mathrm{d}^3\mathbf{k}}{2\omega_{\mathbf{k}}}.$$

We have then found the Lorentz invariant measure $\frac{\mathrm{d}^3\mathbf{k}}{2\omega_{\mathbf{k}}}$ *for integrating over the momentum space of on-shell particles, which is crucial for ensuring that the theory is consistent with special relativity. The previous normalization of the single particle states is consistent with this Lorentz invariant measure, since it ensures that the inner product between single particle states is Lorentz invariant, and it also allows us to compute probabilities and cross sections in a way that is consistent with relativistic invariance:*

$$\int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{k}}} |\mathbf{k}\rangle \langle \mathbf{k}| = 1,$$

which is the completeness relation for the single particle states in the theory. This ensures that we can express any state in the Fock space as a superposition of single particle states, and it also allows us to compute transition amplitudes and probabilities in a way that is consistent with relativistic invariance.

1.1.1 | Feynman's Propagator

1.1.2 | Noether's Theorem

1.2 | Spin 1/2

1.3 | Spin 1

2 Interacting Fields

Free theories are a good starting point to understand the basic concepts and techniques of quantum field theory, but to capture the full complexity of the physical world, we need to introduce interactions between fields. Interactions allow us to describe processes such as scattering, particle decays, and the formation of bound states, which are essential for understanding the behavior of matter and energy at a fundamental level. In a free theory, the fields do not interact with each other, and the equations of motion are linear: they can exhibit *at most quadratic terms in the fields* (in the lagrangian $\mathcal{L}$). We can then build interacting theories by adding non-linear terms to the Lagrangian, which will lead to non-linear equations of motion. These non-linearities are what give rise to the rich phenomenology of interacting quantum field theories.

The most important tools in the developing and understanding of interacting quantum field theories are:

- **Dimensional analysis**, which allows us to determine the relevance of different interaction terms at different energy scales, and to classify interactions as relevant, irrelevant, or marginal.
- **Renormalization**, which is a systematic procedure to deal with the infinities that arise in perturbative calculations of interacting theories, and to extract meaningful physical predictions from them.
- **Symmetry principles**, which play a crucial role in constraining the form of the interactions and in determining the properties of the particles and fields in the theory. For example, gauge symmetries lead to the introduction of gauge fields, which mediate the fundamental forces of nature.

Moreover **locality** is a fundamental principle in quantum field theory, which states that interactions should occur at the same point in spacetime. This principle ensures that the theory respects causality and allows us to construct a consistent framework for describing the dynamics of fields and particles:

$$\mathcal{L}(x) = \mathcal{L}\left(\phi(x), \partial_\mu \phi(x)\right), \quad \mathcal{L}(x) \neq \mathcal{L}\left(\phi(x+y), \partial_\mu \phi(x+y)\right),$$

and the interaction terms we will introduce will be local, meaning that they will involve products of fields and their derivatives evaluated at the same spacetime point x; this ensures that the theory is consistent with the principles of special relativity and quantum mechanics, and it will guide us in constructing the interaction terms in the Lagrangian.

In the next section we will start by performing a dimensional analysis of the fields and the interaction terms, which will help us to understand the behavior of the theory at different energy scales and to identify the most relevant interactions for describing physical phenomena. This will set the stage for our subsequent discussions on renormalization and symmetry principles in interacting quantum field theories.

2.1 | Dimensional Analysis

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2.1.1 | Examples of Interaction Terms

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ϕ^3 Interaction

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ϕ^4 Interaction

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ϕ^n General Interaction

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Complex Scalar Field Interactions

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Quantum Electrodynamics (QED) Interaction

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Yukawa Theory Interaction

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Scalar QED (sQED) Interaction

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2.1.2 | Master Plan

In the next sections, we will introduce interactions in the theory by adding interaction terms to the Lagrangian. The simplest way to do this is to add a term that is a polynomial in the field ϕ, such as $\lambda\phi^4$ or $g\phi^3$. These interaction terms will lead to non-linear equations of motion, which will make the theory more complicated to solve. However, they will also allow us to describe a wider range of physical phenomena, such as scattering processes and particle decays.

More precisely, we are going to:

- Define useful concepts and observables in the theory, such as **cross sections** and **decay rates**.
- Express these observables in terms of the **S-matrix**, which encodes the transition amplitudes between initial and final states in scattering processes: the matrix elements of a generic operator represent the probability amplitude of measuring a specific eigenvalue, corresponding to a transition from an initial state $|i\rangle$ to a final state $|f\rangle$.
- We will use theorems and techniques developed in a QFT framework, such as the **LSZ reduction formula**, to relate the S-matrix elements to correlation functions of the fields or their Fourier transforms, which are the fundamental objects of study in QFT. For example, correlation functions of the form

$$\langle 0| T\{\phi(x_1)\phi(x_2)\cdots\phi(x_n)\} |0\rangle ,$$

  will show how the fields interact at different points in spacetime, and can be used to compute scattering amplitudes and other physical observables.
- We will develop in the end procedures and techniques to compute these correlation functions from an interacting lagrangian in **perturbation theory**; using **Feynman diagrams** and **Feynman rules** we will be able to systematically calculate the contributions of different interaction terms to the physical observables.

2.2 | Scattering Processes

Scattering processes are currently the most important tool to probe the fundamental interactions of nature, and thus to test our theories of particle physics. They let us infer and understand the properties of particles and their interactions at a microscopic level by analyzing the outcomes of experiments and collisions at macroscopic distances.

We are forced to cite some of the most important results of scattering theory, as well as experiments, which is a vast and complex subject, in order to understand its importance in particle physics. Some of the most important results and experiments in scattering theory include:

- **Rutherford scattering**, which was the first experiment to probe the structure of the atom and led to the discovery of the nucleus. Scattering alpha particles off a thin gold foil, Rutherford observed that most of the particles passed through the foil with little deflection, while a small fraction were scattered at large angles, leading to the conclusion that the atom has a small, dense nucleus at its center. This experiment lead to our modern understanding of the atomic structure of matter at a subatomic level, and it was a crucial step in the development of quantum mechanics and quantum field theory.
- **Lowrence Berkeley National Laboratory**, which was the first particle accelerator and allowed for the discovery of many new particles and interactions: in 1931 Lowrence was the first to accelerate particles to relativistic energies, firstly only around 80 keV, and then with Livingston he built the first cyclotron, which was able to accelerate protons to energies of about 1.22 MeV, leading to the discovery of the neutron in 1932 by James Chadwick.
- **Large Hadron Collider (LHC)**, which is the largest and most powerful particle accelerator in the world, located at CERN, and has been instrumental in the discovery of the Higgs boson in 2012, as well as in probing the properties of the Standard Model and searching for new physics beyond it. It has a circumference of about 27 km and can accelerate protons to energies of up to 7 TeV per beam, allowing for collisions at unprecedented energies and luminosities (actually around 13 TeV, with future upgrades planning further increases).
- **Future colliders**, such as the proposed Future Circular Collider (FCC) at CERN, which aims to reach even higher energies and luminosities than the LHC, and could potentially discover new particles and interactions that are beyond the reach of current experiments. Basically all we know about particle physics and the fundamental interactions of nature has been inferred from the observations and measurements of scattering processes, either in natural phenomena (such as cosmic rays) or in controlled experiments.

Such great energies and luminosities allow us to probe the fundamental interactions of nature at a microscopic level, and to test our theories of particle physics with high precision: the **indetermination principle** allows us to probe smaller and smaller distances by increasing the energy of the particles, thus allowing us to explore the structure of matter and the fundamental forces at a deeper level. By analyzing the outcomes of scattering experiments, we can infer the properties of particles and their interactions, and test our theoretical models against experimental data.

Also the **decay rate** of unstable particles is an important observable that can be measured in experiments and computed in theory, providing insights into the properties of the particles and their interactions. In these decay processes, an unstable particle decays into other smaller and lighter particles, and the decay rate quantifies the probability per unit time that this decay will occur.

By studying the decay rates of particles, we can gain insights into their properties, such as their masses, lifetimes, and interaction strengths, and test our theoretical models against experimental data.

2.2.1 | S-Matrix

We prepare a state $|i,\, t_i\rangle$ at an initial time t_i which will transition after some interaction (or decay, scatter...) into a state $|f,\, t_f\rangle$, measured at a final time t_f with a probability given by the square of the amplitude $|\langle f,\, t_f|i,\, t_i\rangle|^2$. Here we can make the following observations:

- The initial and final states are defined in the limit of infinite past and future, where the particles are free and non-interacting. This is a common assumption in scattering theory, as it allows us to define **asymptotic states** that can be used to compute scattering amplitudes and cross sections.
- The process happens on microscopic time and length scales, while being measured on macroscopic scales, thus we can treat the interaction as an instantaneous event, letting us ignore the detailed time dependencies of the process and focus on the transition between the initial and final states.
- The transition from the initial to the final state can be described by a unitary operator called the **S-matrix**, which encodes the **transition amplitudes** between the asymptotic states.

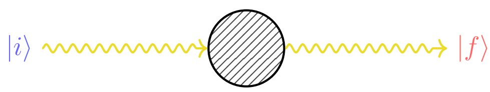


Figure 2.1: Schematic representation of a generic process, where an initial state $|i\rangle$ evolves into a final state $|f\rangle$ through some interaction described by the S-matrix. We are considering asymptotic states, which are defined in the limit of infinite past and future, where the particles are free and non-interacting. The S-matrix encodes the transition amplitudes between these asymptotic states describing the scattering process happening on microscopic time scales.

We tipically use **Heisenberg picture**, so if we want to compute the amplitude of a determined process, we can write it as

$$\langle f|\,\hat{S}\,|i\rangle = \langle f, \infty|i, -\infty\rangle\,,$$

where $|i\rangle$ and $|f\rangle$ are the asymptotic states of the system (time independent and directly related to the free particle states), respectively. *The S-matrix element $\langle f|\,\hat{S}\,|i\rangle$ gives us the amplitude for the transition* from the initial state $|i\rangle$ to the final state $|f\rangle$ due to the interaction described by the S-matrix itself. The S-matrix is a powerful tool in quantum field theory, as it allows us to compute **scattering amplitudes** and **cross sections** for a wide range of processes, and to make predictions that can be tested against experimental data. By analyzing the S-matrix elements for different processes, we can gain insights into the properties of particles and their interactions, testing our theoretical models against experimental observations.

2.3 | Cross Sections

If we analyze a scattering process, such as the Rutherford scattering, we can see how a beam of particles is scattered by a target; in the case of Rutherford scattering, it were alpha particles being scattered by a thin gold foil, where the alpha particles are much smaller than the nucleus of the gold atoms, thus we can treat the nucleus as a still sphere and the alpha particles as point-like particles. We will make the same assumption for the following scattering processes, thus treating the particles as point-like and the interactions as instantaneous events.

In this case, we can define the **cross section** of the target, which quantifies the probability of a scattering event occurring when a particle from the beam interacts with the target. The cross section is an effective area that represents the likelihood of a scattering event, and it can be computed using the S-matrix elements for the process.

The **geometric cross section** of the target, which encodes the effective area of the target for scattering, is given by

$$\sigma \equiv \pi R^2, \tag{2.3.1}$$

where R is the radius of the target. One can compute the number of particles that are scattered by the target, which is given by

$$N = n_B \cdot \sigma, \quad n_B = \frac{N_B}{A},$$

where n_B is the number density of the beam, N_B is the number of particles in the beam, and A is the area of the target. The cross section σ can be interpreted as an effective area that quantifies the likelihood of a scattering event occurring when a particle from the beam interacts with the target. The power of this approach resides in its simplicity: with few assumptions it allows us to compute the probability of scattering events for a wide range of processes, while linking macroscopical observables, such as the number of scattered particles, to the microscopic dynamics of the interactions encoded in the S-matrix elements.

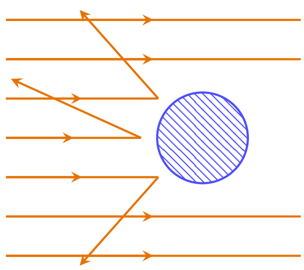


Figure 2.2: Schematic representation of a scattering process, where a beam of particles (orange arrows) is scattered by a target (blue circle). The geometric cross section of the target is given by $\sigma = \pi R^2$, with R being the radius of the target, while the number of scattered particles depends on the interaction probability. We are going to generalize this model to quantum field theory, computing the probability of scattering processes and expressing them in terms of the S-matrix elements.

Now we want to take this model and generalize it to quantum field theory, where we can compute the probability of scattering processes; we want to extend the model

- for any type of scattering process, or interaction;
- for a probability of scattering, not just a number of particles.

2.3.1 | Impact Parameter

If we assume a beam of particles along the z-axis, we can define the **impact parameter b**, which represents the perpendicular distance among the trajectory of the incoming particle and the center of the target:

$$N = \int \mathrm{d}^2\mathbf{b}\, n_B\, P(\mathbf{b}) = n_B \int \mathrm{d}^2\mathbf{b}\, P(\mathbf{b}),$$

thus we can now define the **cross section** as

$$\sigma = \int \mathrm{d}^2\mathbf{b}\, P(\mathbf{b}), \tag{2.3.2}$$

where $P(\mathbf{b})$ is the probability of scattering for a given impact parameter $\mathbf{b}$, generalizing the concept of a geometric cross section. This expression allows us to compute the cross section for a scattering process by integrating the probability of scattering over all possible impact parameters, thus providing a more general and flexible framework for analyzing scattering processes in quantum field theory.

`img/impact_parameter_scattering.png`

Figure 2.3: Schematic representation of a scattering process in terms of the impact parameter $\mathbf{b}$: the probability of scattering can be expressed as a function of the impact parameter, allowing us to compute the cross section by integrating the probability of scattering over all possible impact parameters.

We can also express it in its **differential form**, which ideally counts only a subset of events, with respect to the **solid angle** Ω as

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \int \mathrm{d}^2\mathbf{b}\, \frac{P(\mathbf{b})}{\mathrm{d}\Omega}, \tag{2.3.3}$$

where we are considering only the events that are scattered in a specific direction defined by the solid angle $\mathrm{d}\Omega$, thus providing a more detailed description of the scattering process. Alternatively, we can express it with respect to the measure of the space of **final states** $|f\rangle$ as

$$\frac{\mathrm{d}\sigma}{\mathrm{d}f} = \int \mathrm{d}^2\mathbf{b}\, \frac{P(\mathbf{b})}{\mathrm{d}f}. \tag{2.3.4}$$

The consequent step towards the computation of the cross section in QFT are:

- Express the probability of scattering $P(\mathbf{b})$ in terms of the S-matrix elements for the process

  $$\langle f| \hat{S} |i\rangle = \langle f, \infty | i, -\infty\rangle\,,$$

  which encode the transition amplitudes between the initial and final states.

- Integrate the probability of scattering over all possible impact parameters $\mathbf{b}$ to compute the cross section for the scattering process, using the expressions in equations (2.3.2), (2.3.3), or (2.3.4); then express the result in terms of observables that can be measured in experiments, such as the scattering angle or the energy or momenta of the scattered particles.

2.3.2 | Wave Packets and Regularization

Now one of the natural questions that arises is: what are the initial and final states $|i\rangle$ and $|f\rangle$ that we should use to compute the S-matrix elements for the scattering process? In typical scattering processes, the quantum dispersion of the particle momenta Δp is smaller than the experimental resolution, thus it makes sense to use **eigenstates of the momentum operator**, such as plane wave states. However, there are some issues:

- If the quantum dispersion of the particle momenta Δp is comparable to zero, for the Heisenberg uncertainty principle, the particle is completely delocalized in space

  $$\Delta p \Delta x \geq \frac{\hbar}{2}, \quad \Delta p = 0 \implies \Delta x \to \infty,$$

  thus it can interact with the target at any point in space, leading to divergences in the scattering amplitudes.

- We will show that S-matrix elements for plane wave states are proportional to delta functions, which leads to divergences when we compute the probability of scattering by taking the square of the amplitude: the probability of scattering is given by the square of the amplitude $\langle f| \hat{S} |i\rangle \sim \delta^{(4)}(p_i - p_f)$, but then

  $$P \sim |\langle f| \hat{S} |i\rangle|^2 \sim |\delta^{(4)}(p_i - p_f)|^2 = \delta^{(4)}(p_i - p_f)\delta^{(4)}(0).$$

  The term $\delta^{(4)}(0)$ is divergent, and it arises from the fact that we are using plane wave states, which are not normalizable and lead to divergences in the scattering amplitudes. This is a common issue in quantum field theory, and it requires regularization techniques to handle these divergences and obtain finite results for physical observables.

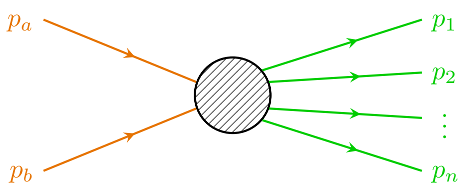
TODO: Add reference to book by schwarz.

One way to overcome the last problem is to consider a finite volume V and a finite time T, so that the delta function can be regularized and then take only at the end the limit $V \to \infty$ and $T \to \infty$. This is a common technique in quantum field theory to handle divergences and make sense of scattering amplitudes. By working in a finite volume and time, we can ensure that the calculations are well-defined and then take the appropriate limits to recover physical results. However, this approach can be obscure and cumbersome, and it can lead to complications in the calculations.

TODO: add reference to Peskin and Schroeder.

We will instead follow the Peskin and Schroeder approach, which is to consider the states as **wave packets** in the momentum space, which are **localized in both position and momentum space**, thus avoiding the issues with the delta function. This approach allows us to compute scattering amplitudes and cross sections without encountering the divergences associated with plane wave states. By using wave packets, we can ensure that the initial and final states are well-defined and that the scattering probabilities are finite and physically meaningful.

We can consider a general process in which the initial state is a two particle state, with momenta p_a and p_b, while the final state is a N particle state:



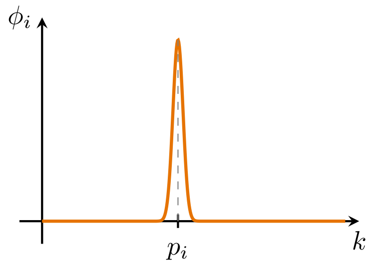
For incoming states we replace $|p_i\rangle$ with $|\phi_i\rangle$, which is a **wave packet** centered around the momentum p_i:

$$|p_i\rangle \to |\phi_i\rangle = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2E_\mathbf{k}}} \phi_i(\mathbf{k}) \, |\mathbf{k}\rangle \, ,$$

with

$$\langle \phi_i | \phi_i \rangle = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} |\phi_i(\mathbf{k})|^2 = 1,$$

remembering that $\langle \mathbf{k} | \mathbf{k}' \rangle = 2E_\mathbf{k} \, (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}')$; all of this indicates that $\phi_i(\mathbf{k})$ is a very narrow normalized function centered around p_i.



Example (LHC collider). We can assume, very conservatively, that the LHC collider works with $\Delta p \sim 1\,\mathrm{MeV}$, which is unrealistic (since it would correspond to a resolution of about 10^{-6}, not realistic) and leads to a delocalization of the particles of about

$$\Delta x \sim \frac{\hbar}{2\Delta p} \sim \frac{1}{2\,\mathrm{MeV}} \sim 100\,\mathrm{fm} = 10^{-13}\,\mathrm{m},$$

which, compared to the size of the particles bunches in the LHC, which is of the order of $1\,\mathrm{\mu m} = 10^{-6}\,\mathrm{m}$, is negligible, thus we can safely treat the particles as localized in space, and thus we can use wave packets to compute the scattering amplitudes and cross sections without encountering divergences.

This is a common assumption in high-energy physics, with typically well-localized particles and small quantum dispersion of their momenta compared to the experimental resolution; this allows us to use wave packets to describe the initial and final states in scattering processes.

Now we are left with the task of computing the S-matrix elements for the scattering process using wave packets, and then integrating the probability of scattering over all possible impact parameters to compute the cross section for the scattering process. But how are we going to model the wave packets? We can consider different types of wave packets, such as **Gaussian wave packets**, which are commonly used in quantum mechanics and quantum field theory due to their nice mathematical properties and their ability to represent *localized states in both position and momentum space*. By choosing appropriate wave packet profiles, we can ensure that the initial and final states are well-defined and that the scattering probabilities are finite and physically meaningful.

Example (Gaussian wave packet). We can consider a Gaussian wave packet centered around p_i with a width Δp:

$$\phi(\mathbf{k}) = N \exp\left[-\frac{1}{4} \left(\frac{(\mathbf{k} - \mathbf{p})^2}{\Delta p} \right)^2 - i\mathbf{k} \cdot \langle \mathbf{x} \rangle \right], \tag{2.3.5}$$

where N is a normalization constant, Δp is the width of the wave packet in momentum space, and $\langle \mathbf{x} \rangle$ is the average position of the wave packet in real space: using $\hat{x}^\mu = i\frac{\partial}{\partial p_\mu}$, one can check that the uncertainty principle is satisfied with $\Delta x \Delta p = \frac{1}{2}$.

Thus the wave packet is normalized and has a well-defined momentum distribution, with the properties of a momentum eigenstate while being relatively well localized in space for finite Δp. By using Gaussian wave packets, we can compute the S-matrix elements for the scattering process without encountering divergences, and we can ensure that the initial and final states are well-defined and physically meaningful.

Initial states. Sticking to the Gaussian wave packet example, we can write the initial states as the following superposition of momentum eigenstates:

$$|p_a\rangle \to |\phi_a\rangle = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2E_\mathbf{k}}} \phi_a(\mathbf{k}) \, |\mathbf{k}\rangle \,,$$
$$|p_b\rangle \to |\phi_b\rangle = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2E_\mathbf{k}}} \phi_b(\mathbf{k}) \, e^{-i\mathbf{k}\cdot\mathbf{b}} \, |\mathbf{k}\rangle \,,$$

where we shifted the second wave packet by the impact parameter $\mathbf{b}$, defined as

$$\mathbf{b} = \langle\phi_b| \, \hat{\mathbf{x}}^\perp \, |\phi_b\rangle \,,$$

with $\hat{\mathbf{x}}^\perp$ being the position operator in the plane perpendicular to the beam axis. Looking at the expression for the gaussian wave packet in (2.3.5), we can think of having took out of ϕ_b the exponential factor $e^{-i\mathbf{k}\cdot\langle\hat{\mathbf{x}}^\perp\rangle}$, whereas in ϕ_a it can be set to zero by choosing the average position of the wave packet to be at the origin. In this way we can model the initial states as wave packets that are localized in both position and momentum space, with the second wave packet shifted by the impact parameter $\mathbf{b}$ with respect to the first one, to account for the spatial separation of the incoming particles in the scattering process.

Outgoing states. For the outgoing states we can consider just the momentum eigenstates, since the scattering has already happened, thus we can write

$$|\phi_i\rangle = |p_i\rangle \,, \quad \forall i = 1, \ldots, N.$$

Since the outgoing particles are detected at macroscopic distances, we can treat them as free particles and thus we can use momentum eigenstates to describe them: the scattering already happened, thus we can treat the outgoing particles as free particles.

Now we can compute the **S-matrix elements** for the scattering process using the initial and final states defined above. We can write the S-matrix elements for momentum eigenstates as

$$\langle f, \infty|i, -\infty\rangle = \langle f| \, \hat{S} \, |i\rangle = \langle p_1, \ldots, p_N| \, \hat{S} \, |p_a, p_b\rangle \,,$$

where we are considering the initial state as a two particle state with momenta p_a and p_b, and the final state as a N particle state with momenta $p_1, \ldots, p_N$. The S-matrix can be expressed as

$$\hat{S} = \mathbb{I} + i\hat{T},$$

where the identity term corresponds to the case where no scattering occurs, while the $i\hat{T}$ term corresponds to the case some interactions among the fields, i.e. the scattering occurs. Thus we can write[1]

$$\langle p_1, \ldots, p_N| \, i\hat{T} \, |p_a, p_b\rangle = (2\pi)^4 \delta^{(4)}(p_a + p_b - \sum_{i=1}^{N} p_i) i\mathcal{M},$$

where the delta function ensures the conservation of energy and momentum, and $\mathcal{M}$ is the **invariant matrix element** or **scattering amplitude** for the process, which encodes the dynamics

[1] Note that we evaluate the transition amplitude here using exact momentum eigenstates $|p_a, p_b\rangle$ rather than the initial wave packets $|\phi_a, \phi_b\rangle$. This is done to formally define the invariant scattering amplitude $\mathcal{M}$ and cleanly factor out the energy-momentum conserving Dirac delta $\delta^{(4)}$. The physical wave packets enter the calculation in the subsequent step: exploiting the linearity of the $\hat{T}$ operator, the realistic transition amplitude is obtained by integrating these exact-momentum matrix elements over the momentum distributions $\phi_a(\mathbf{k})$ and $\phi_b(\mathbf{k})$ of the incoming wave packets.

of the interaction (we omitted its dependencies for simplicity, but it relies on the initial and final states, thus $\mathcal{M} = \mathcal{M}(p_a, p_b \to p_1, \ldots, p_N)$). The invariant amplitude can be computed using Feynman diagrams and the rules of quantum field theory, and it is a key quantity in determining the cross section for the scattering process.

Thus we can now compute the **differential cross section** in the space of final configurations as in (2.3.4):

$$\frac{\mathrm{d}\sigma}{\mathrm{d}f} = \int \mathrm{d}^2\mathbf{b}\, \frac{\mathrm{d}P}{\mathrm{d}f},$$

where the differential probability of scattering in the space of final states is given by

$$\frac{\mathrm{d}P}{\mathrm{d}f} = |\, \langle p_1,\, \ldots,\, p_N,\, +\infty|\, i\hat{T}\, |\phi_a,\, \phi_b,\, -\infty\rangle\, |^2,$$

and the measure of the space of final states for N particles is normalized as[2]

$$\mathrm{d}f = \prod_{i=1}^{N} \frac{\mathrm{d}^3\mathbf{p}_i}{(2\pi)^3}\frac{1}{2E_{p_i}}.$$

Now we can put everything back together and after some algebra (full computation in Appendix B) we can write the final expression for the differential cross section

TODO: write it.

$$\begin{aligned}
\mathrm{d}\sigma &= \frac{1}{\Phi}|\mathcal{M}(p_a, p_b \to p_1, \ldots, p_N)|^2 \mathrm{d}\Pi_N,\\
\mathrm{d}\Pi_N &= \left[\prod_{i=1}^{N} \frac{\mathrm{d}^3\mathbf{p}_i}{(2\pi)^3 2E_{p_i}}\right](2\pi)^4\delta^{(4)}(p_a + p_b - \sum_{i=1}^{N} p_i),\\
\Phi &= 2E_a 2E_b|v_a - v_b| = 4|E_b p_a^z - E_a p_b^z|,
\end{aligned}$$

where:

- $\mathrm{d}\Pi_N$ is the Lorentz-invariant phase space measure for N final state particles, which accounts for the density of final states and the conservation of energy and momentum through the delta function.
- Φ is the flux factor, which accounts for the incoming particles and their relative velocity, ensuring that the cross section is properly normalized with respect to the initial state of the scattering process.
- $\mathcal{M}$ is again the invariant matrix element or scattering amplitude, which encodes the dynamics of the interaction and can be computed using Feynman diagrams and the rules of quantum field theory.

Φ is the only non Lorentz invariant term: we have indeed

$$\Phi = 4|E_b p_a^z - E_a p_b^z|,$$

where z is the beam axis, thus it is not invariant. But we can recognize its invariance under:

[2] We can convince ourself with the resolution of the identity for 1 particle states, which reads

$$\mathbb{I} = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\frac{1}{2E_{\mathbf{k}}}\, |\mathbf{k}\rangle\langle\mathbf{k}|\,,$$

where $\mathrm{d}f = \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\frac{1}{2E_{\mathbf{k}}}$ is the measure of the space of final states for 1 particle.

1. boosts along the beam axis, since E_a and p_a^z transform in the same way, and E_b and p_b^z also transform in the same way, thus the combination $E_b p_a^z - E_a p_b^z$ is invariant under boosts along the beam axis.

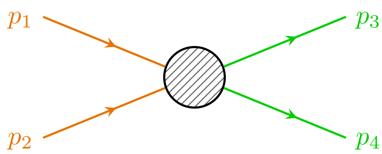
2. rotations around the beam axis (in the perpendicular plane), since E_a, E_b, p_a^z and p_b^z are scalars under rotations around the beam axis, thus the combination $E_b p_a^z - E_a p_b^z$ is invariant under rotations around the beam axis.

Thus the flux factor Φ transforms as the inverse of an area in the plane perpendicular to the beam axis, behaving as a cross section (which is consistent with the interpretation of the cross section as an effective area that quantifies the likelihood of a scattering event occurring when a particle from the beam interacts with the target).

Even if we focus on the case of a beam of particles scattering on a target, the computed expression for the differential cross section remains symmetric for exchange of beam and target, thus it can be applied to any scattering process, regardless of the specific roles of the particles involved. We can also imagine to use it for collisions of two beams of particles, as in e^+e^- collisions in LEP, $p\bar{p}$ collisions in the Tevatron, or more recently pp collisions in the LHC.

We are basically performing a change of reference frame in order to reverse the problem: we can consider the target as the beam and the beam as the target, it all depends on the symmetries and the choice of the reference frame. This is a common technique in scattering theory, where we can choose a reference frame that simplifies the calculations (even center of mass, it depends) and makes it easier to analyze the scattering process. By considering different reference frames, we can gain insights into the dynamics of the interaction and compute cross sections more efficiently.

Example (2 → 2 scattering in center of mass frame). We can consider a simple scattering process where two particles collide and produce two particles in the final state. In the center of mass frame, the initial momenta of the particles are equal and opposite, and we can analyze the scattering process using the invariant amplitude $\mathcal{M}$ and the phase space factor Φ to compute the differential cross section for this process.



The incoming particles have momenta p_1 and p_2, and the outgoing particles have momenta p_3 and p_4, and their spatial parts is conserved: $\mathbf{p}_1 + \mathbf{p}_2 = \mathbf{p}_3 + \mathbf{p}_4 = 0$ (since we are in the center of mass frame). The energy of the particles is given by the energy momentum relation $E_i = \sqrt{\mathbf{p}_i^2 + m_i^2}$, where m_i is the mass of the particle. We can write

$$\mathrm{d}\Pi_2 = \frac{\mathrm{d}^3\mathbf{p}_3}{(2\pi)^3 2E_3}\frac{\mathrm{d}^3\mathbf{p}_4}{(2\pi)^3 2E_4}(2\pi)^4\delta^{(4)}(p_1 + p_2 - p_3 - p_4),$$

which, after integrating in $\mathrm{d}^3\mathbf{p}_4$ using the delta function, gives $\mathbf{p}_4 = -\mathbf{p}_3$; using then polar coordinates for $\mathrm{d}^3\mathbf{p}_3 = p^2\,\mathrm{d}p\,\mathrm{d}\Omega$, we can write

$$E_3 = \sqrt{p^2 + m_3^2}, \quad E_4 = \sqrt{p^2 + m_4^2}, \quad p = |\mathbf{p}_3| = |\mathbf{p}_4|,$$

and thus

$$\int \mathrm{d}p\, \delta(E_1 + E_2 - E_3 - E_4) = \left| \frac{p}{E_3} + \frac{p}{E_4} \right|^{-1} = \frac{E_3 E_4}{p(E_3 + E_4)} = \frac{E_3 E_4}{pE},$$

where the last energy E is the total energy of the system in the center of mass frame, which is given by $E = E_1 + E_2 = E_3 + E_4$. Thus we can write

$$\mathrm{d}\Pi_2 = \mathrm{d}\Omega \frac{p}{16\pi^2 E},$$

with a flux $\Phi = 2E_1 2E_2 |v_1 - v_2|$, with now

$$|v_1 - v_2| = \left| \frac{|\mathbf{p}_1|}{E_1} + \frac{|\mathbf{p}_2|}{E_2} \right| = p_{in} \frac{E}{E_1\, E_2},$$

where p_{in} is the magnitude of the incoming momenta (which are equal in the center of mass frame) $p_{in} = |\mathbf{p}_1| = |\mathbf{p}_2|$. Thus we can write the final expression for the differential cross section as

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{1}{64\pi^2 E^2} \frac{p}{p_{in}} |\mathcal{M}|^2,$$

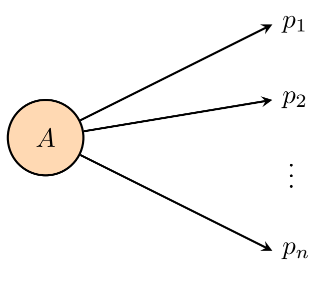
where we used the fact that $E_1 E_2 = \frac{E^2}{4}$ in the center of mass frame. This expression allows us to compute the *differential cross section for the 2 → 2 scattering process only in the center of mass frame*, given the invariant amplitude $\mathcal{M}$ and the kinematic variables of the process.

Remark. *Most of the time this problem is independent of the azimuthal angle ϕ, thus we can integrate over it and write $\mathrm{d}\Omega = 2\pi \mathrm{d}\cos\theta$, thus simplifying the expression for the differential cross section.*

Furthermore $\mathrm{d}\sigma$ is invariant under boosts along the beam axis, but the solid angle $\mathrm{d}\Omega$ is not, thus the expression for the differential cross section $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}$ is not invariant.

2.3.3 | Decay Rates

We can similarly deal with **unstable particles**, which after a certain lifetime can decay into other lighter particles. Systems composed by unstable particles can be analyzed using the S-matrix formalism in a similar way to how we compute scattering amplitudes and cross sections for stable particles. By analyzing the transition amplitudes for the decay process, we can determine the decay rate and the branching ratios for different decay channels, which provide insights into the properties of the unstable particle and its interactions with other particles.



The **decay rate** Γ of an unstable particle can be computed as

$$\Gamma = \frac{\text{Number of decays per unit time}}{\text{Number of particles}},$$

and we can define the **lifetime** τ of the particle as the inverse of the decay rate:

$$\tau = \frac{1}{\Gamma}.$$

Decay rates and lifetimes are defined in the rest frame of the particle, where the particle is at rest and its energy is equal to its mass $E = m$ and its momentum is zero $\mathbf{p} = 0$.

The decay rate can be computed using the S-matrix element for the transition from the initial state (the unstable particle) to the final state (the decay products), and it is related to the **invariant**

amplitude $\mathcal{M}$ for the decay process. We cannot directly apply the limit for $t_i \to -\infty$, since the particle is unstable and it would have decayed before, it cannot exist for an infinite amount of time. We therefore need to consider a finite time interval for the decay process, but we can make a simplification: if we assume the particle to have a lifetime way longer than the time scale of the interaction (always true for most practical cases), we can still apply the S-matrix formalism and compute the decay rate using the invariant amplitude for the decay process, and we will have to take into account the instability of the particle only in one step of the calculation.[3]

TODO: check footnote.

Defining the **differential decay rate** as

$$\frac{\mathrm{d}\Gamma}{\mathrm{d}f} = \frac{1}{T}\mathrm{d}P, \tag{2.3.6}$$

where T is the time interval during which the decay process occurs $T = t_f - t_i$, and $\mathrm{d}P$ is the probability of decaying from the initial state to the final state within that time interval; we can compute the total decay rate by integrating over the phase space of the final states; let us first write the expression for the delta function in the limit of finite and large T:

$$\delta(0) = \int_{t_i}^{t_f} \mathrm{d}t\, e^{i(E_i - E_f)t}|_{t_f \to \infty, t_i \to -\infty},$$

which is the critical step in the calculation, since it allows us to regularize the delta function and make sense of the transition amplitude for the decay process; the complete calculation is shown in Appendix B. Having solved the subtleties of defining the asymptotic states for unstable particles, the final expression for the decay rate after integrating over the phase space of the final states is

TODO: write it.

$$\mathrm{d}\Gamma = \frac{1}{2m}|\mathcal{M}_{p_i \to p_f}|^2 \mathrm{d}\Pi_N, \tag{2.3.7}$$

where m is the mass of the unstable particle, $\mathcal{M}_{p_i \to p_f}$ is the invariant amplitude for the decay process, and $\mathrm{d}\Pi_N$ is the phase space factor for the N-particle final state, which accounts for the kinematics of the decay products and ensures that energy and momentum are conserved in the decay process. It is very similar to the expression for the differential cross section, with some differences concerning the flux factor term $\Phi = 2E_a 2E_b|v_a - v_b|$:

- $2E_a \to 2m$ since we are in the rest frame of the unstable particle, thus $E_a = m$;
- $2E_b|v_a - v_b|$ is not present since we are not dealing with a scattering process, but rather with a decay process, thus there is no incoming particle b and no relative velocity to consider.

This expression allows us to compute the decay rate for an unstable particle given the invariant amplitude and the kinematic variables of the decay process.

Identical particles. When dealing with identical particles among the decay products, the previous formula are still valid, with the appropriate modifications to take into account the symmetrization (for bosons) or antisymmetrization (for fermions) of the wave function. This means that the scattering amplitude must be modified to include the appropriate symmetry factors, which can affect the calculation of cross sections and decay rates in order to consider the spin statistics theorem

$$|p_3, p_4\rangle = \pm |p_4, p_3\rangle.$$

[3] We will consider a finite time interval for the decay process and then take the limit of large time intervals to recover the usual expression for the decay rate.

This can be accomplished in two ways: one can restrict the integration over the phase space to a specific region that accounts for the indistinguishability of the particles (thus over all final inequivalent states), or one can include a symmetry factor in the calculation of the amplitude (such as the permutation number $\frac{1}{N!}$) to account for the number of identical particles in the final state. Both approaches ensure that the physical predictions are consistent with the principles of quantum mechanics and the statistics of identical particles.

2.4 | LSZ reduction and Green's functions

In this section we aim to show how the S-matrix elements can be computed from the Green's functions of the interacting theory. This is done by the LSZ reduction formula, which relates the S-matrix elements to the time-ordered correlation functions of the fields.

The Green's functions of the interacting theory are defined as the time-ordered correlation functions of the field operators in the interacting theory, evaluated in the vacuum state of the interacting theory:

$$\langle f| \hat{S} |i\rangle \to \langle\Omega| T\{\hat{\phi}(x_1)\hat{\phi}(x_2)\cdots\hat{\phi}(x_n)\} |\Omega\rangle \tag{2.4.1}$$

where

- $\hat{\phi}$ is the field operator of the interacting theory,
- $|\Omega\rangle$ is the vacuum state of the interacting theory,

where we highlighted the difference among the vacuum state of the interacting theory and the free theory one: interactions can modify the structure of the vacuum and the Green's functions of the interacting theory encode all the information about its dynamics; they are the building blocks for computing physical observables such as scattering amplitudes and cross sections.

We now start from the structure analysis of the two point functions $\langle\Omega| \hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\} |\Omega\rangle$ in an interacting theory, investigating its physical content and how it relates to the S-matrix elements $\langle f| \hat{S} |i\rangle$, finally relating it to the LSZ reduction formula for connecting the scattering matrix elements to the n-point Green's functions. We will start from a **real scalar theory** before generalizing to more complicated theories.

2.4.1 | Two Point Functions

TODO: Add reference to Peskin 7.1

To analyze the structure of the two point function, we start from the definition of the time-ordered two point function in the interacting theory:

$$\langle\Omega| T\{\hat{\phi}(x)\hat{\phi}(y)\} |\Omega\rangle\,,$$

the idea is now to insert a complete set of physical states $\langle\lambda|$[4] as a resolution of the identity $\mathbb{I} = \sum_\lambda |\lambda\rangle\langle\lambda|$ between the two field operators; but we have to be more precise, since the vacuum state $|\Omega\rangle$ is a physical state of the theory, we have to separate it from the other states in the spectrum of the theory, so that we can write the resolution of the identity as

$$\mathbb{I} = |\Omega\rangle\langle\Omega| + \sum_{\lambda\neq\Omega} |\lambda\rangle\langle\lambda|\,,$$

where $|\lambda\rangle$ are seen as eigenstates of the momentum operator $\hat{\mathbf{P}}$ plus other quantum numbers, so that we can write them as $|\lambda_{\mathbf{p}}\rangle$ to make the momentum dependence explicit. They are normalized according to the relativistic normalization convention, so that we have

$$\langle\lambda_{\mathbf{p}}|\lambda'_{\mathbf{p}'}\rangle = (2\pi)^3 2E_{\mathbf{p}}(\lambda)\delta^{(3)}(\mathbf{p}-\mathbf{p}')\delta_{\lambda,\lambda'},$$

[4] Here we can interpret λ as the set of quantum numbers characterizing the different states of our theory.

where $E_{\mathbf{p}}(\lambda) = \sqrt{\mathbf{p}^2 + m_\lambda^2}$ is the energy of the state $|\lambda_{\mathbf{p}}\rangle$ with momentum $\mathbf{p}$ and mass m_λ. The sum over λ includes all the physical states of the theory, such as the one particle states, the two particle states, the bound states, etc.

We cited the mass m_λ of the state $|\lambda_{\mathbf{p}}\rangle$ as a label for the state, but it is important to note that the mass m_λ is not a free parameter, but it is determined by the dynamics of the theory and it is related to the eigenvalue of the momentum operator $\hat{P}^\mu$ on the state $|\lambda_{\mathbf{p}}\rangle$ by the relativistic dispersion relation:

$$\hat{P}^\mu |\lambda_{\mathbf{p}}\rangle = p^\mu |\lambda_{\mathbf{p}}\rangle \,, \quad p^2 = p_\mu p^\mu = m_\lambda^2,$$

so that the mass m_λ is determined by the eigenvalue of $\hat{P}^2$ on the state $|\lambda_{\mathbf{p}}\rangle$:

$$\hat{P}^2 |\lambda_{\mathbf{p}}\rangle = m_\lambda^2 |\lambda_{\mathbf{p}}\rangle \,.$$

Putting everything together, we can write the resolution of the identity as

$$\mathbb{I} = |\Omega\rangle \langle\Omega| + \sum_{\lambda \neq \Omega} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} |\lambda_{\mathbf{p}}\rangle \langle\lambda_{\mathbf{p}}| \,,$$

with the factor of $1/(2E_{\mathbf{p}}(\lambda))$ coming from the relativistic normalization of the states $|\lambda_{\mathbf{p}}\rangle$.

If we now assume $x^0 > y^0$, the time-ordering operator T does not change the order of the fields, and we can neglect it in order to write the two point function as:

$$\begin{aligned}
\langle\Omega| \hat{\phi}(x)\hat{\phi}(y) |\Omega\rangle &= \langle\Omega| \hat{\phi}(x) \left(|\Omega\rangle \langle\Omega| + \sum_{\lambda \neq \Omega} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} |\lambda_{\mathbf{p}}\rangle \langle\lambda_{\mathbf{p}}| \right) \hat{\phi}(y) |\Omega\rangle \\
&= \langle\Omega| \hat{\phi}(x) |\Omega\rangle \langle\Omega| \hat{\phi}(y) |\Omega\rangle + \sum_{\lambda \neq \Omega} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} \langle\Omega| \hat{\phi}(x) |\lambda_{\mathbf{p}}\rangle \langle\lambda_{\mathbf{p}}| \hat{\phi}(y) |\Omega\rangle \,,
\end{aligned}$$

and if we now relate $\hat{\phi}(x)$ to $\hat{\phi}(0)$ by a spacetime translation, using the unitary operator $\hat{U}_{\mathbf{p}} = e^{i\hat{P}\cdot x}$ that implements the translation, so that

$$\hat{U}_{\mathbf{p}} = e^{i\hat{P}\cdot x} \implies \hat{\phi}(x) = e^{i\hat{P}\cdot x}\, \hat{\phi}(0)\, e^{-i\hat{P}\cdot x} = \hat{U}_{\mathbf{p}}\, \hat{\phi}(0)\, \hat{U}_{\mathbf{p}}^\dagger,$$

we can compute the terms by assuming that the vacuum is *translationally invariant*, and since we said that $|\lambda_{\mathbf{p}}\rangle$ are eigenstates of the momentum operator $\hat{P}$ with eigenvalue p, we can write:

$$\begin{aligned}
\langle\Omega| \hat{\phi}(x) |\lambda_{\mathbf{p}}\rangle &= \langle\Omega| e^{i\hat{P}\cdot x} \hat{\phi}(0) e^{-i\hat{P}\cdot x} |\lambda_{\mathbf{p}}\rangle \\
&= e^{-ip\cdot x} \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{p}}\rangle \,,
\end{aligned}$$

from vacuum invariance and momentum eigenstate properties. We can now assume a Lorentz transformation Λ and the corresponding unitary operator $\hat{U}(\Lambda)$ to exist such that $\Lambda p = (m_\lambda, \mathbf{0})$ and $\hat{U}(\Lambda) |\lambda_{\mathbf{p}}\rangle = |\lambda_{\mathbf{0}}\rangle$. Hence, inserting the identity operator $\hat{U}(\Lambda)^\dagger \hat{U}(\Lambda) = \mathbb{I}$ two times, we can write

$$\langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{p}}\rangle = \left(\langle\Omega| \hat{U}(\Lambda)^\dagger \right) \left(\hat{U}(\Lambda) \hat{\phi}(0) \hat{U}(\Lambda)^\dagger \right) \left(\hat{U}(\Lambda) |\lambda_{\mathbf{p}}\rangle \right) = \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{0}}\rangle \,,$$

since the vacuum is invariant under Lorentz transformations, and the field operator transforms trivially: $\hat{\phi}(\Lambda 0) = \hat{\phi}(0)$. Thus we can write

$$\langle\Omega| \hat{\phi}(x) |\lambda_{\mathbf{p}}\rangle = \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{0}}\rangle\, e^{-ip\cdot x} = e^{-ip\cdot x} F(\lambda),$$

where we have defined the function $F(\lambda) = \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{0}}\rangle$ which encodes the information about the coupling of the field operator to the state $|\lambda\rangle$. In the particular case of $|\lambda_{\mathbf{p}}\rangle = |\Omega\rangle$, we get

$$\langle\Omega| \hat{\phi}(x) |\Omega\rangle = v = \text{constant},$$

which is a consequence of the translational invariance of the vacuum state and represents the vacuum expectation value of the field operator. We can then always redefine the field operator by subtracting the vacuum expectation value

$$\phi(x) = v + \tilde{\phi}(x), \quad \langle\Omega|\, \tilde{\phi}(x)\, |\Omega\rangle = 0,$$

since the vacuum expectation value of the field operator does not affect the dynamics of the theory, and it can be absorbed into a redefinition of the field operator. This means that we can always choose a field operator such that its vacuum expectation value is zero, which simplifies the analysis of the two point function and the LSZ reduction formula. Thus the two point (after the redefinition of the field operator) function can be written as

$$\langle\Omega|\, \hat{\phi}(x)\hat{\phi}(y)\, |\Omega\rangle = \sum_\lambda |F(\lambda)|^2 \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} e^{-ip\cdot(x-y)},$$

valid for $x^0 > y^0$ and where the last term

$$D(x,\, y,\, m_\lambda) = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} e^{-ip\cdot(x-y)} = \langle 0|\, \hat{\phi}_{0,\lambda}(x)\hat{\phi}_{0,\lambda}(y)\, |0\rangle\,, \tag{2.4.2}$$

is the **free propagator** of a particle with mass m_λ, with $\hat{\phi}_{0,\lambda}$ the free field operator of a particle with mass m_λ: it is the amplitude for a free scalar particle to propagate from point y to point x. We can now reintroduce the time-ordering to write the two point function in a more general form

$$\begin{aligned}\langle\Omega|\, T\{\hat{\phi}(x)\hat{\phi}(y)\}\, |\Omega\rangle &= \Theta(x^0 - y^0)\, \langle\Omega|\, \hat{\phi}(x)\hat{\phi}(y)\, |\Omega\rangle + \Theta(y^0 - x^0)\, \langle\Omega|\, \hat{\phi}(y)\hat{\phi}(x)\, |\Omega\rangle \\ &= \Theta(x^0 - y^0) \sum_\lambda |F(\lambda)|^2 D(x,\, y,\, m_\lambda) + \Theta(y^0 - x^0) \sum_\lambda |F(\lambda)|^2 D(y,\, x,\, m_\lambda).\end{aligned}$$

In this last expression we can recognize the **Feynman propagator** of a particle with mass m_λ:

$$\begin{aligned}D_F(x - y,\, m_\lambda) &= \Theta(x^0 - y^0) D(x,\, y,\, m_\lambda) + \Theta(y^0 - x^0) D(y,\, x,\, m_\lambda) \\ &= \langle 0|\, \hat{T}\{\hat{\phi}_{0,\lambda}(x)\hat{\phi}_{0,\lambda}(y)\}\, |0\rangle = \int \frac{\mathrm{d}^4 p}{(2\pi)^4} \frac{i}{p^2 - m_\lambda^2 + i\epsilon} e^{-ip\cdot(x-y)},\end{aligned} \tag{2.4.3}$$

which is the amplitude for a free scalar particle to propagate from point y to point x in a time-ordered manner, and it is given by the Fourier transform of the free propagator with the appropriate $i\epsilon = i0^+$ prescription to ensure causality.

With this last definition, we can now write the final expression, the **Källén-Lehmann spectral representation** of the two point function, as:

$$\langle\Omega|\, T\{\hat{\phi}(x)\hat{\phi}(y)\}\, |\Omega\rangle = \int \frac{\mathrm{d}M}{2\pi} g(M^2) D_F(x,\, y,\, M), \tag{2.4.4}$$

where we have defined the **spectral density** function $g(M^2)$ as

$$g(M^2) = \sum_\lambda (2\pi) |F(\lambda)|^2 \delta(M^2 - m_\lambda^2). \tag{2.4.5}$$

The spectral density function encodes the information about the spectrum of the theory,[5] and it is a positive function since it is defined as a sum of positive terms. The Källén-Lehmann representation of the two point function shows that the two point function can be written as a superposition of free propagators with different masses, weighted by the spectral density function.

[5]The spectral density function is a measure of the number of states with a given mass: the spectrum of the theory is the set of all possible masses of physical particles.

This spectral representation is very useful for analyzing the properties of the theory, and it will be crucial for deriving the LSZ reduction formula in the next section: it will allow us to relate the S-matrix elements to the poles of the two point function (which correspond to the physical particles in the spectrum of the theory) and to the residues of these poles (which correspond to the field strength renormalization constants of these particles).

Let us make some observations about the Källén-Lehmann representation to understand better the look and physical meaning of the spectral density function $g(M^2)$:

- If $M^2 = m^2$ with m the physical mass of a **one particle state** (i.e. m^2 eigenvalue of $\hat{P}^2$), then the spectral density function has a delta function contribution

$$g(M^2)\Big|_{M^2=m^2} = Z(2\pi)\delta(M^2 - m^2),$$

  with $Z = |F(\lambda = \text{1-part. state})|^2$ which represents the *field strength renormalization constant* of the one particle state. This means that the two point function has a pole at $p^2 = m^2$ with residue Z, which is a consequence of the presence of a one particle state in the spectrum of the theory.

- For a **two particle bound state** we have two cases:
  - If $M^2 \leq (2m)^2$ then the spectral density function has a delta function contribution

$$g(M^2)\Big|_{M^2\leq(2m)^2} = \sum_{\lambda\in\text{2-part. states}} Z_B(2\pi)\delta(M^2 - m_\lambda^2),$$

    with $Z_B = |F(\lambda = \text{2-part. state})|^2$ which represents the *field strength renormalization constant* of the two particle bound state. This means that the two point function has a pole at $p^2 = M_B^2$ with residue Z_B, which is a consequence of the presence of a two particle bound state in the spectrum of the theory.
  - If $M^2 > (2m)^2$ then the spectral density function has a continuous contribution given by the two particle continuum states, which represent the scattering states of two particles with mass m

$$g(M^2) = Z_C \sum_{\lambda\in\text{2-part. cont. states}} (2\pi)\delta(M^2 - m_\lambda^2),$$

    with $Z_C = |F(\lambda = \text{2-part. cont. state})|^2$ which represents the *field strength renormalization constant* of the two particle continuum states. This means that the two point function has a branch cut starting at $p^2 = (2m)^2$, which is a consequence of the presence of the two particle continuum states in the spectrum of the theory.

- The spectral density function $g(M^2)$ is a positive function of the **bare mass** M, since it is defined as a sum of positive terms. This means that the two point function is a positive function, which is a consequence of the unitarity of the theory.

- The physical mass m^2 is the eigenvalue of $\hat{P}^2$, so that it represents the mass of the one particle state in the spectrum of the theory. The physical mass is different from the bare mass M^2 which appears in the Lagrangian of the theory, and it is related to the bare mass by a renormalization procedure.

Thus we can now understand a typical look of the spectral density function $g(M^2)$ for a theory with a one particle state, some two particle bound states and a two particle continuum states:

img/spectral_density_function.jpg

img/analytic_structure_2pt_func_fourier_p2.png

Figure 2.4: On the left image we can see the spectral density function for a theory with a one particle state and a two particle bound and continuum states, with the delta function contributions and the continuous part. On the right image we can see the analytic structure in the complex p^2-plane of the two point function Fourier transform, with the pole at $p^2 = m^2$ corresponding to the one particle state, the poles at $p^2 = M_B^2 \leq (2m)^2$ corresponding to the two particle bound states, and the branch cut starting at $p^2 = (2m)^2$ corresponding to the two particle continuum states.

We can now study again the two point function in momentum space, by performing a Fourier transform of the Källén-Lehmann representation:

$$\begin{aligned}\int \mathrm{d}^4x\, e^{ip\cdot x} \langle\Omega| T\{\hat{\phi}(x)\hat{\phi}(0)\} |\Omega\rangle &= \int \mathrm{d}^4x\, e^{ip\cdot x} \int \frac{\mathrm{d}M}{2\pi} g(M^2) D_F(x,\, 0,\, M) \\ &= \frac{iZ}{p^2 - m^2 + i0^+} + \int_{\sim(2m)^2}^{\infty} \frac{\mathrm{d}M^2}{2\pi} \frac{ig(M^2)}{p^2 - M^2 + i\epsilon},\end{aligned}$$

where we have divided the integral over M^2 into the contribution from the one particle state, which gives a pole at $p^2 = m^2$ with residue iZ, and the contribution from the two particle states, which gives a branch cut starting at $p^2 = (2m)^2$ (these are regular terms for $p^2 \sim m^2$). The contribution from the two particle bound states, if present, would give poles at $p^2 = M_B^2 \leq (2m)^2$ with residue iZ_B (as shown in the above figure), but we have not explicitly written them in the last expression for simplicity. To sum up, in momentum space the two point function has the following analytic structure in the complex p^2-plane:

1. A branch cut starting at $p^2 = (2m)^2$ due to the presence of the two particle continuum states in the spectrum of the theory, which represent the scattering states of two particles with mass m.

2. A pole at $p^2 = m^2$ with residue iZ, which is a consequence of the presence of a one particle state in the spectrum of the theory.

3. Poles at $p^2 \leq (2m)^2$ for each two particle bound state in the spectrum of the theory, with residue iZ_B, which is a consequence of the presence of the two particle bound states in the spectrum of the theory.

For a free theory we would have only the one particle state in the spectrum of the theory, so that the Källén-Lehmann representation of the two point function would reduce to

$$\int \mathrm{d}^4x\, e^{ip\cdot x} \langle 0| T\{\hat{\phi}_0(x)\hat{\phi}_0(0)\} |0\rangle = \frac{i}{p^2 - m^2 + i0^+},$$

with $Z = 1$ since the field operator creates a one particle state with unit probability, and the physical mass m^2 coincides with the bare mass M^2 since there are no interactions in the theory.

For an interacting theory we will have corrections to the physical mass and to the field strength renormalization constant, so that we can write

$$\begin{cases} m = m_\lambda + O(g), \\ Z = 1 + O(g), \end{cases}$$

where g is the coupling constant of the theory, so that the physical mass m^2 and the field strength renormalization constant Z receive corrections from the interactions in the theory.

Example (Dirac Field). In the end we will give rules for the generalization of the LSZ reduction formula to more complicated theories, such as theories with fermions and gauge fields, but let us first analyze the structure of the two point function for a Dirac field. If we compute these last results for a Dirac field, we would find

$$\int \mathrm{d}^4x\, e^{ip\cdot x} \langle\Omega| \hat{T}\{\hat{\psi}(x)\hat{\bar{\psi}}(0)\} |\Omega\rangle = \frac{iZ(\not{p}+m)}{p^2 - m^2 + i\epsilon},$$

such that

$$\langle\Omega| \hat{\psi}(0) |\lambda_{\mathbf{p},s}\rangle \Big|_{\lambda=1\text{-part}} = u_s(\mathbf{p})F(\lambda = 1\text{-part}) = u_s(\mathbf{p})\sqrt{Z},$$

where $u_s(\mathbf{p})$ is the Dirac spinor for a particle with momentum $\mathbf{p}$ and spin s, with

$$(\not{p} + m) = \sum_s u_s(\mathbf{p})\overline{u}_s(\mathbf{p}),$$

which is the completeness relation for the Dirac spinors. The spectral density function for the Dirac field would have a delta function contribution at $M^2 = m^2$ with residue Z, which corresponds to the one particle state in the spectrum of the theory, and a branch cut starting at $M^2 = (2m)^2$ due to the presence of the two particle continuum states in the spectrum of the theory, which represent the scattering states of two particles with mass m. We would not have two particle bound states in the spectrum of the theory, since the Dirac field does not have self-interactions, so that the spectral density function would not have delta function contributions at $M^2 \leq (2m)^2$ corresponding to two particle bound states. **(?)** Is this last statement true?

Example (Unstable Particle State). If we are considering unstable particles, i.e particles with a decay rate $\Gamma \neq 0$, then the last results are not valid anymore, since we adopted eigenvalues of the momentum operator $\hat{P}^\mu$, and in particular of $\hat{H}$, which cannot be defined for unstable particles:

$$\hat{H} |\psi\rangle = E |\psi\rangle, \hat{S}(t, t_0) |\psi\rangle = e^{i\hat{H}(t-t_0)} |\psi\rangle = e^{iE(t-t_0)} |\psi\rangle,$$

since an eigenstate by itself can only get a phase and does not decay, while an unstable particle state has to decay in order to be unstable.

Thus a rigorous definition of the physical mass for an unstable particle is rather non-trivial; however the decay rate Γ of the unstable particle state is usually very small compared to its lifetime, so that we can consider those as approximated eigenstates of the Hamiltonian.

We have to take into account the next facts:

- If $\Gamma \neq 0$ the pole of the two point function is not anymore on the real axis, but it is shifted in the complex plane, so that it acquires an imaginary part proportional to the decay rate Γ of the unstable particle state.
- If $\Gamma \ll m$, we may still define the physical mass m of the unstable particle state as the real part of the location of the pole of the two point function, since the imaginary part is

small compared to the real part, and it represents a small correction to the physical mass of the unstable particle state.

Thus in the end we can define the physical mass m of the unstable particle state as

$$m^2 = \mathrm{Re}\,[\text{propagator pole}]\,,$$

which gives us a well-defined physical mass for the unstable particle state, and we can write the two point function in momentum space as

$$\int \mathrm{d}^4x\, e^{ip\cdot x} \langle\Omega| T\{\hat{\phi}(x)\hat{\phi}(0)\} |\Omega\rangle = \frac{iZ}{p^2 - m^2 + im\Gamma} + \Lambda(p^2),$$

where $\Lambda(p^2)$ is a regular function for $p^2 \sim m^2$ which encodes the regular contributions from the other states in the spectrum of the theory.

2.4.2 | LSZ Reduction Formula

TODO: Add reference to Schwartz 6.1

Let us start from the definition of the S-matrix elements in terms of the asymptotic in and out states:

$$\langle f| \hat{S} |i\rangle\,,$$

where $|i\rangle$ and $|f\rangle$ are the in and out states of the theory, which are defined as the asymptotic states of the theory at $t \to -\infty$ and $t \to +\infty$ respectively. Asymptotically, the interacting theory reduces to a free theory, so that the in and out states can be approximated by the free particle states of the free theory. This means that the field operators of the interacting theory can be approximated by the field operators of the free theory in this limit, in which they act as creation and annihilation operators for the free particle states (or 1-particle):

$$\hat{\phi}(x)\Big|_{t\to\pm\infty} \sim \sqrt{Z} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \left(\hat{a}_{\mathbf{p}} e^{-ip\cdot x} + \hat{a}^\dagger_{\mathbf{p}} e^{ip\cdot x}\right),$$

where Z is the field strength renormalization constant of the one particle state in the spectrum of the theory, and $\hat{a}_{\mathbf{p}}$ and $\hat{a}^\dagger_{\mathbf{p}}$ are the annihilation and creation operators for the free particle states of the free theory. The energy appearing in the denominator is the physical energy of the one particle state, which is related to the physical mass m by the relativistic dispersion relation $E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2} = p^0$.

Remark. *Note that the annihilation and creation operators $\hat{a}_{\mathbf{p}}$ and $\hat{a}^\dagger_{\mathbf{p}}$ are time independent in a free theory. We will assume that they are also time independent in the interacting theory, when considering the asymptotic limit $t \to \pm\infty$, even if in general*

$$\hat{a}_{\mathbf{p}}(t\to\infty) \neq \hat{a}_{\mathbf{p}}(t \to -\infty),$$

since the interactions can change the number of particles in the theory, so that the annihilation and creation operators at $t \to +\infty$ can be different from those at $t \to -\infty$ (the free-like field configurations may be different after the scattering process).

If we consider a scattering process with 2 incoming particles and n outgoing particles, we can write the in and out states as

$$\begin{aligned} |i\rangle &= \sqrt{2E_a}\sqrt{2E_b}\hat{a}^\dagger_{\mathbf{p}_a}\hat{a}^\dagger_{\mathbf{p}_b} |\Omega\rangle\,, \\ |f\rangle &= \sqrt{2E_1}\sqrt{2E_2}\cdots\sqrt{2E_n}\,\hat{a}^\dagger_{\mathbf{p}_1}\hat{a}^\dagger_{\mathbf{p}_2}\cdots\hat{a}^\dagger_{\mathbf{p}_n} |\Omega\rangle\,, \end{aligned}$$

where we have properly normalized the states by the factors of $\sqrt{2E}$ according to the relativistic normalization convention. It is now time to compute the S-matrix element $\langle f|\,\hat{S}\,|i\rangle$ by inserting the expressions for the in and out states and recalling that $\hat{S} = \mathbb{I} + i\hat{T}$, such that we can write

$$\begin{aligned}\langle f|\,\hat{S}\,|i\rangle = \langle f|\,\mathbb{I} + i\hat{T}\,|i\rangle = \langle f|\,i\hat{T}\,|i\rangle = \sqrt{2E_a}\sqrt{2E_b}\sqrt{2E_1}\sqrt{2E_2}\cdots\sqrt{2E_n}\times \\ \times \langle\Omega|\,\hat{a}_{\mathbf{p}_1}(+\infty)\hat{a}_{\mathbf{p}_2}(+\infty)\cdots\hat{a}_{\mathbf{p}_n}(+\infty)\left(i\hat{T}\right)\hat{a}^\dagger_{\mathbf{p}_a}(-\infty)\,\hat{a}^\dagger_{\mathbf{p}_b}(-\infty)\,|\Omega\rangle\,.\end{aligned}$$

The idea is now to rewrite everyting in terms of Green's functions of the theory, so let us first prove the following identity:

$$\begin{aligned}&\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x\, e^{ip\cdot x}(\partial_x^2 + m^2)\hat{T}\{\hat{\phi}(x)\hat{O}(y_1,\ldots,y_n)\} \\ &= \sqrt{2E_{\mathbf{p}}}\left(\hat{a}_{\mathbf{p}}(+\infty)\hat{T}\hat{O}(y_1,\ldots,y_n) - \hat{T}\hat{O}(y_1,\ldots,y_n)\hat{a}_{\mathbf{p}}(-\infty)\right),\end{aligned}$$

where $\hat{O}(y_1,\ldots,y_n)$ is an arbitrary product of operators of the theory, and we have used the fact that the field operator $\hat{\phi}(x)$ can be approximated by the free field operator in the asymptotic limit.

In order to prove this identity, we now define $\hat{\mathcal{O}}(x) = \hat{T}\{\hat{\phi}(x)\hat{O}(y_1,\ldots,y_n)\}$ as the time ordered product of the field operator and the arbitrary product of operators, so that we can write

$$\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x\, e^{ip\cdot x}(\partial_x^2 + m^2)\hat{\mathcal{O}}(x) = \frac{i}{\sqrt{Z}}\int \mathrm{d}^4x\, e^{ip\cdot x}(\partial_t^2 - \partial_i^2 + m^2)\hat{\mathcal{O}}(x),$$

where we have explicitly written the derivatives in terms of the time and spatial derivatives. We can now integrate by parts the spatial derivatives, so that it changes sign and start acting on the left $\overrightarrow{\partial}_i \to \overleftarrow{\partial}_i$,[6] getting to

$$\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x\, e^{ip\cdot x}(\partial_t^2 + \mathbf{p}^2 + m^2)\hat{\mathcal{O}}(x),$$

where we can recognize the relativistic dispersion relation $E_{\mathbf{p}}^2 = \mathbf{p}^2 + m^2$ for a particle with mass m and momentum $\mathbf{p}$, so that we can write

$$\frac{1}{\sqrt{Z}}\int \mathrm{d}^4x\, i\, e^{ip\cdot x}(\partial_t^2 + E_{\mathbf{p}}^2)\hat{\mathcal{O}}(x) = \frac{1}{\sqrt{Z}}\int \mathrm{d}^4x\,\partial_t\left(e^{ip\cdot x}(i\partial_t + E_{\mathbf{p}})\hat{\mathcal{O}}(x)\right).$$

Now we can split the integral in a spatial and a time integral, so that

$$\frac{1}{\sqrt{Z}}\int \mathrm{d}t\,\partial_t e^{iE_{\mathbf{p}}t}\int \mathrm{d}^3\mathbf{x}\, e^{-i\mathbf{p}\cdot\mathbf{x}}(i\partial_t + E_{\mathbf{p}})\hat{\mathcal{O}}(x),$$

which in this form is a total time derivative (the first integral), so that we can evaluate it at the boundaries $t = \pm\infty$, where we know the field operators to behave as free field operators, and the ladder operators to be time independent; so we can now work on the second integral and substitute the expression for $\hat{\mathcal{O}}(x)$ and the asymptotic behavior of the field operator, getting to

$$\int\frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\int \mathrm{d}^3\mathbf{x}\left[\frac{E_{\mathbf{p}} + E_{\mathbf{k}}}{\sqrt{2E_{\mathbf{k}}}}e^{-iE_{\mathbf{k}}t}e^{-i(\mathbf{p}-\mathbf{k})\cdot\mathbf{x}}\hat{T}\{\hat{a}_{\mathbf{k}}\hat{O}\} + \frac{E_{\mathbf{p}} - E_{\mathbf{k}}}{\sqrt{2E_{\mathbf{k}}}}e^{iE_{\mathbf{k}}t}e^{i(\mathbf{p}-\mathbf{k})\cdot\mathbf{x}}\hat{T}\{\hat{a}^\dagger_{\mathbf{k}}\hat{O}\}\right],$$

in which a spatial integration gives us a delta function $\delta^{(3)}(\mathbf{p}-\mathbf{k})$ which allows us to perform the integral over $\mathbf{k}$, getting to

$$\begin{aligned}\frac{1}{\sqrt{Z}}\int \mathrm{d}^4x\, i\, e^{ip\cdot x}(\partial_t^2 + E_{\mathbf{p}}^2)\hat{\mathcal{O}}(x) &= \int \mathrm{d}t\,\partial_t\left[e^{iE_{\mathbf{p}}t}\sqrt{2E_{\mathbf{p}}}\left(e^{-iE_{\mathbf{p}}t}\hat{T}\{\hat{a}_{\mathbf{p}}(t)\hat{O}\}\right)\right] \\ &= \sqrt{2E_{\mathbf{p}}}\left[\hat{a}_{\mathbf{p}}(+\infty)\hat{T}\hat{O}(y_1,\ldots,y_n) - \hat{T}\hat{O}(y_1,\ldots,y_n)\hat{a}_{\mathbf{p}}(-\infty)\right],\end{aligned}$$

[6] We are assuming that the fields vanish at spatial infinity, so that the boundary terms from the integration by parts vanish: then the spatial derivatives will change sign and start acting on the left.

yielding the desired result. We can obtain a similar relation holding for its hermitian conjugate, which is

$$\begin{aligned}&\frac{1}{\sqrt{Z}}\int \mathrm{d}^4x\,(-i)\,e^{-ip\cdot x}(\partial_x^2+m^2)\hat{T}\{\hat{\phi}(x)\hat{O}(y_1,\dots,y_n)\}\\&=\sqrt{2E_{\mathbf{p}}}\left[\hat{a}^\dagger_{\mathbf{p}}(+\infty)\hat{T}\hat{O}(y_1,\dots,y_n)-\hat{T}\hat{O}(y_1,\dots,y_n)\hat{a}^\dagger_{\mathbf{p}}(-\infty)\right].\end{aligned}$$

Now we can apply the derived result to the expression for the S-matrix element $\langle f|\,\hat{S}\,|i\rangle$ that we wrote above, so that we can write

$$\begin{aligned}\langle f|\,\hat{S}\,|i\rangle &= \sqrt{2E_a 2E_b 2E_1 2E_2\cdots 2E_n}\\ &\times\langle\Omega|\,\hat{T}\hat{a}_{\mathbf{p}_1}(+\infty)\hat{a}_{\mathbf{p}_2}(+\infty)\cdots\hat{a}_{\mathbf{p}_n}(+\infty)\left(i\hat{T}\right)\hat{a}^\dagger_{\mathbf{p}_a}(-\infty)\hat{a}^\dagger_{\mathbf{p}_b}(-\infty)\,|\Omega\rangle\,,\end{aligned}$$

where we inserted a time ordering operator T to use the derived result (the time ordering does not change the result since the ladder operators are already time-ordered). We can now apply the derived result for each of the ladder operators in the above expression: while we consider $\hat{a}_{\mathbf{p}_1}$ all the other ladder operators can be grouped in the arbitrary product of operators $\hat{O}$ in the derived result, so that we can apply it to each of the ladder operators in the above expression, getting to the famous **LSZ reduction formula**, which is essentially

$$\begin{aligned}\langle f|\,\hat{S}\,|i\rangle =& \left[\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x_a\,e^{-ip_a\cdot x_a}(\partial^2+m^2)\phi(x_a)\right]\left[\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x_b\,e^{-ip_b\cdot x_b}(\partial^2+m^2)\phi(x_b)\right]\\ &\times\left[\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x_1\,e^{ip_1\cdot x_1}(\partial^2+m^2)\phi(x_1)\right]\cdots\left[\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x_n\,e^{ip_n\cdot x_n}(\partial^2+m^2)\phi(x_n)\right]\\ &\times\langle\Omega|\,T\{\phi(x)\phi(y)\dots\phi(x_1)\phi(x_2)\dots\phi(x_n)\}\,|\Omega\rangle\,,\end{aligned}\tag{2.4.6}$$

which is a way to relate the S-matrix elements to the time-ordered correlation functions of the fields, which are the Green's functions of the theory. The LSZ reduction formula is a powerful tool for computing the S-matrix elements from the Green's functions, and it is widely used in quantum field theory to compute scattering amplitudes and cross sections for various processes.

The LSZ reduction formula can be interpreted as a way to amputate the external legs of the Green's functions, which correspond to the free propagators of the theory, and to put them on shell by applying the operator (∂^2+m^2) to them, which gives zero when acting on the free propagators. The factors of $1/\sqrt{Z}$ in the formula are necessary to cancel the field strength renormalization constants that appear in the residues of the poles of the Green's functions, so that we can obtain finite results for the S-matrix elements.

Substantially, it is a bunch of Fourier transforms of the field operators, where the only difference is in the phase signs: incoming particles have a negative phase sign, while outgoing particles have a positive phase sign.

If we rewrite it in the momentum space, integrating by parts the derivatives so that they act on the exponentials on their left (which corresponds to the substitution $i(\partial_i^2+m^2)\to -i(p_i^2-m^2)$) we can write then the LSZ reduction formula as

$$\begin{aligned}\langle f|\,\hat{S}\,|i\rangle =& \prod_{i=a,b,1,\dots,n}\left[\frac{i}{\sqrt{Z}}(p_i^2-m^2)\right]\\ &\times\int \mathrm{d}^4x_a\,\mathrm{d}^4x_b\,\mathrm{d}^4x_1\,\cdots\,\mathrm{d}^4x_n\,e^{i(p_1\cdot x_1+\cdots+p_n\cdot x_n-p_a\cdot x_a-p_b\cdot x_b)}\\ &\times\langle\Omega|\,T\{\phi(x_1)\cdots\phi(x_n)\phi(x_a)\phi(x_b)\}\,|\Omega\rangle\,.\end{aligned}\tag{2.4.7}$$

We have assumed asymptotic particles as the final state, with on shell momenta $p_i^2=m^2$, so that the factors of (∂^2+m^2) in the LSZ reduction formula will give zero when acting on the external legs of the Green's functions, which are the free propagators of the theory. This means that the only contribution to the S-matrix elements will come from the poles of the Green's functions, which correspond to the physical particles in the spectrum of the theory:

$$\frac{iZ}{p_i^2 - m^2}.$$

The residues of these poles will give the field strength renormalization constants Z for each particle, which will cancel with the factors of $1/\sqrt{Z}$ in the LSZ reduction formula, giving a finite result for the S-matrix elements. Indeed, as $p_i \to \pm m$, the factors of $(p_i^2 - m^2)$ in the LSZ reduction formula will cancel with the poles of the Green's functions, giving a finite result for the S-matrix elements, since namely its just a series of regular terms for $p_i^2 \sim m^2$ multiplied by $\sqrt{Z}$.

2.4.3 | Generalization to Other Fields

If we repeat the same analysis for different types of fields, such as Dirac fields or gauge fields, we will find similar results for the two point functions and the LSZ reduction formula, with some modifications due to the different spin and statistics of the fields:

- For a **complex scalar field**, we would have two types of particles, the particle and the antiparticle, with different creation and annihilation operators, and the LSZ reduction formula would involve both types of operators for the in and out states:
  - $\phi^\dagger$ for *incoming particles* and *outgoing antiparticles*, with a negative phase sign in the Fourier transform.
  - ϕ for *outgoing particles* and *incoming antiparticles*, with a positive phase sign in the Fourier transform.

  This would lead to a more complicated LSZ reduction formula, but the general structure would be the same as for the real scalar field, and with this prescription we could also compute the results for Dirac theory ($\phi \to \psi$ and $\phi^\dagger \to \bar{\psi}$).

- If we are considering a **theory of several fields**, then the Green function will contain all the different fields along with the different masses of the quanta, and the LSZ reduction formula will involve the corresponding field operators for each type of particle in the in and out states, with the appropriate phase signs in the Fourier transforms.

- When considering **particle with different spins**, we have to take into account the different transformation properties of the fields under Lorentz transformations, which will affect the structure of the two point functions and the LSZ reduction formula. For example, for a Dirac field we would have the following field operators for the in and out states:
  - $\overline{u}_{s,\alpha}$ for *outgoing particles*;
  - $u_{s,\alpha}$ for *incoming particles*;
  - $v_{s,\alpha}$ for *outgoing antiparticles*;
  - $\overline{v}_{s,\alpha}$ for *incoming antiparticles.*

  If we instead consider a spin-1 gauge field, we would have the following field operators for the in and out states:

- $\epsilon_\mu(p,\,\lambda)$ for *outgoing particles*;
- $\epsilon^*_\mu(p,\,\lambda)$ for *incoming particles.*

The spinorial/vectorial indices (α, μ etc.) of the field operators will be contracted with the corresponding indices of the Green's functions in the LSZ reduction formula, and the final expression for the S-matrix elements will involve the appropriate spinor/vector structures for each type of particle in the in and out states:

$$\langle\Omega|\,T\{\dots A^\mu(x)\dots\psi_\alpha(y)\dots\}\,|\Omega\rangle\,.$$

- For a Dirac fermion the factor (∂^2-m^2) in the LSZ reduction formula will be replaced by the Dirac operator $(\not\partial-m)$, which gives zero when acting on the free propagator of the Dirac field, and the field strength renormalization constant Z will be replaced by the corresponding field strength renormalization constant for the Dirac field. Furthermore external factors $\not p-m$ will cancel out the poles of the Green's functions, leaving a factor of $\sqrt{Z}$ as $p_i\to$ *on-shell.*

We have to make one final observation: we did not make any assumption on the behaviour of the field operators of the interacting theory, apart from the fact that they can be approximated by the field operators of the free theory in the asymptotic limit (as ladder operat), which is roughly equivalent to the statement

$$\langle\Omega|\,\hat\phi(x)\,|\lambda_{\mathbf{p}}\rangle=\sqrt{Z}e^{-ip\cdot x},$$

whose validity was argued when we derived the Källén-Lehmann representation of the two point function. This means that the LSZ reduction formula is valid for any interacting theory, as long as the field operators of the interacting theory can be approximated by the field operators of the free theory in the asymptotic limit, which is a very general condition that is satisfied by most of the theories we are interested in.

Hence, our derivation is still valid when replacing the one particle state $|\mathbf{p}\rangle$ to any state $|\lambda_{\mathbf{p}}\rangle$ and the field operator $\hat\phi(x)$ to any operator $\hat O(x)$ of the theory, as long as the following condition is satisfied:

$$\langle\Omega|\,\hat O(x)\,|\lambda_{\mathbf{p}}\rangle\sim e^{-ip\cdot x},$$

which is valid as long as $\langle\Omega|\,\hat O(x)\,|\lambda_{\mathbf{p}}\rangle\neq 0$, so that the operator $\hat O(x)$ creates a one particle state from the vacuum with non-zero probability, and the LSZ reduction formula can be applied to any such operator. The LSZ reduction formula is a very general result that can be applied to a wide range of theories and operators, and it is a powerful tool for computing the S-matrix elements from the Green's functions of the theory.

Finally, inserting any operator $\hat O(x)$ among asymptotic states in the LSZ reduction formula is allowed in the form of the following substitutions:

$$\begin{cases}\langle f|\,\hat S\,|i\rangle=\langle f,t=+\infty|i,t=-\infty\rangle\\ \langle\Omega|\,\hat T\{\hat\phi(x_1)\dots\hat\phi(x_n)\}\,|\Omega\rangle\end{cases}\quad\Longrightarrow\quad\begin{cases}\langle f,t=+\infty|\,\hat O(x)\,|i,t=-\infty\rangle\,,\\ \langle\Omega|\,\hat T\{\hat\phi(x_1)\dots\hat\phi(x_n)\hat O(x)\}\,|\Omega\rangle\,,\end{cases}$$

respectively for the l.h.s and the r.h.s of the LSZ reduction formula. This means that we can compute matrix elements of any operator $\hat O(x)$ between asymptotic states by using the LSZ reduction formula, which is a very powerful result that allows us to compute a wide range of physical observables.

3 Feynman Diagrams

In the previous chapters we have seen how to calculate Green's functions in QFT and, as we have notice, the number of terms in the expansion of the Green's function grows very quickly as we increase the order of the correction, making it very difficult to keep track of all the terms and their contributions. This is where Feynman diagrams come in handy, as they provide a **visual representation of the interactions** and allow us to easily keep track of the different terms in the expansion.

In this chapter we will see how to derive a set of practical rules, known as Feynman rules, that allow us to compute Green's functions and scattering amplitudes in a systematic way for an *interacting theory in perturbation theory*, without having to calculate all the contractions explicitly. This method is based on Tylor expansions in the coupling constants g_j, assuming a weak enough interaction, and it is a very powerful tool for calculating physical quantities in QFT, as it allows us to write down the contribution of each diagram without having to calculate all the contractions explicitly, which is a very powerful tool for calculating physical quantities in QFT.

We will start by reviewing the Hamiltonian formalism and the interaction picture, which are essential for understanding how to calculate Green's functions in an interacting theory, and then we will see how to derive the Feynman rules for a simple interacting theory, and how to use them to calculate physical quantities. Through some examples, we will see how to apply the Feynman rules in practice, and then we will see how to generalize them to more complicated theories and to momentum space, which is the most common way to use Feynman diagrams in QFT.

3.1 | Hamiltonian Formalism and Interaction Picture

In QFT is often more convenient to work in the Hamiltonian formalism, where we express the dynamics of the system in terms of a Hamiltonian operator. The Hamiltonian can be split into a free part, which describes non-interacting particles, and an interaction part, which describes the interactions between particles.

The **Heisenberg picture** is often used in QFT, where operators evolve in time while states remain constant. However, for perturbative calculations, it is more convenient to use the **interaction picture**, where both operators and states evolve in time. Let us start from the Heisenberg equation of motion for an operator $\hat{O}(x)$:

$$i\partial_t \hat{O}(x) = [\hat{O}(x), \hat{H}(x)],$$

where $\hat{H}(x)$ is the Hamiltonian operator. We can write the solution to this equation through the time evolution operator $\hat{S}(t, t_0)$ as

$$\hat{O}(\mathbf{x}, t) = \hat{S}^\dagger(t, t_0)\hat{O}(\mathbf{x}, t_0)\hat{S}(t, t_0),$$

which can be inserted back into the Heisenberg equation of motion to get an equation for the time evolution operator:

$$i\partial_t \hat{S}(t, t_0) = \hat{H}(t)\hat{S}(t, t_0).$$

Now let us split the Hamiltonian into a free part and an interaction part:

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t),$$

and the Lagrangian into a free part and an interaction part:

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_{\text{int}}.$$

In practice, $\hat{H}_0$ will be the Hamiltonian of a free theory, which describes non-interacting particles whose dynamics can be solved exactly, while $\hat{V}(t)$ will be the interaction Hamiltonian, which will be treated as a perturbation and encodes all the interaction terms. The free part of the Lagrangian $\mathcal{L}_0$ will describe the dynamics of the free fields, while the interaction part $\mathcal{L}_{\text{int}}$ will describe the interactions between the fields, indeed we can write the Legendre transform of the Lagrangian to get the Hamiltonian as

$$H = \int \mathrm{d}^3\mathbf{x}\, \mathcal{H}(\mathbf{x}), \qquad \begin{cases} \mathcal{H} = \pi\dot{\phi} - \mathcal{L}, \\ \pi = \partial\mathcal{L}/\partial\dot{\phi}, \end{cases}$$

and if $\mathcal{L}_{\text{int}}$ does not contain derivatives, then we simply have $\mathcal{H}_{\text{int}} = -\mathcal{L}_{\text{int}}$, and thus we can write the interaction Hamiltonian as

$$V = -\int \mathrm{d}^3\mathbf{x}\, \mathcal{L}_{\text{int}}.$$

If $\mathcal{L}_{\text{int}}$ does indeed contain derivatives, then similar results can be obtained, but it will be easier to derive them in the Lagrangian formalism, which is more convenient for theories with derivative interactions, and we will see how to do that in the next chapters.

3.1.1 | Interaction Picture

We are now moving from the Heisenberg picture to the interaction picture, where both operators and states evolve in time. In the interaction picture, the *fields evolve according to the free Hamiltonian H_0*, while the *states evolve according to the total Hamiltonian V*: the interaction is encoded in the evolution of the states, while the fields evolve as if they were free fields. This is a very important point, as it allows us to use the free fields to calculate the interactions, which is much easier than dealing with the full interacting fields.

Then of course the *matrix elements* we are interested in computing *will be independent from the chosen picture*, since they are physical quantities, and thus we can choose the picture that is more convenient for our calculations. Schrödinger, Heisenberg and interaction picture are just different ways of describing the same physics, and they are identically equivalent equivalent at a conventional reference time $t = t_0$.

Thus we can start from the Schrödinger picture, where the fields are time-independent, and then we can move to the interaction and Heisenberg picture, where the fields evolve in time according to the free Hamiltonian and the total Hamiltonian respectively. We can write the fields in the different pictures and see how they are related to each other as

$$\begin{cases} \hat{\phi}_0 = \hat{\phi}_I = e^{i\hat{H}_0(t-t_0)}\hat{\phi}_S e^{-i\hat{H}_0(t-t_0)}, \quad \hat{\phi}_S = \hat{\phi}_I(t = t_0) \\ \hat{\phi} = \hat{\phi}_H = \hat{S}^\dagger(t,t_0)\hat{\phi}_S\hat{S}(t,t_0) = \hat{U}^\dagger(t,t_0)\hat{\phi}_I\hat{U}(t,t_0), \end{cases}$$

where

$$\hat{U}(t,t_0) = e^{i\hat{H}_0(t-t_0)}\hat{S}(t,t_0),$$

is the time evolution operator connecting the interaction picture to the Heisenberg picture, while $\hat{S}(t,t_0)$ is the time evolution operator connecting the Schrödinger picture to the Heisenberg picture.

Now having $\hat{H}_H = \hat{H}_S$ and $\hat{H}_0 = \hat{H}_{0,S}$ (i.e. the Heisenberg Hamiltonian is the same as the Schrödinger Hamiltonian, and the free Hamiltonian is the same in both pictures) we can write the equation of motion for $\hat{U}$ by inserting the fields in the Heisenberg equation of motion, which gives

$$i\partial_t\hat{U}(t,t_0) = \hat{V}_I(t,t_0)\hat{U}(t,t_0),$$

with

$$\hat{V}_I = e^{i\hat{H}_0(t-t_0)}\hat{V}_S e^{-i\hat{H}_0(t-t_0)},$$

which is the **interaction potential** (potential in the interaction picture). In practice, since $\hat{V}$ is given by the integral of the interaction Lagrangian, we can write it as

$$\hat{V}_I = \hat{V}_S(\hat{\phi})\big|_{\hat{\phi}\to\hat{\phi}_I} = \hat{V}(\hat{\phi}_0),$$

which means that the potential in the interaction picture is the same as the potential in the Schrödinger picture, but with the fields replaced by their interaction picture counterparts (which is the same as the free fields). This is a very important result, as it allows us to use the free fields to calculate the interactions, which is much easier than dealing with the full interacting fields.

Dyson Series

The solution to the evolution equation for $\hat{U}$ can be written as the following well-known time-ordered exponential:

$$\hat{U}(t,t_0) = \hat{T}\exp\left(-i\int_{t_0}^t \mathrm{d}t'\hat{V}_I(t')\right),$$

which can be expanded in a power series as

$$\hat{U}(t,t_0) = \mathbb{I} + (-i)\int_{t_0}^t \mathrm{d}t'\hat{V}_I(t') + \frac{(-i)^2}{2!}\int_{t_0}^t \mathrm{d}t'\int_{t_0}^t \mathrm{d}t''\hat{T}\{\hat{V}_I(t')\hat{V}_I(t'')\} + \dots$$

where if we now partition the integration domain of the n-th term of the series in a set of $n!$ subdomains, with the idea to apply the time-ordering operator: up to renaming the integration variables, we can reorder the integration limits as

$$t' > t'' > \cdots > t^{(n)} > t_0,$$

making the integral of the n-th term of the series as

$$\hat{U}(t,t_0) = \mathbb{I} + (-i)\int_{t_0}^{t} \mathrm{d}t' \hat{V}_I(t') + (-i)^2 \int_{t_0}^{t} \mathrm{d}t' \int_{t_0}^{t'} \mathrm{d}t'' \hat{V}_I(t')\hat{V}_I(t'') + \ldots$$

which is known as the **Dyson series**, and it is the solution to the evolution equation for $\hat{U}$. Thus we can write the Dyson series in such a way that the time-ordering operator is not needed, since the integration limits ensure that the operators are ordered correctly in time. In the end we have

$$\hat{U}(t,t_0) = \mathbb{I} - i\int_{t_0}^{t} \mathrm{d}t' \hat{V}_I(t')\hat{U}(t',t_0),$$

which is an integral equation for $\hat{U}$ that can be solved iteratively, and we can see that the Dyson series is the solution to this integral equation. This is a very important result, as it allows us to calculate the time evolution operator in the interaction picture through

$$i\partial_t \hat{U}(t,t_0) = \hat{V}_I(t)\hat{U}(t,t_0),$$

which is essential for calculating scattering amplitudes and other physical quantities in QFT, even if the computation remains quite complicated.

3.1.2 | Computing Green's Functions

TODO: Complete from the note, professor did not finish the topic.

Consider a Green's function of the form

$$\langle\Omega|\, \hat{T}\{\hat{\phi}(x_1)\ldots\hat{\phi}(x_n)\}\, |\Omega\rangle\,,$$

where $|\Omega\rangle$ is the vacuum state of the interacting theory, and $\hat{\phi}(x) = \hat{\phi}_H(x)$ are the interacting fields in the Heisenberg picture. Since the matrix elements are independent from the chosen picture, we can write it in the interaction picture as

$$\langle\Omega|\, \hat{T}\{\hat{\phi}(x_1)\ldots\hat{\phi}(x_n)\}\, |\Omega\rangle = \frac{\langle 0|\, \hat{T}\{\hat{\phi}(x_1)\ldots\hat{\phi}(x_n)\exp\left[-i\int_{-\infty}^{\infty}\mathrm{d}t\, \hat{V}_I(t)\right]\}\, |0\rangle}{\langle 0|\, \hat{T}\{\exp\left[-i\int_{-\infty}^{\infty}\mathrm{d}t\, \hat{V}_I(t)\right]\}\, |0\rangle},$$

where we have used the fact that the vacuum state in the interaction picture is the same as the vacuum state in the Schrödinger picture, and we have also used the fact that the time evolution operator in the interaction picture can be written as a time-ordered exponential of the interaction Hamiltonian.

We can expand the computation considering that the interaction Hamiltonian is given by the integral of the interaction Lagrangian, which is the case for a large class of theories, and thus we can write

$$\mathcal{H}_{int} = -\mathcal{L}_{int}, \qquad \int \mathrm{d}t\, V_I(t) = -\int \mathrm{d}^4x\, \mathcal{L}_{int}(x),$$

so that the Green's function can be written as

$$\langle\Omega|\, \hat{T}\{\hat{\phi}(x_1)\ldots\hat{\phi}(x_n)\}\, |\Omega\rangle = \frac{\langle 0|\, \hat{T}\{\hat{\phi}(x_1)\ldots\hat{\phi}(x_n)\}\exp\left[i\int \mathrm{d}^4x\, \hat{\mathcal{L}}_{int}\right]\, |0\rangle}{\langle 0|\, \hat{T}\{\exp\left[i\int \mathrm{d}^4x\, \hat{\mathcal{L}}_{int}\right]\}\, |0\rangle},$$

which is a very important result, as it allows us to calculate Green's functions in the interaction picture, which is much easier than calculating them in the Heisenberg picture, since we can use the free fields to calculate the interactions. This is the basis for perturbation theory in QFT, where we can expand the exponential of the interaction Lagrangian and calculate the Green's functions order by order in perturbation theory.

3.1.3 | Wick's Theorem

Wick's theorem is a fundamental result in quantum field theory that allows us to express the time-ordered product of field operators as a sum of normal-ordered products and contractions. It is particularly useful for calculating Green's functions and scattering amplitudes in perturbation theory. Let us consider a time-ordered Green's function of the form

$$\langle 0| \hat{T}\{\hat{\phi}(x_1)\dots\hat{\phi}(x_n)\} |0\rangle ,$$

where $\hat{\phi}(x)$ is a free scalar field operator, thus behaving as a ladder operator. We can manipulate the fields as

$$\hat{\phi} = \hat{\phi}^+ + \hat{\phi}^-, \qquad \begin{cases} \hat{\phi}^+ |0\rangle = 0, \\ \langle 0| \hat{\phi}^- = 0, \end{cases}, \qquad \begin{cases} \hat{\phi}^+ = \displaystyle\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_\mathbf{p}}} \hat{a}_\mathbf{p} e^{-ip\cdot x}, \\ \hat{\phi}^- = \displaystyle\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_\mathbf{p}}} \hat{a}^\dagger_\mathbf{p} e^{ip\cdot x}, \end{cases}$$

where $\hat{\phi}^+$ is the annihilation part of the field, and $\hat{\phi}^-$ is the creation part of the field. We can then use the normal ordering of the product of $\hat{a}$ and $\hat{a}^\dagger$ to simplify the expression: it puts all the creation operators to the left of the annihilation operators, thus ensuring that the vacuum expectation value of any normal-ordered product is zero. We will use the notation $:\hat{O}:$ to denote the normal ordering of an operator (or a general product of operators) $\hat{O}$.

Example. Let us consider the product of four ladder operators:

$$\hat{a}(p_1)\hat{a}^\dagger(p_2)\hat{a}(p_3)\hat{a}^\dagger(p_4),$$

then we can normal order this product as

$$:\hat{a}(p_1)\hat{a}^\dagger(p_2)\hat{a}(p_3)\hat{a}^\dagger(p_4) := \hat{a}^\dagger(p_2)\hat{a}^\dagger(p_4)\hat{a}(p_1)\hat{a}(p_3),$$

and we can see that the vacuum expectation value of this normal-ordered product is zero:

$$\langle 0| : \hat{a}(p_1)\hat{a}^\dagger(p_2)\hat{a}(p_3)\hat{a}^\dagger(p_4) : |0\rangle = 0.$$

If we now consider the time-ordered product of two fields, a **2-point T-product**, assuming $x^0 > y^0$, we can write it as

$$\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}_{x^0>y^0} = \hat{\phi}^+(x)\hat{\phi}^+(y) + \hat{\phi}^+(x)\hat{\phi}^-(y) + \hat{\phi}^-(x)\hat{\phi}^+(y) + \hat{\phi}^-(x)\hat{\phi}^-(y),$$

where the only term that is not normal-ordered is the second term. We can then write the normal-ordered product of the two fields as

$$:\hat{\phi}(x)\hat{\phi}(y) := \hat{\phi}^-(x)\hat{\phi}^-(y) + \hat{\phi}^-(x)\hat{\phi}^+(y) + \hat{\phi}^-(y)\hat{\phi}^+(x) + \hat{\phi}^+(x)\hat{\phi}^+(y),$$

which enforces

$$\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}_{x^0>y^0} =: \hat{\phi}(x)\hat{\phi}(y) : +[\hat{\phi}^+(x), \hat{\phi}^-(y)].$$

Since a symmetric expression exists for the case $x^0 < y^0$, we can write in general that the time-ordered product of two fields can be written as:

$$\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\} =: \hat{\phi}(x)\hat{\phi}(y) : +\theta(x^0 - y^0)[\hat{\phi}^+(x), \hat{\phi}^-(y)] + \theta(y^0 - x^0)[\hat{\phi}^+(y), \hat{\phi}^-(x)].$$

If we now take the vacuum expectation value of this time-ordered product (the commutator is not an operator, it is just a function multiplied by the identity), we can see that the normal-ordered part will vanish, since it contains only properly ordered creation and annihilation operators, and thus we are left with[1]

$$\langle 0| \hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\} |0\rangle = \Theta(x^0 - y^0)[\hat{\phi}^+(x), \hat{\phi}^-(y)] + \Theta(y^0 - x^0)[\hat{\phi}^+(y), \hat{\phi}^-(x)] = D_F(x, y),$$

where $D_F(x, y)$ is the Feynman propagator, describing the propagation of particles between two points in spacetime:

$$\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\} =: \hat{\phi}(x)\hat{\phi}(y) : +D_F(x, y).$$

With these results in hand, we can now define the contraction of two fields, which is a very important concept in Wick's theorem and in the calculation of Green's functions in perturbation theory.

Definition 3.1 (Contraction of two fields)**.** The **contraction of two fields** is defined as *the vacuum expectation value of the time-ordered product* of those two fields, which is the Feynman propagator

$$\underbrace{\hat{\phi}_1\hat{\phi}}_{}{}_2 = \langle 0| \hat{T}\{\hat{\phi}_1\hat{\phi}_2\} |0\rangle = D_F(x_1, x_2).$$

This time-ordered product of two fields can be rewritten in a more convenient form as the normal-ordered product of the same two fields summed to their contraction:

$$\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\} =: \hat{\phi}(x)\hat{\phi}(y) + \underbrace{\hat{\phi}(x)\hat{\phi}}_{}(y) :$$

where the contraction is denoted by a line connecting the two fields, and it represents the Feynman propagator between those two fields.

This is the simplest form of Wick's theorem, which allows us to express the time-ordered product of two fields as a sum of a normal-ordered product and a contraction; we already proven it in the previous steps, since it is the basis for the more general form of Wick's theorem, which we will see next.

This last result can be generalized by induction to the time-ordered product of n fields, which is the most general form of Wick's theorem:

Theorem 3.1 (Wick's theorem)**.** *The time-ordered product of n fields can be written as the normal-ordered product of those n fields plus all possible contractions between those fields:*

$$\hat{T}\{\hat{\phi}(x_1)\ldots\hat{\phi}(x_n)\} =: \hat{\phi}(x_1)\ldots\hat{\phi}(x_n) + \text{ all possible contractions } :$$

where a contraction between two fields is defined as the vacuum expectation value of the time-ordered product of those two fields, which is the Feynman propagator between those two fields:

$$\underbrace{\hat{\phi}_1\hat{\phi}}_{}{}_2 = \langle 0| \hat{T}\{\hat{\phi}_1\hat{\phi}_2\} |0\rangle = D_F(x_1, x_2).$$

[1] Here we simplified the product $\langle 0|0\rangle = 1$, since the commutator is just a function multiplied by the identity operator; we could have recognized the Feynman propagator even without taking the vacuum expectation value, but it is important to see how the normal-ordered part vanishes, since it is a crucial step in the proof of Wick's theorem.

Example (4-point function). Let us consider the following 4-point function

$$\langle 0|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\}\,|0\rangle\,,$$

and let us apply Wick's theorem to it. We can write the time-ordered product of the four fields as the normal-ordered product of those four fields plus all possible contractions between those fields:

$$\begin{aligned}\hat{T}\{\hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4\} = :\hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4 + \\ &+ \wick{\c1{\hat{\phi}}_1\c1{\hat{\phi}}_2\hat{\phi}_3\hat{\phi}_4} + \wick{\c1{\hat{\phi}}_1\hat{\phi}_2\c1{\hat{\phi}}_3\hat{\phi}_4} + \dots \\ &+ \wick{\c1{\hat{\phi}}_1\c2{\hat{\phi}}_2\c1{\hat{\phi}}_3\c2{\hat{\phi}}_4} + \wick{\c1{\hat{\phi}}_1\c1{\hat{\phi}}_2\c2{\hat{\phi}}_3\c2{\hat{\phi}}_4} + \wick{\c1{\hat{\phi}}_1\c2{\hat{\phi}}_2\c2{\hat{\phi}}_3\c1{\hat{\phi}}_4}\,:\end{aligned}$$

where we denoted $\hat{\phi}(x_i) \equiv \hat{\phi}_i$. Note that when contracting two fields in a product of more than two fields, we do not need to worry about having adjacent fields, since the contraction is well defined as

$$:\wick{\c1{\hat{\phi}}_1\hat{\phi}_2\c1{\hat{\phi}}_3\hat{\phi}_4}: = :\hat{\phi}_2\hat{\phi}_4: D_F(x_1,x_3),$$

and thus we can contract any two fields in the product, regardless of their position, and we will always get a well-defined result.

Using this last observation, we can notice how all the contractions of only two fields will give us a normal-ordered product of the remaining two fields, which will have zero vacuum expectation value, thus the only terms that will contribute to the vacuum expectation value of the 4-point function are the terms with two contractions, which will give us a product of two Feynman propagators: taking all the possible contractions of two pairs of fields, we get

$$\begin{aligned}\langle 0|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\}\,|0\rangle &= D_F(x_1,x_2)D_F(x_3,x_4) \\ &+ D_F(x_1,x_3)D_F(x_2,x_4) + D_F(x_1,x_4)D_F(x_2,x_3),\end{aligned}$$

which shows the extent to which the number of terms grows as we increase the number of fields, since we have to sum over all possible contractions.

Proof. Wick's theorem can be easily proven by induction. Starting from the case of two fields, which we already proved, we can assume that the theorem holds for $n-1$ fields, and then we can write the time-ordered product of n fields, assuming without loss of generality that $x_1^0 > x_j^0$ for $j > 1$, as

$$\hat{T}\{\hat{\phi}_1\hat{\phi}_2\cdots\hat{\phi}_n\} = \hat{\phi}_1\hat{T}\{\hat{\phi}_2\cdots\hat{\phi}_n\} = (\hat{\phi}_1^+ + \hat{\phi}_1^-) : \hat{\phi}_2\cdots\hat{\phi}_n + \left(\substack{\text{all contractions of}\\ \hat{\phi}_2\cdots\hat{\phi}_n}\right):,$$

where we note how one can put $\hat{\phi}_1^-$ right back in the normal ordering, since it is a creation operator and thus it does not change the normal ordering of the remaining fields, while $\hat{\phi}_1^+$ is an annihilation operator and thus it has to be commuted with the remaining fields to be put in normal order, and thus it will give us a contraction with each of the remaining fields. Thus we can write

$$\begin{aligned}\hat{T}\{\hat{\phi}_1\hat{\phi}_2\cdots\hat{\phi}_n\} = :\hat{\phi}_1\hat{\phi}_2\cdots\hat{\phi}_n + \hat{\phi}_1 \times \left(\substack{\text{all contractions of}\\ \hat{\phi}_2\cdots\hat{\phi}_n}\right): \\ + \left[\hat{\phi}_1^+\,, :\hat{\phi}_2\cdots\hat{\phi}_n + \left(\substack{\text{all contractions of}\\ \hat{\phi}_2\cdots\hat{\phi}_n}\right):\right],\end{aligned}$$

where in order to write the expression for the normal ordered product of the n fields, we had to sum the commutator of $\hat{\phi}_1^+$ with the normal-ordered product and contractions of the remaining fields, exactely as done in the case of two fields. We can now develop the commutator, which in general behaves as

$$[A, B_1\cdots B_n] = [A,B_1]B_2\cdots B_n + B_1[A,B_2]\cdots B_n + \ldots + B_1\cdots B_{n-1}[A,B_n],$$

where the sum of commutators between $\hat{\phi}_1^+$ and the remaining fields generates all the possible contractions of $\hat{\phi}_1$ with those fields; combining these terms with the normal-ordered product gives us all possible contractions of the n fields, and thus we have proven that the time-ordered product of n fields can be written as the normal-ordered product of those n fields plus all possible contractions between those fields

$$\hat{T}\{\hat{\phi}_1\hat{\phi}_2\cdots\hat{\phi}_n\} =: \hat{\phi}_1\hat{\phi}_2\cdots\hat{\phi}_n + \text{ all possible contractions } :,$$

which is exactly what we wanted to prove: □

Let us now see some important consequences of Wick's theorem, which will be crucial for the calculation of Green's functions in perturbation theory.

Corollary 3.2. *As we said before, the vacuum expectation value of any normal-ordered product is zero, thus we have*

$$\langle 0| : \hat{\phi}(x_1)\ldots\hat{\phi}(x_n) : |0\rangle = 0,$$

and thus we can write the vacuum expectation value of the time-ordered product of n *fields as*

$$\langle 0|\, \hat{T}\{\hat{\phi}(x_1)\ldots\hat{\phi}(x_n)\}\, |0\rangle = \textit{all possible contractions},$$

where all possible contractions means that we need to sum over all possible ways of contracting the fields, and each contraction gives us a Feynman propagator. This is a very important result, as it allows us to calculate Green's functions in terms of Feynman propagators, which are much easier to calculate than the full time-ordered product of fields.

Corollary 3.3. *If in a free theory we have an odd number of fields, then there are no possible contractions, since we cannot contract an odd number of fields, thus we have*

$$\langle 0|\, \hat{T}\{\hat{\phi}(x_1)\ldots\hat{\phi}(x_{2n+1})\}\, |0\rangle = 0,$$

which is a very important result, as it tells us that the vacuum expectation value of the time-ordered product of an odd number of fields is always zero, which is a consequence of the fact that we cannot contract an odd number of fields.

This last result can be proven using the fact that the normal-ordered product of an odd number of fields is zero, since it contains an odd number of creation and annihilation operators, and thus its vacuum expectation value is zero, and thus we are left with only the contractions, which are also zero since we cannot contract an odd number of fields. For an interacting theory, this is not necessarily true, since the interaction can create particles and thus we can have non-zero vacuum expectation values for the time-ordered product of an odd number of fields, but in a free theory this is always zero.

3.2 | Feynman Diagrams and Rules

Let us start to put all the pieces together with an example. We will first derive the Feynman rules for a simple interacting theory in the coordinate space, and then we will see how to generalize them to more complicated theories and to momentum space.

3.2.1 | Coordinates Space

Example (ϕ^3 theory). Let us study a ϕ^3 theory, which is a simple interacting theory of a scalar field with a cubic interaction term in the Lagrangian:

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{3!}\phi^3,$$

where λ is the coupling constant that controls the strength of the interaction. We want to calculate the 2-point function in this theory, which is given by

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\}\,|\Omega\rangle\,,$$

where $|\Omega\rangle$ is the vacuum state of the interacting theory and $\hat{\phi}$ are the field operators. Using the formula we derived before for the Green's function in the interaction picture, we can write this as

$$\frac{\langle 0|\,\hat{T}\{\hat{\phi}_0(x_1)\hat{\phi}_0(x_2)\exp\left[i\int \mathrm{d}^4x\,\mathcal{L}_{\text{int}}\right]\}\,|0\rangle}{\langle 0|\,\hat{T}\{\exp\left[i\int \mathrm{d}^4x\,\mathcal{L}_{\text{int}}\right]\}\,|0\rangle},$$

with the free field operators $\hat{\phi}_0$; now the idea is to expand the exponential in a power series in λ, assuming it small. Let us start from the numerator, which we can write as

$$\begin{aligned}\langle 0|\,\hat{T}\{\hat{\phi}_0(x_1)\hat{\phi}_0(x_2)\}\,|0\rangle + (-i\frac{\lambda}{3!})\,\langle 0|\,\hat{T}\{\hat{\phi}_0(x_1)\hat{\phi}_0(x_2)\int \mathrm{d}^4x\,\hat{\phi}_0(x)^3\}\,|0\rangle +\\ +(-i\frac{\lambda}{3!})^2\,\langle 0|\,\hat{T}\{\hat{\phi}_0(x_1)\hat{\phi}_0(x_2)\int \mathrm{d}^4x\,\hat{\phi}_0(x)^3\int \mathrm{d}^4y\,\hat{\phi}_0(y)^3\}\,|0\rangle + \dots\end{aligned}$$

where we have expanded the exponential of the interaction Lagrangian in a power series, and we have also used the fact that the fields in the interaction picture are the same as the free fields, which we have denoted with a subscript 0. With $\lambda = 0$ we get the free theory back, with the usual Feynman propagator as we expected.

The first term in the expansion thus corresponds to the free theory, which is just a single line connecting the two external points x_1 and x_2, representing the propagation of a particle from x_1 to x_2. The second term corresponds to a correction to the free theory, which can be represented as a diagram with a loop, where the particle propagates from x_1 to x, interacts at x with another particle, and then propagates from x to x_2. The third term corresponds to a correction with two loops, where the particle propagates from x_1 to x, interacts at x with another particle, then propagates from x to y, interacts at y with another particle, and then propagates from y to x_2. And so on for higher-order corrections.

If λ remains small, we can *truncate the expansion* at a certain order and get an approximation for the 2-point function. This is the basis for perturbation theory in QFT, where we can calculate physical quantities as a power series in the coupling constant, and each term in the series corresponds to a Feynman diagram that represents a particular process or interaction.

Let us highlight the fact that, for the Wick's theorem, the correction of the order $O(\lambda)$ vanishes, since it contains an odd number of fields, and thus there are no possible contractions, and thus its vacuum expectation value is zero. The first non-vanishing correction to the free theory is thus the second order correction, which is of order $O(\lambda^2)$, and it corresponds to a **diagram with a loop**, as we have seen before.

Let us introduce a simpler **shortcut notation** for the following computations, since we will have to deal with a large number of fields and propagators, and it will be easier to keep track of them in this way. We will denote the Feynman propagator between two external points x_i and x_j as D_{ij}, the Feynman propagator between an external point x_i and an internal point x as D_{ix}, and the free field operator at a point x_i as ϕ_i. Thus we have

$$\begin{cases} D_{ij} = D_F(x_i, x_j), \\ D_{xi} = D_F(x, x_i), \\ \phi_i = \phi_0(x_i), \end{cases}$$

so that we can write the second order correction to the free theory as

$$(-\frac{i\lambda}{3!})^2 \frac{1}{2!} \int \mathrm{d}^4 z_1 \, \mathrm{d}^4 z_2 \, \langle 0 | \, \hat{T}\{\hat{\phi}_x \hat{\phi}_y \hat{\phi}_{z_1} \hat{\phi}_{z_1} \hat{\phi}_{z_1} \hat{\phi}_{z_2} \hat{\phi}_{z_2} \hat{\phi}_{z_2}\} \, |0\rangle \, ,$$

where now we can use Wick's theorem to calculate the vacuum expectation value of this time-ordered product, which will give us a sum of all possible contractions between the fields:

$$(-\frac{i\lambda}{3!})^2 \frac{1}{2!} \int \mathrm{d}^4 x \, \mathrm{d}^4 y \, (D_{1x} D_{2y} D_{xy}^2 + \dots),$$

where we have to note that the number of contractions grows very quickly and different contractions may give us the same contribution, since they may be equivalent under permutations of the fields.

But maybe there is a way in which we can **list all the possible contractions** without having to write them all down, since as we can see, the number of contractions grows very quickly as we increase the order of the correction, and they may also be equivalent. This is where **Feynman diagrams** come in handy, as they provide a visual representation of the interactions and allow us to easily keep track of the different terms in the expansion.

Now we can start to derive the Feynman rules for this theory, which will allow us to write down the contribution of each diagram without having to calculate all the contractions explicitly. The Feynman graphical representation can be derived from this example, where we have the following elements:

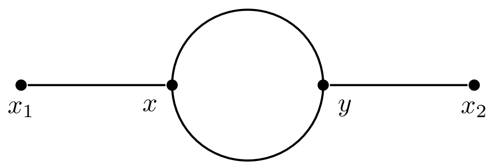


Figure 3.1: Feynman diagram for the second order correction to the 2-point function in a ϕ^3 theory: each internal point corresponds to an interaction vertex, connected to three lines, each corresponding to a propagator.

- x_1, x_2 external points, represented as dots.
- x, y internal points, represented as vertices.
- D_{x1} and D_{y2} propagators, represented as lines connecting the external points to the internal points.
- D_{xy}^2 propagators, represented as two lines connecting the internal points x and y.

More generally, we can write the **Feynman rules** for an n-point function as

- N external points $x_1, x_2, \dots, x_n$, each connected to only one propagator (line) in every theory.
- Any number of vertexes, each corresponding to an interaction term in the Lagrangian

$$i \int \mathrm{d}^4 z \, \mathcal{L}_{\text{int}}(z),$$

  where in particular for our ϕ^3 theory, we had

$$i \int \mathrm{d}^4 z \, (-\frac{\lambda}{3!}\phi^3(z)) = -i\frac{\lambda}{3!} \int \mathrm{d}^4 z \, \phi^3(z),$$

  which means that each vertex has to be connected to three lines, since the interaction term is cubic in the field.
- Every vertex is of order $O(\lambda)$, since it comes from the interaction Lagrangian, and thus the order of the correction is given by the number of vertexes in the diagram.
- The propagators as lines connects ALL the points, since the contractions has to be complete, since we cannot have uncontracted fields in the vacuum expectation value.

These rules telling us how to draw the Feynman diagrams for a given process, and they also allow us to write down the mathematical expression corresponding to each diagram without having to calculate all the contractions explicitly, which is a very powerful tool for calculating physical quantities in QFT.

The procedure of calculating a physical quantity using Feynman diagrams can be summarized as follows:

1. Draw all the possible Feynman diagrams for the process you want to calculate, following the Feynman rules.
2. For each vertex, we have an expression to write down the corresponding factor from the interaction Lagrangian $\to -i\alpha \int \mathrm{d}^4 z$.
3. For each line, write down the corresponding Feynman propagator $D_F(x, y)$.

Diagram Prefactors

We have to understand how to put the correct prefactors α in front of the integrals. Let us start from the interaction Lagrangian, which is given by

$$\mathcal{L}_{\text{int}} = \frac{\lambda^N}{N!}\phi^N,$$

then a diagram with M vertexes (obtained considering corrections up to order $O(\lambda^M)$) and any number of external points will have

- $M!$ ways of permuting the vertexes, since they are indistinguishable.
- $N!$ ways of permuting the fields in each vertex, since they are also indistinguishable, thus a total of $(N!)^M$ ways of permuting the fields in all the vertexes.

Putting all these factors together, we get a factor of $(N!)^M \times M!$ only from the counting of the permutations of the vertexes and fields, and thus we need to divide by this factor to avoid overcounting, since we are summing over all the possible contractions, which already takes into account all the possible permutations of the vertexes and fields.

An example of naive counting of the factors for a diagram with M vertexes and N fields in each vertex would give us a factor of

$$(N!)^M \times M! \times (\frac{\lambda}{N!})^M \times \frac{1}{M!},$$

where the first factor comes from the permutations of fields in each vertex, the second factor comes from the permutations of the vertexes, the third factor comes from the interaction Lagrangian, and the last factor comes from the expansion of the exponential of the interaction Lagrangian. Is it possible that all these factors cancel out and we are left with a factor of λ^M for a diagram with M vertexes?

No, this counting is not correct, since we have to take into account that some diagrams may be equivalent under permutations of the vertexes and fields, and thus we need to divide by the number of symmetries of the diagram: *the number of permutations of the vertexes and fields that leave the diagram unchanged.* This is known as the **symmetry factor** of the diagram, and it can be calculated by counting the number of ways to permute the vertexes and fields without changing the diagram.

Thus the correct factor for a diagram with M vertexes and N fields in each vertex is given by the inverse of the symmetry factor of the diagram, which can be calculated by counting the number of ways to permute the vertexes and fields without changing the diagram:

Definition 3.2 (Symmetry factor)**.** The symmetry factor of a Feynman diagram is the **number of independent transformations** (permutations of the vertexes and fields in the diagram) **mapping the diagram onto itself** (with external points fixed), and it is denoted by S. The contribution of a diagram to a physical quantity is given by the inverse of the symmetry factor, which is $1/S$. The identity has to be counted as a symmetry, while swapping external points is not a symmetry, since they are fixed. One can sum the symmetries of the drawing by hand, or one can multiply the symmetry prefactors of the subdiagrams (D_{xy}, D^2_{xx}...).

Example (Back to the cubic interaction). Let us go back to the second order correction to the 2-point function in the ϕ^3 theory, which corresponds to the following diagram:

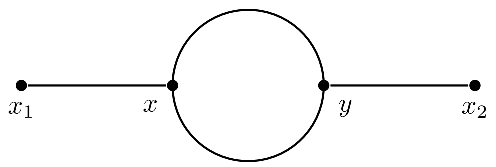


Figure 3.2: Feynman diagram for the second order correction to the 2-point function in a ϕ^3 theory: each internal point corresponds to an interaction vertex, connected to three lines, each corresponding to a propagator.

This diagram has two vertexes, each connected to three lines, and thus it corresponds to a correction of order $O(\lambda^2)$. It can be obtained from 18 different contractions:

- 3 choices of ϕ_x in $\underbrace{\phi_1\phi_x}$;
- 3 choices of ϕ_y in $\underbrace{\phi_2\phi_y}$;
- 2 choices of the remaining ϕ_y to be contracted with the first remaining ϕ_x.

These 18 contractions have to be multiplied by 2 for the swap of the two vertexes $x \leftrightarrow y$. Thus we get a total of 36 contractions which has to be divided by $(3!)^2 \times 2! = 72$, which is the number of permutations of the fields in the vertexes and the number of permutations of the vertexes, thus we get a symmetry factor of $S = 2$ for this diagram, and thus its contribution to the 2-point function is given by

$$\frac{1}{2} \times (-i\frac{\lambda}{3!})^2 \int \mathrm{d}^4 z_1 \, \mathrm{d}^4 z_2 \, D_{xz_1} D_{yz_1} D^2_{z_1 z_2},$$

where the factor of $1/2$ comes from the symmetry factor of the diagram, and the rest of the factors come from the interaction Lagrangian and the Feynman propagators.

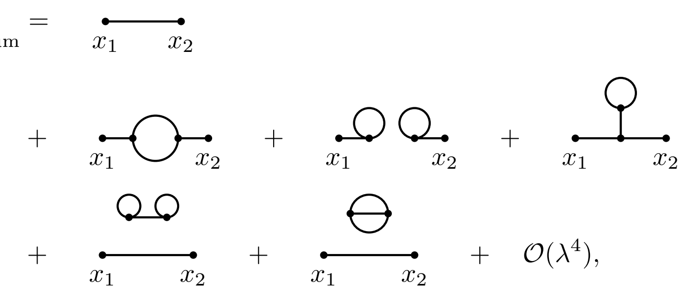
In order to understand the symmetry factor of this diagram, we can also look at it from a different perspective; if we rewrite the contraction, we can see that permuting the propagators between the internal points x and y does not change the diagram, since they are indistinguishable:

$$\phi_1\phi_2\phi_x\phi_x\phi_x\phi_y\phi_y\phi_y \quad = \quad \phi_1\phi_2\phi_x\phi_x\phi_x\phi_y\phi_y\phi_y,$$

and if we compare it to the graphical representation of the diagram, we can see that this contraction symmetry corresponds to the invariance of the diagram under the **vertical flip symmetry**, which is a symmetry of the diagram that leaves the external points fixed and swaps the internal points x and y. This symmetry is the only non-trivial symmetry of the diagram and thus we have a total of 2 symmetries (do not forget the identity), which gives us a symmetry factor of $S = 2$, and thus a contribution of $1/2$ for this diagram.

It can even happen to have an internal subdiagram that is disconnected from the external points, which is called a *vacuum bubble*, and it is possible to show that the contribution of a vacuum bubble cancels out with the denominator of the Green's function. For example, if we were to compute the next order corrections to the 2-point function, we would get the following diagrams:

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}\,|\Omega\rangle\Big|_{\mathrm{num}} =$$



where the fifth and sixth diagrams have vacuum bubble. The complete expression associated to these diagrams is given by

$$\begin{aligned}\langle\Omega|\,\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}\,|\Omega\rangle\Big|_{\mathrm{num}} &= D_{12} + (-i\lambda)^2 \int \mathrm{d}^4x\,\mathrm{d}^4y\,\big(\tfrac{1}{2}D_{1x}D_{2y}D^2_{xy} + \tfrac{1}{4}D_{1x}D^2_{xx}D_{2y}D^2_{yy} \\ &\quad + \tfrac{1}{2}D_{1x}D_{2x}D_{xy}D_{yy} + D_{12}\big(\tfrac{1}{8}D_{xx}D_{xy}D_{yy} + \tfrac{1}{12}D^3_{xy}\big)\big) + \mathcal{O}(\lambda^4),\end{aligned}$$

where the contribution of the vacuum bubbles can be canceled out with the denominator of the Green's function, which is given by

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}\,|\Omega\rangle\Big|_{\mathrm{den}} = \frac{1}{\langle 0|\,\hat{T}\{\exp\big[i\int \mathrm{d}^4z\,\mathcal{L}_{int}\big]\}\,|0\rangle}.$$

Thus the denominator is exactly a Green's function with no external points, and thus it contains only **vacuum diagrams**, which are disconnected from the external points, and thus they do not contribute to the physical quantity we are calculating, and thus they can be canceled out with the denominator. This is a very important result, as it tells us that we can ignore the vacuum bubbles when calculating physical quantities in QFT: they do not contribute to the final result, and thus we can focus only on the connected diagrams, which are the ones that contribute to the physical quantity we are calculating. Indeed we have:

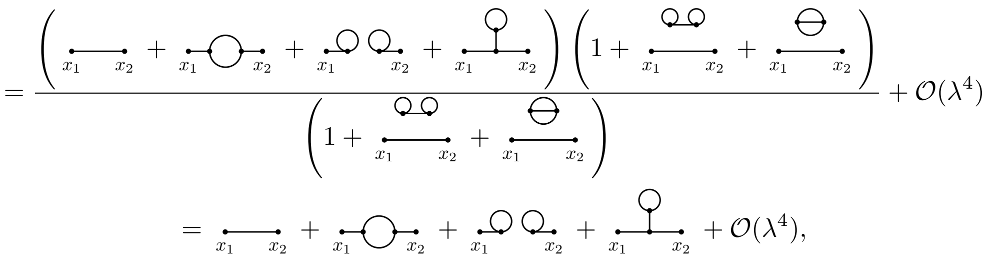
- Numerator: ($\sum$ connected diagrams) ($1 + \sum$ vacuum diagrams).
- Denominator: ($1 + \sum$ vacuum diagrams).

Thus in the end, the contribution of the vacuum diagrams cancels out, and we are left with only the connected diagrams, which are the ones that contribute to the physical quantity we are calculating:

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\cdots\hat{\phi}(z)\}\,|\Omega\rangle = \sum \text{connected diagrams.}$$

Example (Still on the cubic interaction). If we were to show the previous result explicitly for the ϕ^3 theory, we would have to compute the green's function up to order $O(\lambda^4)$, which would give us the following diagrams:

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}\,|\Omega\rangle$$



which shows explicitly that the contribution of the vacuum bubbles cancels out with the denominator, and thus we are left with only the connected diagrams, which are the ones that contribute to the physical quantity we are calculating.

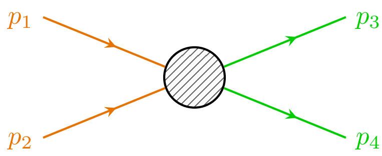
Summary of Feynman Rules in Position Space

Let us group and summarize all the Feynman rules we have derived so far for a scalar field theory in position space:

- **External point**: x_i, represented as a dot, and it is connected to only one propagator.
- **Vertex**: $i\int \mathrm{d}^4z\,\mathcal{L}_{\text{int}}(z)$, represented as a dot and connected to a number of lines equal to the number of fields in the interaction term. It is specific to the theory we are considering, since it depends on the interaction Lagrangian, and thus it can be different for different theories.
- **Propagator**: $D_F(x, y)$, represented as a line connecting two points, and corresponding to the Feynman free propagator between those two points.
- **Symmetry factor**: $1/S$, where S is the number of symmetries of the diagram, which can be calculated by counting the number of ways to permute the vertexes and fields without changing the diagram. It contains the identity as a symmetry, while swapping external points is not a symmetry, since they are fixed.
- $\langle\Omega|\,\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\cdots\hat{\phi}(z)\}\,|\Omega\rangle = \sum$ connected diagrams, since the contribution of the vacuum diagrams cancels out with the denominator of the Green's function, and we can focus only on the connected diagrams, which are the ones contributing to the physical quantity we are calculating.

3.2.2 | Momentum Space

Let us study the same ϕ^3 theory $\mathcal{L}_{int} = -\frac{\lambda}{3!}\phi^3$ for a $2 \to 2$ scattering process.



We work up to the **leading order** (LO) correction, which means up to the first non zero matrix element. It is not a general a priori order, such as $\mathcal{O}(\lambda^0)$ or $\mathcal{O}(\lambda^2)$; the leading order is determined by the structure of the interaction term, and is given by the lowest power of the coupling constant λ which gives a non zero matrix element.

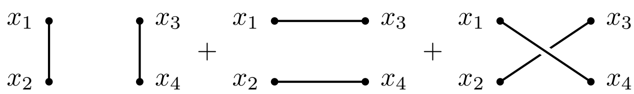
The Green's function we are considering can be computed as

$$\begin{aligned}\langle f|\,\hat{S}\,|i\rangle = \langle p_3, p_4|\,\hat{S}\,|p_1, p_2\rangle = &\prod_{j=1}^{4}\left(-\frac{i}{\sqrt{Z}}(p_j^2 - m^2)\right) \\ &\times \int \mathrm{d}^4x_1 \,\dots\, \mathrm{d}^4x_4 e^{i(p_3x_3 + p_4x_4 - p_1x_1 - p_2x_2)} \\ &\times \langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\}\,|\Omega\rangle\,,\end{aligned}$$

where we have used the LSZ reduction formula to relate the S-matrix element to the Green's function. Now for a free theory $Z = 1$, while for an interacting theory $Z = 1 + O(\lambda)$, thus at the leading order we can set $Z = 1$ (since we have lower order terms), and thus we write the Green's function as

$$\langle f|\,\hat{S}\,|i\rangle = \int \mathrm{d}^4x_1 \,\dots\, \mathrm{d}^4x_4 e^{i(p_3x_3 + p_4x_4 - p_1x_1 - p_2x_2)} \times \langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\}\,|\Omega\rangle\,.$$

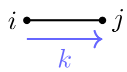
Now we can use the Feynman rules in position space to write down the contribution of the leading order diagram, which is given by

$\mathcal{O}(\lambda^0) =$ 

$$= D_{12}D_{34} + D_{13}D_{24} + D_{14}D_{23},$$

where we have three possible contractions between the four fields, and thus we have three possible diagrams. Now we can write down the contribution of each diagram in momentum space by using the Fourier transform of the Feynman propagator, which is given assigning a momentum k to the internal line (with definite direction $i \to j$), and thus we have

$$D_{ij} = \int \frac{\mathrm{d}^4k}{(2\pi)^4}\frac{e^{-ik(x_i - x_j)}}{k^2 - m^2 + i\epsilon} \quad =$$



Doing this for the first diagram, we get

$$\begin{aligned}\text{Diag I: } &\prod_{j=1}^{4}\left(-i(p_j^2 - m^2)\right) \times \int \mathrm{d}^4x_1 \,\dots\, \mathrm{d}^4x_4 \int \frac{\mathrm{d}^4k_1}{(2\pi)^4}\frac{\mathrm{d}^4k_2}{(2\pi)^4} \\ &\times e^{-ix_1(p_1 - k_1)}e^{-ix_2(p_2 + k_1)}e^{ix_3(p_3 - k_2)}e^{ix_4(p_4 + k_2)} \times \frac{i}{k_1^2 - m^2}\frac{i}{k_2^2 - m^2},\end{aligned}$$

where we can recognize the Fourier transform of the propagators, and thus we can perform the integrals over the external points x_1, x_2, x_3, x_4, which gives us delta functions that set the momenta of the internal lines to be equal to the momenta of the external lines, enforcing momentum conservation at each vertex. Thus simplifing a factor of $(2\pi)^4$ for each delta function, we get a term of the form

$$\delta^{(4)}(p_1-k_1)\delta^{(4)}(p_2+k_1)\delta^{(4)}(p_3-k_2)\delta^{(4)}(p_4+k_2) = \delta^{(4)}(p_1+p_2)\delta^{(4)}(p_3+p_4)\delta^{(4)}(k_1-p_1)\delta^{(4)}(k_2-p_3),$$

where the first two delta functions enforce momentum conservation at the vertices, while the last two delta functions set the internal momenta to be equal to the external momenta. Thus we can perform the integrals over the internal momenta k_1 and k_2, enforcing $k_1 = p_1$ and $k_2 = p_3$, which leads us to

$$\text{Diag I: } \prod_{j=1}^{4} \left(-i(p_j^2 - m^2)\right) \times \delta^{(4)}(p_1+p_2)\delta^{(4)}(p_3+p_4) \times \frac{i}{p_1^2 - m^2}\frac{i}{p_3^2 - m^2} = 0,$$

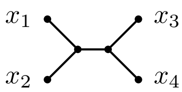
since the delta functions enforce a condition that makes the product vanish: $p_1 + p_2 \neq 0$ as well as $p_3 + p_4 \neq 0$, since they are both positive-energy momenta.

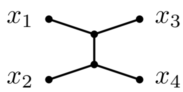
For the second diagram we could repeat the same procedure, and we would get

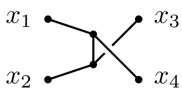
$$\text{Diag II: } \prod_{j=1}^{4} \left(-i(p_j^2 - m^2)\right) \times \delta^{(4)}(p_1-p_3)\delta^{(4)}(p_2-p_4) \times \frac{i}{p_1^2 - m^2}\frac{i}{p_2^2 - m^2},$$

where the delta functions set $p_3 = p_1$ and $p_4 = p_2$ this time, describing the process where the particles p_1 and p_2 do not scatter, leading to a final state that is the same as the initial state (it is the identity process of $\hat{S} = \mathbb{I} + i\hat{T}$). The third diagram gives us the same contribution as the second diagram, since it is just a permutation of the external points, and we can ignore these two processes.

We are interested in the scattering process where the particles p_1 and p_2 interact and scatter into the final state p_3 and p_4 such that $|f\rangle \neq |i\rangle$. The first non trivial diagrams come from the second order correction $\mathcal{O}(\lambda^2)$, which is given by

$\mathcal{O}(\lambda^2) =$ 

$+$ 

$+$ 

,

where we have three possible diagrams, corresponding to the three possible channels of the scattering process: the s-channel, the t-channel and the u-channel (the origin of the names will be cleared in the following sections).

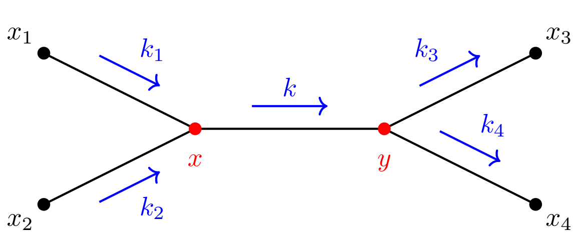


Figure 3.3: s-channel with indication of the exchanged four-momenta at vertices x and y.

Let us focus on the s-channel diagram, the first one, where as done before we assign a momentum k to the internal lines, getting from each vertex and momenta a term of the form $e^{\pm ik\cdot x}$, with a positive sign for outgoing momenta and a negative sign for incoming momenta.

Thus we get the following Fourier transform for the contribution of the first diagram:

$$\begin{aligned}\text{Diag I: } &\prod_{j=1}^{4}\left(-i(p_j^2-m^2)\right)\times\int \mathrm{d}^4x_1\,\ldots\,\mathrm{d}^4x_4\,e^{i(p_3x_3+p_4x_4-p_1x_1-p_2x_2)}\\ &\times(-i\lambda)^2\int \mathrm{d}^4x\,\mathrm{d}^4y\,D_{1x}D_{2x}D_{xy}D_{y3}D_{y4}\end{aligned}$$

where we have used the Feynman rules in position space to write down the contribution of the diagram, and now we can use the Fourier transform of the propagators to write down the contribution of the diagram in momentum space, which is given by

$$\begin{aligned}\text{Diag I: } &\prod(\cdot)\times\int \mathrm{d}^4x_1\,\ldots\,\mathrm{d}^4x_4\,\mathrm{d}^4x\,\mathrm{d}^4y\;\times(-i\lambda)^2\int\frac{\mathrm{d}^4k_1}{(2\pi)^4}\cdots\frac{\mathrm{d}^4k_4}{(2\pi)^4}\frac{\mathrm{d}^4k}{(2\pi)^4}\\ &\times e^{-ix_1(p_1-k_1)}e^{-ix_2(p_2-k_2)}e^{-ix_3(p_3-k_3)}e^{-ix_4(p_4-k_4)}\\ &\times e^{ix(k-k_1-k_2)}e^{iy(k_3+k_4-k)}\times\prod_{q\in\{k_1,\ldots,k_4,k\}}\left(\frac{i}{q^2-m^2}\right).\end{aligned}$$

Paying again attention to factors, we can perform the integrals over the external points x_1, x_2, x_3, x_4, obtaining delta functions that set the momenta of the internal lines to be equal to the momenta of the external lines, enforcing momentum conservation at each vertex. Thus simplifying a factor of $(2\pi)^4$ for each delta function, we get a term of the form

$$\delta^{(4)}(p_1-k_1)\cdots\delta^{(4)}(p_4-k_4)\delta^{(4)}(k-k_1-k_2)\delta^{(4)}(k-k_3-k_4),$$

where the first four delta functions set the internal momenta to be equal to the external momenta, while the last two delta functions enforce momentum conservation at the vertices. We can rewrite the previous expression as

$$\delta^{(4)}(p_1-k_1)\cdots\delta^{(4)}(p_4-k_4)\delta^{(4)}(k-p_1-p_2)\delta^{(4)}(p_1+p_2-p_3-p_4),$$

in order to make the momentum conservation more explicit, setting $k = p_{in}$ and momentum conservation throughout the whole diagram. To recap:

- We rewrote every **propagator in position space as a Fourier transform**, which introduced an integral over the internal momenta $\int\frac{\mathrm{d}^4k_j}{(2\pi)^4}$ for each internal line (assingning to each of them a momentum vector), and a factor of $e^{\pm ik_j\cdot x}$ for each vertex, with a positive sign for outgoing momenta and a negative sign for incoming momenta.

- Each of the six integrals over internal and external points $\int \mathrm{d}^4x_j$ from the LSZ reduction formula give us delta functions that set the momenta $k_j = p_j$.

- The integrals over the internal points $\int \mathrm{d}^4x$ and $\int \mathrm{d}^4y$ give us delta functions that set the momenta of the internal lines to be equal to the momenta of the external lines, **enforcing momentum conservation at each vertex**.

- The combination of the delta functions from the external points and the internal points give us a global delta function that enforces momentum conservation for the whole scattering process, which is given by $(2\pi)^4\delta^{(4)}(p_{in}-p_{out})$, where p_{in} and p_{out} are the total incoming and outgoing momenta, respectively.

If we conclude the calculation for the first diagram, we note how each of the factors $\frac{i}{q^2-m^2}$ cancels with a factor of $-i(p_j^2-m^2)$ from the LSZ reduction formula, since the external momenta are on-shell, thus we are left with only the contribution of the delta enforcing $\delta^{(4)}(k - p_1 - p_2) \implies k = p_1 + p_2$, leaving us with a final contribution of the form

$$\text{Diag I: } \frac{i}{(p_1 + p_2)^2 - m^2}.$$

The delta functions do not always cancel out all the integrals over momenta, and one can show indeed that if the diagram contains a loop, then we are left with an integral over the loop momentum, which is not fixed by the delta functions, and thus we need to integrate over it. For example, if we consider the following diagram, which is a one-loop correction to the 2-point function:

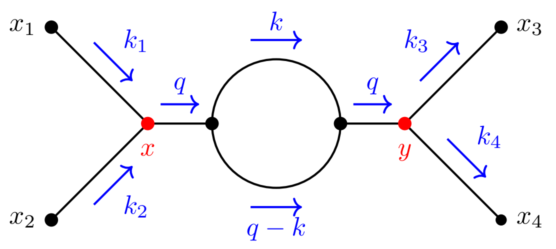


Figure 3.4: 1-loop correction of the s-channel diagram of the $2 \to 2$ scattering process.

we get always the deltas enforcing internal momenta to be equal to the external momenta, but we are left with an integral over the loop momentum k, since among the four internal lines (giving rise to four delta functions, one for each line), only three of the assigned momenta are fixed by the delta functions, while the fourth one remains unfixed for the presence of the global $\delta^{(4)}(p_{in} - p_{out})$, which is always present up to rewriting the delta's arguments: we are left with

$$\int \frac{\mathrm{d}^4 k}{(2\pi)^4}, \qquad k \equiv \text{loop momentum},$$

which is not fixed by the delta functions, and thus we need to integrate over it. This is a general feature of loop diagrams, and it is one of the main reasons why loop diagrams are more difficult to calculate than tree-level diagrams, since they involve integrals over loop momenta that can be divergent, and thus we need to use regularization and renormalization techniques to make sense of them.

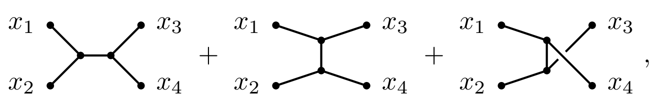
Thus remembering that $\hat{S} = \mathbb{I} + i\hat{T}$, we found that up to $\mathcal{O}(\lambda^0)$ we have the identity process of the scattering, while the first non trivial contribution to the scattering amplitude comes from $\mathcal{O}(\lambda^2)$ where, after performing the integrals over the internal momenta k_1, k_2, k_3, k_4 (enforcing $k_j = p_j$ for $j = 1, 2, 3, 4$ as we said), we are left with a contribution of the form

$$\langle f| i\hat{T} |i\rangle = i\mathcal{M}(2\pi)^4 \delta^{(4)}(p_{in} - p_{out}),$$

where $\mathcal{M}$ is the scattering amplitude, which is a function of the external momenta, and it encodes all the information about the scattering process; this is the quantity that we are interested in calculating, since it encodes the probability amplitude for the scattering process, and we can obtain the cross section and the decay rate from it, which are the physical observables that we can measure in experiments. The Feynman rules in momentum space allow us to calculate $\mathcal{M}$ for any given scattering process, by summing over all the fully connected diagrams that contribute to the process, and applying the rules we have just described.

Finishing the computation for the leading order correction, we had the three channel diagrams, reported again here for clarity:

$\mathcal{O}(\lambda^2) =$



where the contribution of the deltas in the internal lines is respectively given by

- $p_1 + p_2$ for the s-channel diagram;
- $p_1 - p_3$ for the t-channel diagram;
- $p_1 - p_4$ for the u-channel diagram.

Thus in the end we get the following contribution for the leading order correction:

$$i\mathcal{M} = (-i\lambda)^2 \left[\frac{i}{(p_1+p_2)^2 - m^2} + \frac{i}{(p_1-p_3)^2 - m^2} + \frac{i}{(p_1-p_4)^2 - m^2} \right],$$

where we have a contribution from each of the three channels, and we can see that the scattering amplitude is a function of the external momenta, encoding all the information about the scattering process. The $(-i\lambda)^2$ factor comes from the two vertexes, common for all the three diagrams, while inside the parenthesis we have the three propagator contributions of each diagram, which is given by the product of the propagators corresponding to the internal lines, and the delta functions that enforce momentum conservation at each vertex have already been integrated out: only applying the rules we got the result which would have implied a lot of calculations otherwise.

Preliminary Summary of Rules in Momentum Space

Let us summarize the Feynman rules in momentum space for a scalar field theory; for leading order (LO) corrections, this will be plenty enough to understand the scattering process and calculate the scattering amplitude, while for higher order corrections, we will need to take into account also the renormalization of the theory, which is a more advanced topic that we will have to cover in the next chapter. The preliminary Feynman rules in momentum space are the following:

- $i\mathcal{M} = \sum$ fully connected diagrams, since we have seen that only these diagrams contribute to the scattering process , while the diagrams with bubbles are canceled out by the denominator of the green function; $\mathcal{M}$ is the scattering amplitude, which encodes all the information about the scattering process.
- We assign a momentum k to each internal line, with the convention that the momentum flows from the initial state to the final state (thus the direction matters, even if it is a four-vector).
- The external lines are on-shell, thus they satisfy the mass-shell condition $p_j^2 = m^2$ for $j = 1, 2, 3, 4$ and each of them contributes with a factor of $-i(p_j^2 - m^2)$ from the LSZ reduction formula; we have seen that these factors cancel with the propagators of the internal lines, thus external lines contribute with a factor of 1.
- Each of the internal lines corresponds to a propagator, given by $\frac{i}{k^2-m^2}$ in the momentum space, where k is the momentum flowing through the internal line.
- The vertexes correspond to factors from the interaction Lagrangian, which in momentum space are given by $-i\lambda$ for a ϕ^3 theory, and more generally by the Fourier transform of the interaction term in the Lagrangian. They depend on the theory under consideration, and they also depend on the momenta flowing through the vertex.
- **Momentum conservation at each vertex**: the sum of the incoming momenta must equal the sum of the outgoing momenta, which is enforced by delta functions in the mathematical expression corresponding to each diagram.

- Each loop in the diagram corresponds to an unresolved integral over the loop momentum, which is not fixed by the delta functions, and thus we need to integrate over it. The measure of the integral is given by $\int \frac{d^4k}{(2\pi)^4}$ for each unresolved loop momentum k.
- Prefactor: $1/S$, where S is the number of symmetries of the diagram, which can be calculated by counting the number of ways to permute the vertexes and fields without changing the diagram. Same as in position space.

When saying **fully connected diagrams**, we mean that the diagram is connected and that it *does not contain any disconnected subdiagram*, i.e. we do not include in this definition diagrams which are formed by 2 or more connected components, which are not connected to each other, and thus they do not describe a global interaction of the incoming particles scattering into the outgoing particles, but rather they describe separate processes that are not related to each other.

Example (Fully connectedness in $2 \to 4$ scattering). In the following $2 \to 4$ diagram, where we have two incoming particles and four outgoing particles, we have two possible diagrams, one of them is fully connected, while the other one is not fully connected:

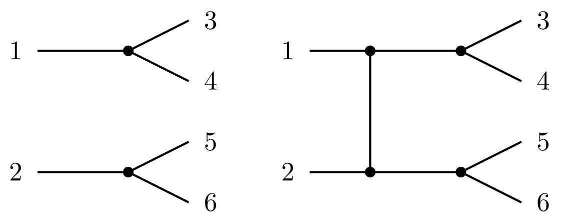


Figure 3.5: In the calculation of scattering amplitudes and fully interacting correlation functions, disconnected sub-diagrams (left) are factored out and excluded. Only fully connected topologies (right) contribute to the non-trivial S-matrix elements.

The non fully connected diagram describes to separate processes more than a global interaction of 2 incoming particles scattering into 4 outgoing particles, and thus it does not contribute to the scattering amplitude of the process we are interested in, while the fully connected diagram describes exactely the process we are interested in.

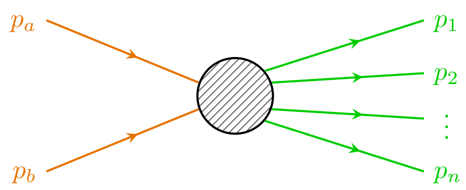
Remark. *At leading order we have set $Z = 1$, and we confused the lagrangian mass with the physical one: in the next chapter we will see that at higher order corrections we need to take into account the renormalization of the theory, which will lead us to distinguish between the lagrangian mass and the physical mass, and to introduce a renormalization factor Z for the fields:*

$$\begin{cases} Z = 1 + \mathcal{O}(\lambda), \\ m_{phys} = m_{\mathcal{L}} + \mathcal{O}(\lambda). \end{cases}$$

this will modify the Feynman rules we have just described, but for now we can ignore these complications and use the rules we have just derived. In our Feynman rules we are using the lagrangian mass $m_{\mathcal{L}}$, but in future it will be convenient to make the substitution $m_{\mathcal{L}} \to m_{phys} + \delta m$, where δm represents additional terms added to the lagrangian.

Back to LSZ Reduction Formula

Now consider a $2 \to n$ scattering process, where we have two incoming particles and n outgoing particles.



If we want to calculate the scattering amplitude for this process, we can use the LSZ reduction formula, which relates the S-matrix element to the Green's function. The Green's function we need to calculate is given by

$$\ldots$$

We have seen thay at leading order (LO) we have $Z = 1$ and in the LSZ reduction formula we have a factor of $\prod_{j=1}^{n+2}\left(-i(p_j^2 - m^2)\right)$, where p_j are the external momenta, which are on-shell, thus they satisfy the mass-shell condition $p_j^2 = m^2$ for $j = 1, 2, \ldots, n+2$. This means that each of the external lines contributing with a factor of $-i(p_j^2 - m^2)$ cancels with a factor of $\frac{i}{p_j^2 - m^2}$ from the propagator of the internal line.

But what happens at higher order corrections? We can schematically represent the contribution of the diagrams to the scattering amplitude as follows:

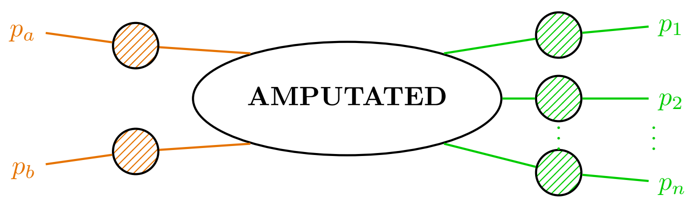
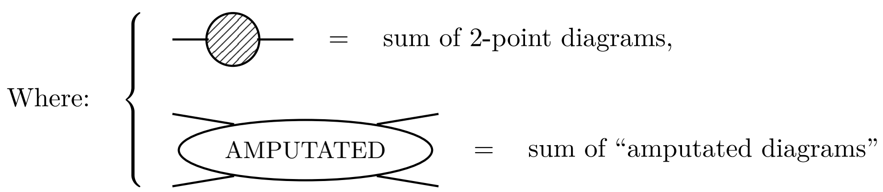


Figure 3.6: Structure of a fully interacting correlation function. To obtain the S-matrix element, the external legs containing the full propagator corrections (hatched circles) are amputated, leaving only the amputated connected Green's function in the center.



Definition 3.3 (Amputated diagram)**.** An amputated diagram is a Feynman diagram where we cannot disconnect any external line by cutting a single internal line. In other words, an amputated diagram is a diagram that does not contain any disconnected subdiagram, and thus it is fully connected.

The contribution of an amputated diagram to the scattering amplitude is given by the *product of the propagators corresponding to the internal lines*, and the *factors from the vertexes*, without including the external propagators, since they are on-shell and thus they do not contribute to the scattering amplitude.

Example. In the following diagram, we have a cut procedure used to amputate a 1-loop self-energy correction from an incoming external leg. Thanks to the LSZ reduction formalism, we know that the external legs of the Green's function contain full propagator corrections, which are given by the sum of all the 2-point diagrams, and thus we can amputate them in order to extract more efficiently the scattering amplitude: we will see how we are ignoring self-deleting terms, since in the computation of the green's function in the LSZ reduction formula these contributions cancel out.

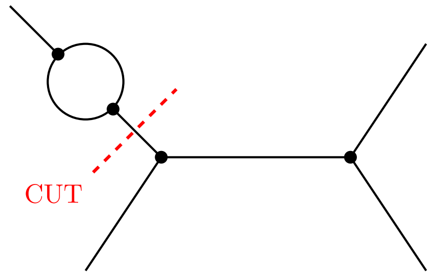
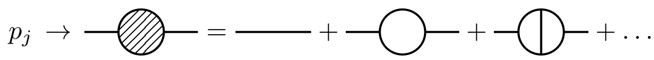


Figure 3.7: Example of a cutting procedure used to amputate a 1-loop self-energy correction from an incoming external leg. In the LSZ reduction formalism, isolating and removing these external full-propagator corrections is required to extract the pure amputated scattering amplitude.

Let us show that the contribution of the external subdiagrams we are amputating is negligible for the scattering amplitude. Each of the external lines in the LSZ reduction contributes with a factor

$$-\frac{i}{\sqrt{Z}}\left(p_j^2-m^2\right),$$

and we know these momenta to be on-shell, thus they satisfy the mass-shell condition $p_j^2=m^2$ for $j=1,2,\dots,n+2$ (m is the physical mass). Thus each of the external subdiagrams we are truncating is the sum of two point functions:



$$\begin{aligned}&= \text{Fourier transform of } \langle\Omega|\, T\hat{\phi}(x_1)\hat{\phi}(x_2)\,|\Omega\rangle\\ &= \frac{iZ}{p_j^2-m^2} + \text{ regular terms at } p_j^2\sim m^2,\end{aligned}$$

which implies that each of the $\frac{i}{p_j^2-m^2}$ terms cancel with the $-i(p_j^2-m^2)$ factor from the LSZ reduction formula, leaving us with a factor of $\sqrt{Z}$ for each external line, which is the pole residue of the full propagator.

Summary of Rules in Momentum Space

- $i\mathcal{M}=\sum$ fully connected amputated diagrams.
- Assign momentum to internal lines, with the convention that the momentum flows from the initial state to the final state, and it is conserved at each vertex.
- For each unresolved momentum, i.e. for each loop, we need to integrate over it with the measure $\int\frac{\mathrm{d}^4k}{(2\pi)^4}$.
- External lines contribute with a factor of 1 each, since they are on-shell and thus they do not contribute to the scattering amplitude.
- Internal lines contribute with a factor of $\frac{i}{p^2-m^2}$ each, where p is the momentum flowing through the internal line: propagator $\equiv$ internal line.
- Vertexes contribute with a factor of $-i\lambda$ for a ϕ^3 theory, and more generally by the Fourier transform of the interaction term in the Lagrangian, which depends on the theory under consideration and on the momenta flowing through the vertex.

- The symmetry factor of the diagram is given by $1/S$, where S is the number of symmetries of the diagram, which can be calculated by counting the number of ways to permute the vertexes and fields without changing the diagram.
- For each external particle, we need to multiply by a factor of $\sqrt{Z}$, which comes from the LSZ reduction formula: Z is the pole residue of the full propagator, and it is equal to 1 at the leading order, but it can receive corrections at higher orders.

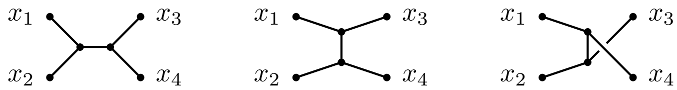
We derived these feynman rules from a real scalar theory with a cubic interaction, but they can be easily generalized to other theories: the only theory specific point is the vertex factor, which depends on the interaction term in the Lagrangian, while the other rules are general and can be applied to any theory.

3.3 | Applications of Feynman Diagrams

Let us now apply the Feynman rules to compute scattering amplitudes for various processes in different quantum field theories. We will start with the simplest case of a scalar field theory and then move on to more complex theories involving fermions and gauge bosons.

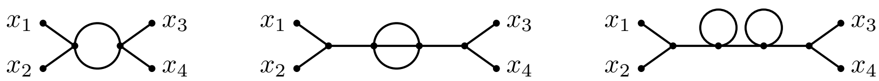
We will divide diagrams in two categories:

1. **The tree-level diagrams**: These are the diagrams that do not contain any loops. They represent the leading order contribution to the scattering amplitude and are often the simplest to compute.



2. **The loop diagrams**: These are the diagrams that contain one or more loops. They represent higher-order corrections to the scattering amplitude and can be more challenging to compute due to the need for regularization and renormalization techniques.

TODO: improve diagrams.



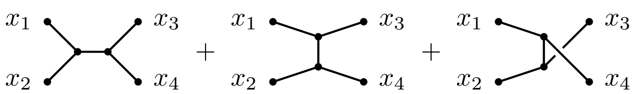
Thus in the loop diagrams, one or more internal lines form a closed loop, which can lead to unresolved integrals like $\int \frac{d^4k}{(2\pi)^4} \frac{1}{k^2-m^2}$, which is divergent and requires regularization and renormalization to obtain a finite result.

Usually the first type of diagrams (tree-level) gives the dominant contribution to the scattering amplitude, while the second type (loop diagrams) provides corrections that can be important for precision calculations. For the first ones, usually some algebra is enough to compute the amplitude, since we have no integrations left, while for the second ones, one needs to use more advanced techniques to handle the divergences and obtain meaningful results. Loop diagrams can also give rise to interesting phenomena such as virtual particles and vacuum polarization, which can have important implications for the behavior of quantum fields and the structure of spacetime, but they are in general more difficult to solve.

Remark. *We said that usually at LO only the tree diagrams appears, but this is nothing general: there exist cases in which we have loops appearing alredy at the leading order, which are more complicated.*

Example ($2 \to 2$ scattering in ϕ^3 theory). The idea is to compute the **differential cross section** $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}$ for the $2 \to 2$ scattering process of identical particles in ϕ^3 theory at tree level, in the center of mass frame, as a function of the total center of mass energy E and the scattering angle θ.

At the leading order, we have already computed the matrix element for the $2 \to 2$ scattering process in ϕ^3 theory: the Feynman rules told us that the relevant diagrams at tree level include the s-channel, t-channel, and u-channel diagrams:

$i\mathcal{M} =$ 

$+\,\mathcal{O}(\lambda^4)$

$$= (-i\lambda)^2 \left[\frac{1}{(p_1+p_2)^2 - m^2} + \frac{1}{(p_1-p_3)^2 - m^2} + \frac{1}{(p_1-p_4)^2 - m^2} \right] + O(\lambda^4).$$

We can define the **scattering angle** θ as the angle between the incoming particle with momentum p_1 and the outgoing particle with momentum p_3:

$$\cos\theta = \frac{\mathbf{p}_1 \cdot \mathbf{p}_3}{|\mathbf{p}_1||\mathbf{p}_3|},$$

and this will allow us to express the scattering amplitude in terms of the Mandelstam variables, which are functions of E and θ. Before proceeding with the calculation, we have to study in more detail the kinematics of the process, which is the subject of the next section.

3.3.1 | 4-Point Kinematics

Consider a more general $2 \to 2$ scattering process, involving two incoming particles with momenta p_1 and p_2, and two outgoing particles with momenta p_3 and p_4. We are not interested in the specific theory describing the interaction or the type of particles involved, but we are requesting that all particles are on-shell, so that

$$p_i^2 = m^2, \quad i = 1, 2, 3, 4.$$

The scattering amplitude is lorentz invariant, so that it is useful to express it in terms of the **Mandelstam variables** s, t, and u.

Definition 3.4 (Mandelstam variables)**.** The Mandelstam variables s, t, and u, are defined as follows:

$$\begin{aligned} s &= (p_1 + p_2)^2 = (p_3 + p_4)^2, \\ t &= (p_1 - p_3)^2 = (p_4 - p_2)^2, \\ u &= (p_1 - p_4)^2 = (p_2 - p_3)^2. \end{aligned}$$

These variables are not independent, since they satisfy the relation

$$s + t + u = \sum_j m_j^2,$$

as we will later prove. These variables are defined in such a way that they are invariant under Lorentz transformations, and provide a convenient way to express the scattering amplitude in terms of the kinematic variables of the process.

Now the idea is to express the scattering amplitude in terms of the Mandelstam variables, which will allow us to compute the differential cross section as a function of the scattering angle θ and the total center of mass energy E.

Starting from their definitions, we can express

$$s = p_1^2 + p_2^2 + 2p_1 \cdot p_2 = m_1^2 + m_2^2 + 2p_1 \cdot p_2,$$

so that it becomes clear that the Mandelstam variables are related to the Lorentz invariants $p_i \cdot p_j$. In particular, we have

$$\begin{cases} p_1 \cdot p_2 = \dfrac{1}{2}\left(s - m_1^2 - m_2^2\right), \\ p_1 \cdot p_3 = \dfrac{1}{2}\left(t - m_1^2 - m_3^2\right), \\ p_2 \cdot p_3 = \dfrac{1}{2}\left(u - m_2^2 - m_3^2\right), \\ p_4 = p_1 + p_2 - p_3, \end{cases}$$

so that every Lorentz invariant $p_i \cdot p_j$ can be expressed in terms of the Mandelstam variables, since p_4 can be always expressed in terms of the other three momenta using momentum conservation. Thus we are going to square the last relation, and we will obtain

$$\begin{aligned} m_4^2 = p_4^2 = (p_1+p_2-p_3)^2 &= m_1^2 + m_2^2 + m_3^2 + 2p_1 \cdot p_2 - 2p_1 \cdot p_3 - 2p_2 \cdot p_3 \\ &= m_1^2 + m_2^2 + m_3^2 + 2\left(\tfrac{1}{2}\right)\left(s - m_1^2 - m_2^2\right) \\ &- 2\left(-\tfrac{1}{2}\right)\left(t - m_1^2 - m_3^2\right) - 2\left(-\tfrac{1}{2}\right)\left(u - m_2^2 - m_3^2\right) \\ &= s + t + u - m_1^2 - m_2^2 - m_3^2. \end{aligned}$$

Thus we have

$$s + t + u = \sum_j m_j^2,$$

which is the relation we wanted to prove. For an N-particle scattering process, we can define the Mandelstam variables as ($N > 4$):

1. $4N$ momentum components p_i^μ for $i = 1, \dots, N$.
2. N on-shell conditions $p_i^2 = m_i^2$.
3. 4 momentum conservation conditions $\sum_{i=1}^N p_i^\mu = 0$ (from translational invariance).
4. 6 independent Lorentz invariants $p_i \cdot p_j$ for $i < j$, i.e. 6 lorentz transformations leaving the amplitude invariant.

Thus in the end we have

$$4N - N - 4 - 6 = 3N - 10 \textbf{ independent invariants},$$

which can be chosen as

$$s_{ij} = (p_i + p_j)^2, \quad i < j,$$

these represent the generalization of the Mandelstam variables for N-particle scattering processes.

Applications

Now for a $2 \to 2$ general process, in the center of mass frame, we have two incoming particles parallel along the z-axis, and two outgoing particles with momenta $\mathbf{p}_3$ and $\mathbf{p}_4$ forming an angle θ with the z-axis

$$p_1 = \begin{pmatrix} E_1 \\ 0 \\ 0 \\ p \end{pmatrix}, \quad p_2 = \begin{pmatrix} E_2 \\ 0 \\ 0 \\ -p \end{pmatrix},$$

where $p = |\mathbf{p}_1| = |\mathbf{p}_2|$ since $\mathbf{p}_1 + \mathbf{p}_2 = 0$ in the center of mass frame. Now considering the outgoing particles, we can rotate the coordinate system so that they lie in the z-y-plane, with momenta

$$\mathbf{p}_4 = \mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}_3 = -\mathbf{p}_3, \quad p' = |\mathbf{p}_3| = |\mathbf{p}_4|,$$

$$\mathbf{p}_3 = \begin{pmatrix} E_3 \\ 0 \\ p' \sin\theta \\ p' \cos\theta \end{pmatrix}, \quad \mathbf{p}_4 = \begin{pmatrix} E_4 \\ 0 \\ -p' \sin\theta \\ -p' \cos\theta \end{pmatrix},$$

since we are still in the center of mass frame; θ is the scattering angle defined previously. Note that the scattering amplitude is invariant under rotations, so that we can always choose the coordinate system in such a way that the outgoing particles lie in the z-y-plane, and this will not affect the final result for the scattering amplitude.

Now let us compute the center of mass energy E and the Mandelstam variables in terms of the scattering angle θ. We have

$$E \equiv E_{cm} = E_1 + E_2 = E_3 + E_4,$$

with the first Mandelstam variable defined as

$$s = (p_1 + p_2)^2 = (E_1 + E_2)^2 = E^2,$$

since the spatial momenta of the incoming particles cancel out. Now

$$(p_1 + p_2) \cdot p_1 = E\, E_1,$$

and we can also express it as

$$(p_1 + p_2) \cdot p_1 = m_1^2 + p_2 \cdot p_1 = m_1^2 + \frac{1}{2}(s - m_1^2 - m_2^2) = \frac{1}{2}(E^2 + m_1^2 - m_2^2).$$

Thus we have

$$\begin{cases} E_1 = \dfrac{E^2 + m_1^2 - m_2^2}{2E}, \\ E_2 = \dfrac{E^2 + m_2^2 - m_1^2}{2E}, \end{cases}$$

and now

$$m_1^2 = p_1^2 = E_1 - |\mathbf{p}_1|^2, \quad p = \frac{\sqrt{[E^2 - (m_1 + m_2)^2]\,[E^2 - (m_1 - m_2)^2]}}{2E},$$

obtaining the expression for the momentum of the incoming particles in the center of mass frame. The problem is symmetric in the incoming and outgoing particles, so that we can do the same calculation for the outgoing particles, and we will obtain

$$\begin{cases} E_3 = \dfrac{E^2 + m_3^2 - m_4^2}{2E}, \\ E_4 = \dfrac{E^2 + m_4^2 - m_3^2}{2E}, \end{cases} \qquad p' = \frac{\sqrt{[E^2 - (m_3 + m_4)^2]\,[E^2 - (m_3 - m_4)^2]}}{2E},$$

where the same calculation enforces the substitutions $1 \to 3$, $2 \to 4$ and $p \to p'$.

For the second Mandelstam variable, we have similarly

$$t = (p_1 - p_3)^2 = m_1^2 + m_3^2 - 2p_1 \cdot p_3 = m_1^2 + m_2^2 - 2(E_1 E_3 - pp' \cos\theta) = t(E, \theta),$$

showing that t is a function of the center of mass energy E and the scattering angle θ. For the third Mandelstam variable, we have

$$u = (p_1 - p_4)^2 = m_1^2 + m_4^2 - 2p_1 \cdot p_4 = m_1^2 + m_4^2 - 2(E_1 E_4 + pp' \cos\theta) = u(E, \theta),$$

which is also a function of the invariants E and θ. Thus we have expressed the Mandelstam variables in terms of the center of mass energy E and the scattering angle θ, which will allow us to compute the scattering amplitude and the differential cross section as a function of these variables.

Special cases

- ($m_1 = m_2$): In this case, we have

$$E_1 = E_2 = \frac{E}{2}, \quad p = \frac{\sqrt{E^2 - 4m^2}}{2}.$$

- ($m_3 = m_4$): In this case, we have

$$E_3 = E_4 = \frac{E}{2}, \quad p' = \frac{\sqrt{E^2 - 4m^2}}{2}.$$

- **Elastic scattering**: In this case, we have $m_1 = m_3$ and $m_2 = m_4$, so that

$$E_1 = E_3, \quad E_2 = E_4, \quad p = p'.$$

- ($m_1 = m_2 = m_3 = m_4$): In this case, we have

$$E_i = \frac{E}{2}, \quad p = p' = \frac{\sqrt{E^2 - 4m^2}}{2}.$$

  This is the combination of the previous two cases, and it is the most common one in practice, since it describes the scattering of identical particles. The mandelstam variables in this case are given by

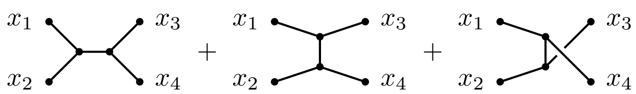
$$\begin{cases} s = E^2, \\ t = -\frac{1}{2}(E^2 - 4m^2)(1 - \cos\theta), \\ u = -\frac{1}{2}(E^2 - 4m^2)(1 + \cos\theta), \end{cases}$$

  with the usual relation $s + t + u = 4m^2$.

Back to The Example

Now we have what we need to compute the scattering amplitude for the $2 \to 2$ scattering process in ϕ^3 theory at tree level, which is given by the sum of the s-channel, t-channel, and u-channel diagrams. Let us come back to this example.

Example ($2 \to 2$ scattering in ϕ^3 theory). Remember that at the leading order, the matrix element for the $2 \to 2$ scattering process in ϕ^3 theory can be computed using the Feynman rules: the relevant Feynman diagrams at tree level include the s-channel, t-channel, and u-channel diagrams, which have the following momentum dependence:

$i\mathcal{M} =$  $+ O(\lambda^4)$,

which contribute to the scattering amplitude as follows:

$$i\mathcal{M} = (-i\lambda)^2 \left[\frac{1}{s - m^2} + \frac{1}{t - m^2} + \frac{1}{u - m^2} \right] + O(\lambda^4),$$

from which it is easy to understand why they are called s-channel, t-channel, and u-channel diagrams; remember that in this case the Mandelstam variables are given by

$$\begin{cases} s = E^2, \\ t = -\frac{1}{2}(E^2 - 4m^2)(1 - \cos\theta), \\ u = -\frac{1}{2}(E^2 - 4m^2)(1 + \cos\theta), \end{cases}$$

since we are considering the scattering of identical particles, so that $m_1 = m_2 = m_3 = m_4 = m$.

Now we can express the scattering amplitude in terms of the Mandelstam variables (expressing them in terms of E and θ as we found before), which will allow us to compute the differential cross section as a function of the scattering angle θ and the total center of mass energy E.

In particular, we have the differential cross section found at the beginning of the section, which in the center of mass frame can be expressed as

$$\begin{aligned}\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} &= \frac{1}{64\pi^2 E^2}\frac{p'}{p}|\mathcal{M}|^2 = \frac{1}{64\pi^2 E^2}\frac{\sqrt{E^2-4m^2}}{\sqrt{E^2-4m^2}}|\mathcal{M}|^2 \\ &= \frac{1}{64\pi^2 s}|\mathcal{M}|^2 = \frac{\lambda^4}{64\pi^2 E^2}\left|\frac{1}{s-m^2}+\frac{1}{t-m^2}+\frac{1}{u-m^2}\right|^2 + O(\lambda^6).\end{aligned}$$

This is the final result for the differential cross section for the $2 \to 2$ scattering process in ϕ^3 theory at tree level, as a function of the total center of mass energy E and the scattering angle θ.

3.3.2 | Feynman Rules for Other Theories

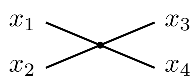
We can generalize some of the Feynman rules we have found for the real scalar field to other theories, such as for different types of fields (complex scalar fields, Dirac fermion fields, vector boson fields) and for different interactions (Yukawa interactions, gauge interactions).

For example, the generalization to a ϕ^n theory is basically trivial:

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{n!}\phi^n,$$

we just have to take into account the fact that each interaction vertex will now involve n lines instead of 3, and the Feynman rule for the vertex will be given by $-i\lambda$ times a combinatorial factor that takes into account the number of ways to connect the lines to the vertex. For example, for a $2 \to 2$ process in a ϕ^4 theory, since the vertexes come from the integrals $\int \mathrm{d}^4x\mathcal{L}_{\text{int}}$ after contracting all the fields, at the tree level we have only one diagram:

$$i\mathcal{M} =$$



$$+\, \mathcal{O}(\lambda^2) = -i\lambda + \mathcal{O}(\lambda^2),$$

which is the one with a single vertex, and the Feynman rule for the vertex will be given by $-i\lambda$.

3.4 | General Approach for Feynman Rules

3.4.1 | Complex Scalar Field

3.4.2 | Dirac Fermion Field

Additional Feynman Rules for Fermions

Example (Yukawa theory).

Example.

3.4.3 | Vector Bosons

3.4.4 | Quantum Electrodynamics (QED)

Example (Two different types of particles in QED).

3.4.5 | Observations on Diagrams

3.5 | Crossing Symmetry

3.5.1 | Crossing Symmetry and Identical Fermions

4 | Quantum Electrodynamics

4.1 | Scattering Processes in QED

Consider QED with two flavors of fermions, namely the electron (e^-) and the muon (μ^-). The Lagrangian for this theory is given by:[1]

$$\begin{aligned}\mathcal{L} = &-\frac{1}{4}F_{\nu\rho}F^{\nu\rho} + \bar{\psi}_e(i\partial\!\!\!/ - m_e)\psi_e + \bar{\psi}_\mu(i\partial\!\!\!/ - m_\mu)\psi_\mu \\ &+ Q_e e\bar{\psi}_e\gamma^\nu\psi_e A_\nu + Q_\mu e\bar{\psi}_\mu\gamma^\nu\psi_\mu A_\nu,\end{aligned}$$

where Q_i (with $i = e, \mu$) are the electric charges of the electron and muon, respectively, in units of the elementary charge e. In this theory, we can study various scattering processes, such as electron-muon scattering, Compton scattering, and more. For what follows, we will set $Q_e = Q_\mu = -1$ for simplicity, as both the electron and muon have the same electric charge in nature, but QED allows for the possibility of different charges.[2]

4.1.1 | Electron - Muon Scattering

Consider the scattering process $e^-e^+ \to \mu^-\mu^+$. The leading order Feynman diagram (Tree level) for this process is given by the exchange of a virtual photon between the electron-positron pair and the muon-antimuon pair. The amplitude for this process can be calculated using the Feynman rules for QED.

TODO: add the Feynman diagram here.

We will not compute explicitly the amplitude here, but we will follow an equivalent simple rule:

Apply the Feynman rules while following the Fermion lines backwards.

Doing so we get the following expression for the amplitude:

$$i\mathcal{M} = \overline{v}_{s_2}(p_2)(-ie\gamma^\nu)u_{s_1}(p_1)\frac{-ig_{\nu\rho}}{(p_1+p_2)^2}\overline{u}_{s_3}(p_3)(-ie\gamma^\rho)v_{s_4}(p_4),$$

where the first term comes from the electron-positron vertex, the second term is the photon propagator, and the third term comes from the muon-antimuon vertex. The indices s_i (with $i = 1, 2, 3, 4$) represent the spin states of the particles involved in the scattering process.

[1] Here μ stands for the muon, it is not a Lorentz index.

[2] This is a non-trivial statement, it is possible to prove that this is true for abelian gauge theories, but not for non-abelian ones.

We can simplify the notation by defining $u_i \equiv u_{s_i}(p_i)$ and $v_i \equiv v_{s_i}(p_i)$ for $i = 1, 2, 3, 4$. With this notation, the amplitude can be written as:

$$i\mathcal{M} = \frac{ie^2}{s} \left(\overline{v}_2 \gamma^\rho u_1\right) \left(\overline{u}_3 \gamma_\rho v_4\right).$$

Since ideally we want to compute the cross section for this process, we need to compute the squared amplitude, averaged over the initial spins and summed over the final spins. This implies that we need to compute $\mathcal{M}^*$ other than $|\mathcal{M}|^2$. Recalling that

$$\left(\overline{\psi}_1 \gamma^{\nu_1} \cdots \gamma^{\nu_n} \psi_2\right) = \left(\overline{\psi}_2 \gamma^{\nu_n} \cdots \gamma^{\nu_1} \psi_1\right),$$

thus the matrix element can be rewritten as

TODO: add the Feynman diagram for the conjugate process here.

$$i\mathcal{M}^* = \frac{ie^2}{s} \left(\overline{u}_1 \gamma^\rho v_2\right) \left(\overline{v}_4 \gamma_\rho u_3\right) =$$

where thus the conjugate process should describe the same diagram but flipped horizontally.

Polarization of the states. Now we could specify the polarization of the incoming and outgoing states, since most of the times we do not have control over the polarization of the particles in the initial state (they are not prepared in any specific spin state, which is therefore random), and we can assume to not measure the polarization of the particles in the final state. This leads us to the definition of an **unpolarized cross section**, which is obtained by averaging over the initial spins and summing over the final spins. In this case, this is obtained by integrating over the phase space the **unpolarized squared scattering amplitude**:

$$|\mathcal{M}|^2_{\text{unpol}} = \frac{1}{2} \sum_{s_1} \frac{1}{2} \sum_{s_2} \sum_{s_3} \sum_{s_4} \left|\mathcal{M}_{s_1\, s_2 \to s_3\, s_4}\right|^2,$$

[... computations ...]

So the square of a fermion line, summed over polarizations, can be expressed as a trace of gamma matrices: it is a powerful result, since it allows us to compute the squared amplitude without having to worry about the spinor structure of the states, but only applying a trace. This is a consequence of the completeness relations for the spinors, which allow us to express the sum over polarizations in terms of traces of gamma matrices.

[... computations ... diagram ...]

Thus we

- Follow the fermion lines backwards to write down the amplitude.
- Replace "cut lines" with a sum over polarizations, which can be expressed as a trace of gamma matrices.

$$|\mathcal{M}|^2_{\text{unpol}} = \frac{e^4}{4s^2} \operatorname{Tr}\left[(\not{p}_1 + m_e)\gamma^\rho(\not{p}_2 + m_e)\gamma^\sigma\right] \operatorname{Tr}\left[(\not{p}_3 + m_\mu)\gamma_\sigma(\not{p}_4 - m_\mu)\gamma_\rho\right].$$

But we need to understand how to compute traces of gamma matrices, which is the topic of the next section, in which we will review some properties of the gamma matrices and of the Dirac algebra in general.

4.1.2 | Dirac Algebra

[...]

Remember that we are tracing with respect to the spinor indices, thus we are tracing over 4×4 matrices, since the gamma matrices are 4×4 matrices. The following identities hold for the traces of gamma matrices:

$$\mathrm{Tr}[\mathbb{I}_4] = 4,$$

$$\mathrm{Tr}[\gamma^\nu] = 0,$$

$$\mathrm{Tr}[\gamma^\nu \gamma^\rho] = 4g^{\nu\rho},$$

Proof.

$$\mathrm{Tr}[\gamma^\nu \gamma^\rho] = \mathrm{Tr}[\gamma^\rho \gamma^\nu] = -\,\mathrm{Tr}[\gamma^\nu \gamma^\rho] + 2g^{\nu\rho}\,\mathrm{Tr}[\mathbb{I}_4] = 4g^{\nu\rho},$$

since we can write

$$2\,\mathrm{Tr}[\gamma^\nu \gamma^\rho] = 2g^{\nu\rho}\,\mathrm{Tr}[\mathbb{I}_4] = 8g^{\nu\rho}.$$

□

Let us use a simplified notation, in which $\mathrm{Tr}[\mu\nu\rho\sigma] \equiv \mathrm{Tr}[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma]$. Then we can proceed in the proof of the following identity:

$$\mathrm{Tr}[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma] = 4\left(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho}\right).$$

Proof.

$$\mathrm{Tr}[\mu\nu\rho\sigma] = \mathrm{Tr}[\nu\sigma\rho\mu] = -\,\mathrm{Tr}[\nu\rho\mu\sigma] + 2g^{\mu\sigma}\,\mathrm{Tr}[\nu\rho],$$

where now we can apply the previous identity to get

$$\mathrm{Tr}[\mu\nu\rho\sigma] = \mathrm{Tr}[\nu\mu\rho\sigma] - 2g^{\mu\rho}\,\mathrm{Tr}[\nu\sigma] + 8g^{\mu\sigma}g^{\nu\rho} = \dots$$

where in the end simplifying we get the desired result:

$$\mathrm{Tr}[\mu\nu\rho\sigma] = 4\left(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho}\right).$$

□

We are not going to prove the followig traces, but they are very useful in the computations involving polirized fermions or chiral theories. Let us define $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$, then we have the following identities:

$$\begin{cases} \mathrm{Tr}\left[\gamma^5\right] = 0, \\ \mathrm{Tr}\left[\gamma^5\gamma^\nu\right] = 0, \\ \mathrm{Tr}\left[\gamma^5\gamma^\nu\gamma^\rho\right] = 0, \\ \mathrm{Tr}\left[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^5\right] = -4i\epsilon^{\mu\nu\rho\sigma}, \\ \mathrm{Tr}\left[\gamma^{\mu_1}\cdots\gamma^{\mu_{2n+1}}\gamma^5\right] = 0, \end{cases}$$

where $\epsilon^{\mu\nu\rho\sigma}$ is the Levi-Civita symbol.

4.1.3 | Back to Electron - Muon Scattering

Let us rewrite the unpolarized squared amplitude for the electron-muon scattering process:

$$|\mathcal{M}|^2_{\text{unpol}} = \frac{e^4}{4s^2} \operatorname{Tr}\left[(\not{p}_1 + m_e)\gamma^\rho(\not{p}_2 + m_e)\gamma^\sigma\right] \operatorname{Tr}\left[(\not{p}_3 + m_\mu)\gamma_\sigma(\not{p}_4 - m_\mu)\gamma_\rho\right].$$

Then let us apply the identities for the traces of gamma matrices to compute the traces in the expression above. From the first trace we get

$$\operatorname{Tr}\left[(\not{p}_1 + m_e)\gamma^\rho(\not{p}_2 + m_e)\gamma^\sigma\right]$$

where computing the products inside the trace we can ignore all the terms that contain an odd number of gamma matrices, since their trace is zero. Thus we are left with

$$\operatorname{Tr}\left[(\not{p}_1 + m_e)\gamma^\rho(\not{p}_2 + m_e)\gamma^\sigma\right] = \operatorname{Tr}\left[\not{p}_1\gamma^\rho\not{p}_2\gamma^\sigma\right] - m_e^2 \operatorname{Tr}[\gamma^\rho\gamma^\sigma],$$

but the last one is a result we already know, while for the first we can take out the four-momentum from the trace, since it is just a number, and get

TODO: check computations

$$\operatorname{Tr}\left[(\not{p}_1 + m_e)\gamma^\rho(\not{p}_2 + m_e)\gamma^\sigma\right] = 4\left(p_1^\rho p_2^\sigma + p_1^\sigma p_2^\rho - g^{\rho\sigma}(p_1 \cdot p_2 - m_e^2)\right).$$

Thus in the end we get

$$= 4(p_1^\nu p_2^\mu + p_1^\mu p_2^\nu - (p_1 \cdot p_2)g^{\mu\nu}) - m_e^2 \cdot 4g^{\mu\nu}.$$

The other trace term is basically identical....

[reasoning about masses...]

Thus we can neglect the mass of the electron, since it will only become importabt in following orders in perturbation theory. Thus we can write the unpolarized squared amplitude as [...]

$$|\mathcal{M}|^2_{\text{unpol}} = \frac{e^4}{4s^2} \cdot 16\left[(p_1 \cdot p_3)(p_2 \cdot p_4) + (p_1 \cdot p_4)(p_2 \cdot p_3) - (p_1 \cdot p_2)(p_3 \cdot p_4) + m_\mu^2(p_1 \cdot p_2)\right].$$

[...]

$$|\mathcal{M}|^2_{\text{unpol}} = \frac{2e^4}{s^2}\left[(t - m_\mu)^2 + (u - m_\mu)^2 + 2m_\mu^2 s\right],$$

where now we have expressed the result in terms of the Mandelstam variables ($s + t + u = 2m_\mu^2 + 2m_e^2 \sim 2m_\mu^2$), which as we have seen are lorentz invariant combinations of the four-momenta of the particles involved in the scattering process. This is a very useful result, since it allows us to express the squared amplitude in terms of Lorentz invariant quantities, which are easier to compute and to interpret physically.

Center of Mass Frame

We move to the center of mass frame, in which the total momentum of the system is zero. In this frame, we can express the four-momenta of the particles in terms of their energies and three-momenta. Let us denote by E_i and $\vec{p}_i$ the energy and three-momentum of the i-th particle, respectively. Then we have

$$E_1 = E_2 = E_3 = E_4 = \frac{E_{cm}}{2} \equiv E_e \geq m_\mu,$$

and we can write

$$p_1 = \begin{pmatrix} E_e \\ 0 \\ 0 \\ E_e \end{pmatrix}, \quad p_2 = \begin{pmatrix} E_e \\ 0 \\ 0 \\ -E_e \end{pmatrix}, \quad p_3 = \begin{pmatrix} E_e \\ 0 \\ p' \sin\theta \\ p' \cos\theta \end{pmatrix}, \quad p_4 = \begin{pmatrix} E_e \\ 0 \\ -p' \sin\theta \\ -p' \cos\theta \end{pmatrix},$$

where $p' = \sqrt{E_e^2 - m_\mu^2}$. Then we can express the Mandelstam variables in terms of the energies and the scattering angle θ:

$$\begin{cases} s = E_{cm}^2 = 4E_e^2, \\ t = (p_3 - p_1)^2 = m_\mu^2 - 2(E_e^2 - E_e p' \cos\theta), \\ u = (p_3 - p_2)^2 = m_\mu^2 - 2(E_e^2 + E_e p' \cos\theta), \end{cases}$$

where we have used the fact that $p_1^2 = p_2^2 = m_e^2 \sim 0$ and $p_3^2 = p_4^2 = m_\mu^2$. Thus we can express the unpolarized squared amplitude in terms of the center of mass energy and the scattering angle:

$$|\mathcal{M}|^2_{\text{unpol}} = \dots$$

which, expanding the squares and using the expression for p' we get to

$$|\mathcal{M}|^2_{\text{unpol}} = e^4 \left[(1 + \cos^2\theta) + \frac{m_\mu^2}{E_e^2}(1 - \cos^2\theta) \right].$$

Now having expressed the unpolarized squared amplitude in terms of the center of mass energy and the scattering angle, we can compute the differential cross section for this process. The differential cross section is given by

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 E_{cm}^2} \frac{p_f}{p_i} |\mathcal{M}|^2_{\text{unpol}},$$

where $E_{cm}^2 = 4E_e^2$ is the center of mass energy squared and the fraction of momenta can be written as

$$\frac{p_f}{p_i} = \frac{p'}{E_e} = \sqrt{1 - \frac{m_\mu^2}{E_e^2}}.$$

Thus we can compute the differential cross section, using the constant

$$\alpha = \frac{e^2}{4\pi} \approx \frac{1}{137},$$

we get

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{16E_e^2} \sqrt{1 - \frac{m_\mu^2}{E_e^2}} \left[(1 + \cos^2\theta) + \frac{m_\mu^2}{E_e^2}(1 - \cos^2\theta) \right].$$

$$\frac{d\sigma}{d\cos\theta} = 2\pi \frac{d\sigma}{d\Omega}.$$

$$\sigma = \int_{-1}^{1} d\cos\theta \, \frac{d\sigma}{d\cos\theta} = \frac{\alpha^2}{E_e^2} \frac{\pi}{3} \sqrt{1 - \frac{m_\mu^2}{E_e^2}} \left(1 + \frac{m_\mu^2}{2E_e^2} \right).$$

Example (Excercise). Use crossing symmetry to compute the unpolarized squared amplitude for the $e^-\mu^- \to e^-\mu^-$ process:

$$|\mathcal{M}|^2_{\text{unpol}}(e^-\mu^- \to e^-\mu^-) = \frac{2e^4}{t^2} \left[(s - m_\mu^2)^2 + (u - m_\mu^2)^2 + 2m_\mu^2 t \right].$$

[...]

Remark (Crossing Symmetry). *[...]*

4.1.4 | Polarized Matrix Elements

There are several formalism to compute the polarized matrix elements, which are the amplitudes for specific spin states of the particles involved in the scattering process. Let us review some of the most common cases:

- **Massless case** In the massless case, we can use the helicity states u_L and u_R, chirality projectors to compute the polarized matrix elements. The chirality projectors are defined as

$$P_L = \frac{1-\gamma_5}{2}, \quad P_R = \frac{1+\gamma_5}{2},$$

and they project the spinors onto their left-handed and right-handed components, respectively:

$$\begin{cases} P_L u_L = u_L, & P_L v_R = v_R, \\ P_R u_R = u_R \dots \end{cases}$$

Let us consider $i\mathcal{M}$ with left-handed fermions

$$u_L(p)\overline{u}_L(p) = \sum_s P_L u_s \overline{u}_s = P_L(\not{p} + m) = \frac{1-\gamma_5}{2}\not{p},$$

Thus we can still use the trace technology to compute the polarized matrix elements, but we need to insert the chirality projectors in the traces, which will make the computations more involved, since we will have to use also the identities for the traces of gamma matrices with γ_5.

...

- **Massive case** In the massive case, we can use the spin states u_s and v_s to compute the polarized matrix elements. [...]

$$P_S = \frac{1}{2}\left(1 + \gamma_5 \not{S}\right),$$

where S^μ is the spin four-vector, which in the rest frame of the particle is given by $S^\mu = (0, \mathbf{s})$, where $\mathbf{s}$ is the three-dimensional spin vector. The spin projectors P_S project the spinors onto their spin states, and they can be used to compute the polarized matrix elements in the massive case. In other frames we can lorentz transform the spin four-vector to get the correct expression for the spin projector.

- **Alternative** The **spinor-helicity formalism** ...

4.1.5 | External Photons and Compton Scattering

TODO: add the Feynman diagram for Compton scattering here.

Let us consider a process where we have an external photon, The amplitude for this process can be computed using the Feynman rules for QED, and it will involve the polarization vector of the external photon.

$$i\mathcal{M} = \epsilon_\mu(p)(i\mathcal{M})^\mu,$$

where the last term is only a way to write the rest of the diagram, which will be contracted with the polarization vector of the photon. The amplitude for this process will depend on the polarization of the external photon, and we will need to sum over the polarizations of the photon to get the unpolarized cross section for this process.

In the rest frame,

$$p^\mu = \begin{pmatrix} E \\ 0 \\ 0 \\ E \end{pmatrix},$$

assuming to be in the Lorenz gauge

$$p_\mu \epsilon^\mu(p) = 0,$$

we can note that this choice does not fix completely the gauge, since we can still perform a gauge transformation of the form

$$\epsilon^\mu(p) \to \epsilon^\mu(p) + \alpha p^\mu,$$

which for example we can use to fix $\epsilon^0(p) = 0$. Thus we have a basis of polarization vectors for the photon, which are given by

$$\epsilon^\mu_\pm(p) = \begin{pmatrix} 0 \\ 1 \\ \pm i \\ 0 \end{pmatrix},$$

[...] and the Ward identity

$$p_\mu (\mathcal{M})^\mu = 0,$$

which tells us how if we replace the polarization vector of the photon with its momentum, we get a vanishing amplitude, which is a consequence of the gauge invariance of the theory. This identity will be very useful in the computations involving external photons, since it allows us to simplify the expressions for the amplitudes by using the fact that we can replace the polarization vector with the momentum of the photon without changing the value of the amplitude.

In other words, the term of the amplitude contracting with the polarization vector of the photon should be a conserved current that couples to the photon, thus it should satisfy the Ward identity, which is a consequence of the gauge invariance of the theory.

$$\partial_\mu J^\mu = 0 \implies p_\mu J^\mu = 0.$$

Polarization Sums

If we look at

$$|\mathcal{M}|^2 = (\epsilon^\mu_s \mathcal{M}_\mu)(\epsilon^{*\nu}_s \mathcal{M}^*_\nu),$$

then for the unpolarized cross section we need to sum over the polarizations of the photon, which means that we need to compute

$$\sum_{s=\pm} \epsilon^\mu_s \epsilon^{*\nu}_s.$$

Assuming again the momentum along the z-axis, we can write this summation as

$$\sum_{s=\pm} \epsilon^\mu_s \epsilon^{*\nu}_s = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

We can rewrite this expression in a more covariant way, by noting that the polarization vectors of the photon satisfy the following completeness relation:

$$\sum_{s=\pm} \epsilon_s^\mu \epsilon_s^{*\nu} = -g^{\mu\nu} + \frac{p^\mu \eta^\nu + p^\nu \eta^\mu}{p \cdot \eta},$$

where η^μ is an arbitrary massless four-vector that satisfies the condition $\epsilon_s \cdot \eta = 0$. This completeness relation allows us to express the sum over the polarizations of the photon in a covariant way, which is very useful in the computations involving external photons, since it allows us to simplify the expressions for the amplitudes by using the fact that we can replace the sum over the polarizations with a covariant expression that involves the metric tensor and the momentum of the photon.

If we chose $\eta^\mu = (1, 0, 0, -1)$, then we can check that the completeness relation reduces to the expression we found before for the sum over the polarizations of the photon:

$$\epsilon_\pm^\mu = \begin{pmatrix} 0 \\ 1 \\ \pm i \\ 0 \end{pmatrix}.$$

[...] [...]

However

$$|\mathcal{M}|^2 = \sum_s \epsilon_s^\mu \epsilon_s^{*\nu} \mathcal{M}_\mu \mathcal{M}_\nu^*,$$

where for the Ward identity to hold we need to have

$$\begin{cases} p_\mu \mathcal{M}^\mu = 0, \\ p_\nu \mathcal{M}^{*\nu} = 0, \end{cases}$$

son in practice we are allowed to replace the sum over the polarizations of the photon with $-g^{\mu\nu}$ without changing the value of the amplitude, since the terms proportional to p^μ and p^ν will vanish when contracted with $\mathcal{M}_\mu$ and $\mathcal{M}_\nu^*$, respectively. Thus we can write

$$\sum_s \epsilon_s^\mu \epsilon_s^{*\nu} \to -g^{\mu\nu}.$$

This derivation is a consequence of Ward Identity in the particular case of just one external photon, but it can be generalized to the case of multiple external photons; indeed we know a stronger version of the Ward identity, which states that if we replace the polarization vector of any external photon with its momentum, we get a vanishing amplitude:

$$p_\mu \mathcal{M}^{\mu\,\nu_1 \cdots \nu_n} = 0,$$

where now we have $n + 1$ external photons, and the amplitude is a tensor with n Lorentz indices, one for each external photon (we removed the polarization vectors of the photons, they are not inside $\mathcal{M}^{\mu\,\nu_1 \cdots \nu_n}$). Thus we can replace the sum over the polarizations of each external photon with $-g^{\mu\nu}$ without changing the value of the amplitude, since the terms proportional to p^μ and p^ν will vanish when contracted with the amplitude due to the Ward identity. This is a very powerful result, since it allows us to simplify the computations involving external photons by replacing the sum over the polarizations with a simple expression that involves only the metric tensor.

We will prove the Ward identity later, but for now we can use it to compute the unpolarized cross section for the Compton scattering process.

Compton Scattering

The compton scattering process is given by $e^-(p_1)\gamma(p_2) \to e^-(p_3)\gamma(p_4)$. Let us set the mass of the electron to zero for simplicity, since it will not play a role in the computations at leading order. Then we can write the amplitude for this process as

TODO: add diagrams

$$
\begin{aligned}
i\mathcal{M}\Big|_{\text{Tree}} &= \dots \textit{diagrams} \dots \\
&= -(ie)^2 \left[\overline{u}_3 \gamma^\nu \frac{i(\not{p}_1 + \not{p}_2)}{(p_1+p_2)^2} \gamma^\nu u_1 + \overline{u}_3 \gamma^\nu \frac{i(\not{p}_3 - \not{p}_2)}{(p_2-p_3)^2} \gamma^\nu u_1 \right] \epsilon_{2\mu} \epsilon^*_{4\nu} \equiv i\mathcal{M}^{\mu\nu} \epsilon_{2\mu} \epsilon^*_{4\nu},
\end{aligned}
$$

and we now want to verify the general Ward identity in this process for $\epsilon_2^\mu \to p_2^\mu$. Thus we need to check the contraction among momentum and the rest of the amplitude:

TODO: check computations

$$
p_{2\mu}(i\mathcal{M})^{\mu\nu} = -(ie)^2 \left[\overline{u}_3 \not{p}_2 \frac{i(\not{p}_1 + \not{p}_2)}{(p_1+p_2)^2} \gamma^\nu u_1 + \overline{u}_3 \gamma^\nu \frac{i(\not{p}_3 - \not{p}_2)}{(p_2-p_3)^2} \not{p}_2 u_1 \right] \epsilon^*_{4\nu}.
$$

$$
\not{p}\not{p} = p^2 \mathbb{I}_4,
$$

$$
\begin{aligned}
\text{I line: } & \not{p}_1\not{p}_2 = -\not{p}_2\not{p}_1 + 2p_1 \cdot p_2 \text{ and } \not{p}_2 u_1 = 0, \\
\text{II line: } & \not{p}_2\not{p}_3 = -\not{p}_3\not{p}_2 + 2p_2 \cdot p_3 \text{ and } \overline{u}_3 \not{p}_3 = 0,
\end{aligned}
$$

[...]

Remark. *We get a zero amplitude independently of the choice of the polarization vector of the outgoing photon ϵ_4^ν, and it should work even for an unphysical state of ϵ_4^ν.*

It is important to note that the Ward identity does not hold for individual Feynman diagrams, but only for the sum of all the diagrams that contribute to a given process. There was indeed a cancellation between the two diagrams that contribute to the Compton scattering process, which is a consequence of the gauge invariance of the theory.

The single diagram is not gauge invariant indeed, but their sum is, and this is a consequence of the fact that the gauge invariance of the theory is a global property that is not manifest in the individual diagrams, but only in their sum.

4.1.6 | Notes on Convergence

4.1.7 | Ward Identity and Lorentz Transformations

4.2 | QED as an Abelian Gauge Theory

4.3 | Born Approximation and Non-Relativistic Scattering

[...]

If

$$
V(\mathbf{x}) = -\int \frac{d^3\mathbf{q}}{(2\pi)^3} \tilde{M}(\mathbf{q}) e^{i\mathbf{q}\cdot\mathbf{x}},
$$

then ...

TODO: draw the Feynman diagram

Considering the same process in Yukawa theory, we have at tree level [*diagram* $= -ig\mathbb{I}$]

$$i\mathcal{M} = (-ig)^2 \overline{u}_3 u_1 \frac{i}{(p_3-p_1)^2 - m_\phi^2} \overline{u}_4 u_2,$$

where $(p_1-p_3)^2 \sim -|\mathbf{q}|^2$, while for the [...] we have

$$\overline{u}_s u_{s'} = u_s^\dagger \gamma^0 u_{s'} \sim 2 \begin{pmatrix} \xi_s^\dagger \\ \xi_s^\dagger \end{pmatrix} \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix} \begin{pmatrix} \xi_{s'} \\ \xi_{s'} \end{pmatrix} = 2m\delta_{ss'}.$$

Then the matrix element is

$$i\mathcal{M} = ig^2 2m_e 2m_p \frac{1}{|\mathbf{q}|^2 + m_\phi^2} \text{ for } \begin{cases} s_3 = s_1, \\ s_4 = s_2, \end{cases}$$

and then

$$\tilde{\mathcal{M}} = \frac{g^2}{|\mathbf{q}|^2 + m_\phi^2}.$$

Then let us recover the formula for the potential:

$$\begin{aligned} V(\mathbf{x}) &= -\int \frac{d^3\mathbf{q}}{(2\pi)^3} \tilde{\mathcal{M}}(\mathbf{q}) e^{i\mathbf{q}\cdot\mathbf{x}} \\ &= -\frac{g^2}{(2\pi)^2} \int_0^\infty \mathrm{d}q \frac{q^2}{q^2+m_\phi^2} \int_{-1}^1 \mathrm{d}(\cos\theta) e^{iqr\cos\theta}, \end{aligned}$$

where we have switched to spherical coordinates and $r = |\mathbf{x}|$ (note that the integral over the azimuthal angle ϕ gives 2π and we already simplified it). Then

$$\begin{aligned} V(\mathbf{x}) &= -\frac{g^2}{(2\pi)^2} \int_0^\infty \mathrm{d}q \frac{q^2}{q^2+m_\phi^2} \frac{e^{iqr} - e^{-iqr}}{iqr} \\ &= -\frac{g^2}{(2\pi)^2 ir} \int_{-\infty}^\infty \mathrm{d}q \frac{qe^{iqr}}{q^2+m_\phi^2}, \end{aligned}$$

where we have extended the integral to negative q since the integrand is an odd function of q (in the second integral perform $q \to -q$). Then we can evaluate the integral using the residue theorem, and we find

TODO: draw the complex plane

[*draw of complex plane with upper circle and the two residues at* $\pm imq$]

$$V(\mathbf{x}) = -\frac{g^2}{4\pi} \frac{e^{-m_\phi r}}{r},$$

which is the Yukawa potential. In the limit $m_\phi \to 0$, we recover the Coulomb potential $V(\mathbf{x}) = -\frac{g^2}{4\pi r}$.

Characteristics of the potential:

- **Short range**: the potential decays exponentially at large distances, with a characteristic length scale $r \sim 1/m_\phi$ (the Compton wavelength of the mediator).
- **Universally attractive**: the potential is always negative, regardless of the signs of the charges (since the coupling g appears squared); thus exchanging particles with antiparticles does not change the sign of $i\mathcal{M}$ and therefore the potential $V(\mathbf{x})$.

Instead for the Coulomb potential, we have

- **Long range**: the potential decays as $1/r$ at large distances, and thus it has an infinite range.
- **Attractive or repulsive**: the potential can be either negative (attractive) or positive (repulsive) depending on the signs of the charges (since the coupling g appears squared, but the charges can be positive or negative); thus exchanging particles with antiparticles changes the sign of $i\mathcal{M}$ and therefore the potential $V(\mathbf{x})$.

Thus QFT yields results that are consistent with the phenomenological potentials used in non-relativistic quantum mechanics, and it also provides a deeper understanding of the underlying physics, such as the role of virtual particles and the nature of the interactions.

5 Path Integrals

Path integrals were invented as an alternative but equivalent quantization procedure to canonical quantization. In this chapter, we will introduce the path integral formulation of quantum field theory and show how it can be used to compute correlation functions and scattering amplitudes.

Applying path integrals to traditional quantum mechanics seems a bit complicated and unnecessary, but it becomes very useful in quantum field theory, where the canonical quantization procedure can be quite cumbersome. Moreover, path integrals provide a more intuitive picture of quantum mechanics and quantum field theory, as they allow us to think of particles as taking all possible paths between two points, rather than just a single path.

Among other things, path integrals are particularly useful:

- **Lorentz invariance is manifest**: the path integral formulation is manifestly Lorentz invariant, since it is a lagrangian formulation of the theory, and thus it does not require a choice of time coordinate or a specific reference frame (we do not need the hamiltonian).
- **Quantization of gauge field theories** is more straightforward: the path integral formulation allows us to easily implement gauge fixing and Faddeev-Popov ghosts, which are necessary for quantizing gauge theories.
- **Numerical simulations for strongly interacting theories** (where perturbation theory fails): path integrals can be evaluated numerically using Monte Carlo methods, which is particularly useful for studying strongly interacting theories such as quantum chromodynamics (QCD) on the lattice (the idea is to discretize spacetime into a lattice and then evaluate the path integral numerically).
- The use of **functional integrals** allows us to easily derive the generating functional for correlation functions, which is a powerful tool for computing correlation functions and scattering amplitudes in quantum field theory. For examples *symmetries* in QFT are much easier to understand in the path integral formulation, since they can be implemented as transformations of the fields in the path integral, and thus we can easily derive Ward identities and other relations between correlation functions.

5.1 | Path Integral Formulation of Quantum Mechanics

The idea of the path integral formulation is to express the transition amplitude between two states as a sum over all possible paths that the system can take between those states. In quantum mechanics, the most used eexample to illustrate the path integral formulation is the **double slit experiment**, where we can think of the particle as taking all possible paths from the source to the detector, and the interference pattern arises from the constructive and destructive interference of these paths.

TODO: Add drawing of double slit experiment

[*drawing of source of particles, slits and detector*]

$$A(x \to y) = A(x \to a \to y) + A(x \to b \to y),$$

where a and b are the two slits. In the path integral formulation, we can express the transition amplitude as a sum over all possible paths that the particle can take from x to y, but we can do more, adding virtually an infinite number of screens with slits in between x and y, and thus we can express the transition amplitude as a sum over all possible paths that the particle can take from x to y through any number of slits. In the limit of infinitely many screens with infinitely many slits, we can express the transition amplitude as a sum over all possible paths that the particle can take from x to y, without any constraints.

TODO: Add drawing of infinite walls and slits among x and y

[*draw of infinite walls and slits among x and y*]

If γ is a specific path from x to y, then we can associate to it a complex number $A(\gamma)$ called the amplitude of the path, and then we can express the transition amplitude as a sum over all possible paths:

$$A(x \to y) = \sum_{\gamma} A(x \to y,\, \gamma).$$

In order to implement this idea, we can discretize the time interval between x and y into N small intervals of size ϵ, and then we can express the transition amplitude as a sum over all possible paths that the particle can take from x to y through these small intervals:

$$\langle f;\, t_f | i;\, t_i \rangle\,, \quad t_i < t_1 < \cdots < t_n < t_f, \quad t_{k+1} - t_k = \epsilon,$$

where we have inserted $N-1$ complete sets of states at intermediate times $t_1, \ldots, t_{N-1}$. Then we can express the transition amplitude as a sum over all possible paths that the particle can take from x to y through these small intervals by adding the resolution of the identity at each intermediate time:

$$\langle f;\, t_f | i;\, t_i \rangle = \int \mathrm{d}q_1 \cdots \mathrm{d}q_n \,\langle f;\, t_f | q_n;\, t_n \rangle \langle q_n;\, t_n | q_{N-1};\, t_{N-1} \rangle \cdots \langle q_1;\, t_1 | i;\, t_i \rangle\,.$$

The result to this expression is

$$\langle f;\, t_f | i;\, t_i \rangle = N \int_{q(t_i)=x_i}^{q(t_f)=x_f} \mathcal{D}q(t) e^{iS[q(t)]},$$

where $S[q(t)] = \int \mathrm{d}t\, L(q(t), \dot{q}(t))$ is the action of the path $q(t)$, and $\mathcal{D}q(t)$ is a measure that assigns a weight to each path. The normalization factor N is chosen such that the transition amplitude is properly normalized. In order to understand this expression more intuitively, we can always go back to its discretized version:

$$\langle f;\, t_f | i;\, t_i \rangle = N \int \mathrm{d}q_1 \cdots \mathrm{d}q_n \exp\left[i \sum_k L(q_k, \dot{q}_k) \cdot \epsilon \right], \quad \dot{q}_k = \frac{q_{k+1} - q_k}{\epsilon}.$$

We can also generalize the last expression using the time ordering operator $\hat{T}$:

$$\langle f;\, t_f|\, \hat{T}\{\hat{q}(t_a)\hat{q}(t_b)\cdots\}\, |i;\, t_i\rangle = N\int_{q(t_i)=x_i}^{q(t_f)=x_f} \mathcal{D}q(t)e^{iS[q(t)]}q(t_a)q(t_b)\cdots\,, \quad t_i < t_a < t_b < \cdots < t_f,$$

where the time ordering operator $\hat{T}$ ensures that the operators are ordered in time from right to left, with the earliest time on the right and the latest time on the left. This expression allows us to compute correlation functions of the form $\langle f;\, t_f|\, \hat{T}\{\hat{q}(t_a)\hat{q}(t_b)\cdots\}\, |i;\, t_i\rangle$ using path integrals.

There exists also a discretized version of the last expression, which is more intuitive and easier to understand:

$$\langle f;\, t_f|\, \hat{T}\{\hat{q}(t_a)\hat{q}(t_b)\cdots\}\, |i;\, t_i\rangle = N\int \mathrm{d}q_1\cdots \mathrm{d}q_n \exp\left[i\sum_k L(q_k,\dot{q}_k)\cdot\epsilon\right]q(t_a)q(t_b)\cdots\,,$$

where we have replaced the operators $\hat{q}(t_a)$ and $\hat{q}(t_b)$ with the corresponding functions $q(t_a)$ and $q(t_b)$ in the path integral as did before. This expression allows us to compute correlation functions of the form $\langle f;\, t_f|\, \hat{T}\{\hat{q}(t_a)\hat{q}(t_b)\cdots\}\, |i;\, t_i\rangle$ using path integrals in a more intuitive way. Time ordering is basically implemented for free, since the path integral already sums over all possible paths, and thus it automatically takes into account all possible time orderings of the operators. **(?)** time ordering.

5.1.1 | Classical Limit

In the classical limit, the path integral is dominated by the path that minimizes the action, which is the classical path. This is because in the classical limit, the action is much larger than $\hbar \to 0$, and thus the contributions from paths that deviate from the classical path are suppressed by a factor of $e^{iS/\hbar}$,[1] which oscillates rapidly and averages out to zero. Therefore, in the classical limit, we can approximate the path integral by evaluating it at the classical path, which gives us the classical equations of motion.

Thus the most important contribution to the path integral comes from the classical path, which is the path that minimizes the action. This is known as the principle of least action, and it is a fundamental principle in classical mechanics that states that the motion of a system is determined by the path that minimizes the action. From these procedure we can easily compute the Euler-Lagrange equations of motion, which are the equations that govern the dynamics of the system in classical mechanics. In the path integral formulation, we can derive the Euler-Lagrange equations by performing a variation of the action with respect to the path, and then setting the variation to zero.

[1] We omitted the $\hbar$ factor for simplicity, but via dimensional analysis it can be easily recovered.

5.2 | Path Integral Formulation of QFT

The path integral formulation can be generalized with the use of some handy substitutions:

$$\begin{cases} q \to \phi(x), \\ t,\, \mathrm{d}t \to x^\mu,\, \mathrm{d}^4x, \end{cases}$$

... green's finctions... $|i\rangle = |f\rangle = |\Omega\rangle$...

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\cdots\hat{\phi}(x_n)\}\,|\Omega\rangle = N\int \mathcal{D}\phi(x)e^{iS[\phi(x)]}\phi(x_1)\phi(x_2)\cdots\phi(x_n),$$

where now the action is $S[\phi(x)] = \int \mathrm{d}^4x\,\mathcal{L}(\phi(x), \partial_\mu\phi(x))$. The normalization N can be fixed by requiring that the vacuum-to-vacuum transition amplitude is equal to one $\langle\Omega|\Omega\rangle = 1$:

$$N = \frac{1}{\int \mathcal{D}\phi(x)e^{iS[\phi(x)]}}.$$

In order to get the discretized version of the last expression, we can discretize spacetime into a lattice of points, $x_i \to \phi_i \equiv \phi(x_i)$ and

TODO: Add drawing of lattice

[*draw of lattice*]

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\cdots\hat{\phi}(x_n)\}\,|\Omega\rangle = N\int \mathrm{d}\phi_1\cdots\mathrm{d}\phi_N \exp\left[i\sum_{k=1}^{N}\Delta^4x\mathcal{L}(\phi_k)\right]\phi_1\phi_2\cdots\phi_N,$$

where again... and in the limit for $n \to \infty$ ...

But how do we evaluate the path integral? In general, it is very difficult to evaluate the path integral exactly, but we have some options based on the context in which we are working:

- **Lattice QFT**: we can evaluate the path integral numerically using Monte Carlo methods, which is particularly useful for studying strongly interacting theories such as quantum chromodynamics (QCD) on the lattice (the idea is to discretize spacetime into a lattice and then evaluate the path integral numerically). In the end we could infere the properties at the continuum limit by taking the limit of the lattice spacing going to zero.
- **Perturbation Theory**: if the lagrangian of the theory is quadratic in the fields, then we can evaluate the path integral exactly using *gaussian integrals*. If the lagrangian is not quadratic, then we can treat the non-quadratic terms as perturbations and evaluate the path integral perturbatively using Feynman diagrams: we can Taylor expand the exponential of the action around the free theory, and then we can use Wick's theorem to evaluate the resulting correlation functions in terms of the free propagators. This is the standard approach for computing scattering amplitudes in quantum field theory, and it is based on the idea that we can treat interactions as small perturbations to the free theory.

5.2.1 | Gaussian Integrals

One dimensional gaussian integrals are given by

$$\int_{-\infty}^{\infty} \mathrm{d}z e^{-\frac{a}{2}z^2} = \sqrt{\frac{2\pi}{a}}, \quad a > 0.$$

Now the generalization to n dimensions is given by $\mathbf{z} = (z_1, z_2, \ldots, z_n)$

$$\int \mathrm{d}z_1 \cdots \mathrm{d}z_n \; \exp\left[-\frac{1}{2}\mathbf{z}^T \hat{A}\mathbf{z} + \mathbf{J}\cdot\mathbf{z}\right] = \sqrt{\frac{(2\pi)^{n/2}}{\sqrt{\det \hat{A}}}}\exp\left[\frac{1}{2}\mathbf{J}^T\hat{A}^{-1}\mathbf{J}\right].$$

As we said cubic or higher order terms in the lagrangian can be treated as perturbations, and thus we can evaluate the path integral perturbatively using Feynman diagrams: we can Taylor expand the exponential of the action around the free theory

$$\int_{-\infty}^{\infty} \mathrm{d}z \exp\left[-\frac{1}{2}\right]\ldots\ldots$$

and

$$\ldots$$

5.2.2 | Computing Green's Functions

Since we are working in a space of functions, we have to define the functional derivative as

$$\frac{\partial}{\partial J(x)}J(y) = \delta^{(4)}(x-y),$$

which implies

$$\frac{\partial}{\partial J(x)}\int \mathrm{d}^4y\, J(y)\phi(y) = \phi(x),$$

which is the continuum version of the discrete case $\frac{\partial}{\partial J_i}\sum_j J_j\phi_j = \phi_i$ (and $\frac{\partial}{\partial J_i}J_j = \delta_{ij}$). This functional derivative obeys the same properties as the ordinary derivative, such as linearity and the product rule. We can use this functional derivative to compute correlation functions by introducing a source term $J(x)$ in the action, and then taking functional derivatives with respect to $J(x)$ to obtain correlation functions of the fields.

Since we can almost treat it as a normal derivative we have

$$\frac{\partial}{\partial J(x)}\int \mathrm{d}^4y\,(\partial_\mu J(y))\phi(y) = -\partial_\mu\phi(x),$$

having integrated by parts and assuming that the fields vanish at infinity. This is a very useful property, since it allows us to compute correlation functions involving derivatives of the fields by simply taking functional derivatives with respect to the source term.

Example (Exercise). Consider a lagrangian $\mathcal{L} = \mathcal{L}(l_k, \partial_\mu l_k)$ show that the equations of motion can ...

We can also define the **generating functional** $Z[J]$ as

$$Z[J] \equiv \int \mathcal{D}\phi e^{iS[\phi(x)]+i\int \mathrm{d}^4x J(x)\phi(x)},$$

where $J(x)$ can be thought as a classical source that couples to the quantum field $\phi(x)$. The generating functional $Z[J]$ is a powerful tool for computing correlation functions, since we can obtain correlation functions by taking functional derivatives of $Z[J]$ with respect to $J(x)$ and then setting $J(x) = 0$:

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\cdots\hat{\phi}(x_n)\}\,|\Omega\rangle = \frac{1}{Z[0]}\left(-i\frac{\partial}{\partial J(x_1)}\right)\left(-i\frac{\partial}{\partial J(x_2)}\right)\cdots\left(-i\frac{\partial}{\partial J(x_n)}\right)Z[J]\bigg|_{J=0},$$

where the normalization factor $1/Z[0]$ is necessary to ensure that the correlation functions are properly normalized. The generating functional $Z[J]$ can be evaluated exactly for free theories, and it can be evaluated perturbatively for interacting theories using Feynman diagrams.

If we can compute the generating functional $Z[J]$ for a given theory, then we can compute all correlation functions of the theory by taking functional derivatives with respect to $J(x)$ and then setting $J(x) = 0$: we basically know everything about the theory, and thus we can compute all scattering amplitudes and other physical observables. This is why the generating functional $Z[J]$ is such a powerful tool in quantum field theory.

In order to compute exactely the generating functional $Z[J]$ for a free theory, we can use the fact that the action is quadratic in the fields, and thus we can evaluate the path integral exactly using gaussian integrals: rewriting the action as

$$i \int \mathrm{d}^4x \, (\mathcal{L} + \phi J) = -\frac{i}{2} \int \mathrm{d}^4x \phi \left(\partial^2 + m^2 - i\epsilon\right) \phi + i \int \mathrm{d}^4x \, J\phi,$$

where we have added a small imaginary part $i\epsilon$ to the mass term in order to ensure that the path integral converges. Then we can evaluate the path integral using gaussian integrals, and we find

$$Z[J] = Z[0] \exp \left[i \int \mathrm{d}^4x \left(-\frac{1}{2}\phi(\partial^2 + m^2 - i\epsilon)\phi + J\phi \right) \right],$$

where we can rename $(\partial^2 + m^2 - i\epsilon) = \hat{A}$ as the matrix used when defining the gaussian integral. We have to note a couple of things:

$$\begin{cases} \hat{A} = \partial^2 + m^2 - i\epsilon, \\ \hat{A} D_F(x-y) = \delta^{(4)}(x-y), \\ \sum_j A_{ij}(D_F)_{jk} = \delta_{ik} \to \int \mathrm{d}^4p \, e^{ip\cdot(x-y)} \Pi(p), \\ i(\partial^2 + m^2 - i\epsilon)\Pi(p) = 1 \implies \Pi(p) = \dfrac{i}{p^2 - m^2 + i\epsilon}. \end{cases}$$

Thus now we can rewrite the generating functional $Z[J]$ as

$$Z[J] = Z_0 \exp \left[-\frac{1}{2} \int \mathrm{d}^4x \int \mathrm{d}^4y J(x) D_F(x-y) J(y) \right],$$

which is the exact expression for the generating functional of a free theory. From this expression, we can compute all correlation functions of the free theory by taking functional derivatives with respect to $J(x)$ and then setting $J(x) = 0$.

Green's Functions of a Free Theory

Example (Two point green's function). The two point green's function is given by

$$\langle 0 | \hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\} |0\rangle = \frac{1}{Z[0]} \left(-i \frac{\partial}{\partial J(x_1)} \right) \left(-i \frac{\partial}{\partial J(x_2)} \right) Z[J] \Big|_{J=0}.$$

Then we can compute the functional derivatives and we find

$$\begin{aligned} \langle 0 | \hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\} |0\rangle &= \frac{1}{Z[0]} \left(-i \frac{\partial}{\partial J(x_1)} \right) \left[-\frac{1}{2} \int \mathrm{d}^4y D_F(x_2 - y) J(y) - \frac{1}{2} \int \mathrm{d}^4x J(x) D_F(x - x_2) \right] \frac{Z[J]}{Z_0} \Big|_{J=0} \\ &= \frac{1}{2} D_F(x_2 - x_1) + \frac{1}{2} D_F(x_1 - x_2) = D_F(x_2 - x_1), \end{aligned}$$

which is the Feynman propagator of the free theory.

Example (Four point green's function). If we rename for convenience

$$\begin{cases} D_F(x-y) = D_{xy}, \\ J(x) = J_x, \\ \int \mathrm{d}^4x \int \mathrm{d}^4y J(x) D_F(x-y) J(y) = \sum_{x,y} J_x D_{xy} J_y = J_x D_{xy} J_y, \\ \dfrac{\partial}{\partial J_z}(J_x D_{xy} J_y) = D_{zy} J_y + J_x D_{xz} = 2 J_x D_{xz}, \end{cases}$$

then we can compute the four point green's function as

$$\begin{aligned} \langle 0| \hat{T}\{\hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4\} |0\rangle &= (-i)^4 \frac{\partial^4}{\partial J_1 \partial J_2 \partial J_3 \partial J_4} \exp\left[-\frac{1}{2} J_x D_{xy} J_y\right]\bigg|_{J=0} \\ &= \frac{\partial}{\partial J_1} \cdots \frac{\partial}{\partial J_3} (-J_x D_{x4}) \exp[\cdots]\bigg|_{J=0} \\ &= \frac{\partial}{\partial J_1} \frac{\partial}{\partial J_2} (-D_{34} + J_x D_{x4} J_y D_{yz}) \exp[\cdots]\bigg|_{J=0} \\ &= \cdots = D_{14} D_{23} + D_{24} D_{13} + D_{12} D_{34} \\ &= \sum \text{ full contractions of } \phi_1\phi_2\phi_3\phi_4, \end{aligned}$$

which is the sum of all possible contractions of the four fields, and thus it is the sum of all possible Feynman diagrams with four external legs.

5.2.3 | Perturbation Theory and Feynman Diagrams

[...]

Since the procedure adopted untill now is valid only for free theories: we were treating the lagrangian without derivatives of the fields as a gaussian integral, and thus we were able to evaluate the path integral exactly. Let us check that also interacting lagrangians can be treated in the path integral formulation. . .

We know that the Feynman rules come from . . . For interacting field the recipe is

- For each field in the lagrangian, we can adopt the substitution $\phi(x) \to \int \mathrm{d}^4k \phi(k) e^{-ik\cdot x}$.
- For incoming particles (or outgoing anti-particles) at the vertex, we have to apply a derivative of the form
$$\frac{\delta}{\delta\phi(\pm p)} \mathcal{L}_{\int},$$
where the minus sign is for incoming particles and the plus sign is for outgoing particles.
- For outgoing particles (or incoming anti-particles) at the vertex, we have to apply a derivative of the form
$$\frac{\delta}{\delta\phi^*(\pm p)} \mathcal{L}_{\int}.$$
- For both the previous cases, we have to evaluate the resulting expression at $\phi = 0$ and then we have to multiply by i.

Example (Interacting Lagrangian in sQED). [...] Computing the fuctional derivatives appeared in the previous recipe, we have to use the following properties of the functional derivative:

$$\frac{\delta}{\delta\phi(p)}\phi(k) = \delta^{(4)}(p-k), \qquad \frac{\partial}{\partial A^\mu(p)} A^\nu(k) = \delta^\nu_\mu \delta^{(4)}(p-k),$$

so that in the end we have found the Feynman rule for the vertex, which is given by

$$\cdot = e\delta_\mu^{\ \nu}(-i(p_1)_\nu - i(p_2)_\nu)\delta^{(4)}(-p_1 + p_2 - p_3) = -ie(p_1 + p_2)_\mu.$$

We can also apply the same procedure to compute the Feynman rule for the other vertex, which is given by ...

5.2.4 | Fermionic Path Integrals

Inside the path integrals, every quantity commutes, since we are treating them as ordinary functions, and thus we cannot use the same procedure for fermionic fields: they anticommute. In order to treat fermionic fields in the path integral formulation, we have to introduce a new type of variables called **Grassmann variables** θ_i, which are anticommuting variables that satisfy the following properties:

- **Anticommutation**: $\theta_i\theta_j = -\theta_j\theta_i$ for all i, j. In particular, this implies that $\theta_i^2 = 0$ for all i.
- **Integration** and **differentiation**: the integral of a Grassmann variable is well defined and identical to differentiation:

  $$\int \mathrm{d}\theta_1 \cdots \mathrm{d}\theta_n X = \frac{\partial}{\partial\theta_n} \cdots \frac{\partial}{\partial\theta_1} X,$$

  This definition is consistent with the anticommutation property, since it implies that the integral of any product of Grassmann variables is zero unless it contains exactly one factor of each variable. Furthermore, obviously we have

  $$\frac{\partial\theta_i}{\partial\theta_j} = \delta_{ij},$$

  which is consistent with the anticommutation property, since it implies that the derivative of any product of Grassmann variables is zero unless it contains exactly one factor of each variable.
- **Gaussian integrals**: the gaussian integral over Grassmann variables is given by

  $$\int \mathrm{d}\theta_1 \mathrm{d}\overline{\theta}_1 \cdots \mathrm{d}\theta_n \mathrm{d}\overline{\theta}_n \exp\left(-\sum_{i,j} \overline{\theta}_i A_{ij} \theta_j + \sum_j \left(\overline{\eta}_j \theta_j + \overline{\theta}_j \eta_j\right)\right) = \det A \exp\left(\sum_{i,j} \overline{\eta}_i (A^{-1})_{ij} \eta_j\right),$$

  where A is an $n \times n$ antisymmetric matrix and η_j and $\overline{\eta}_j$ are Grassmann variables that serve as sources for the fermionic fields. This formula is the fermionic analogue of the gaussian integral for bosonic fields, and it is a crucial tool for evaluating path integrals involving fermionic fields. Note that differently from the bosonic case, the determinant appears in the numerator rather than in the denominator.

We have now the tools needed to evaluate path integrals involving fermionic fields, and thus we can compute correlation functions and scattering amplitudes for theories that include fermions, such as quantum electrodynamics (QED) and the Standard Model of particle physics.

Let us then derive the generating functional for a free fermionic theory, which is given by

$$Z[\eta, \overline{\eta}] = \int \mathcal{D}\psi \mathcal{D}\overline{\psi} \exp\left[i \int \mathrm{d}^4x \left(\overline{\psi}(i\partial\!\!\!/ - m)\psi + \overline{\eta}\psi + \overline{\psi}\eta\right)\right],$$

and after applying the substitution $m \to m - i\epsilon$ in order to ensure the formal convergence of the path integral, we can evaluate it using the formula for gaussian integrals over Grassmann variables, and we find

$$Z[\eta,\overline{\eta}] = Z[0,0]\exp\left[-\int \mathrm{d}^4x \int \mathrm{d}^4y\,\overline{\eta}(x) S_F(x-y)\eta(y)\right],$$

where the fermionic propagator $S_F(x-y)$ appeared: it is defined as the inverse of the operator $i\not{\partial} - m + i\epsilon$:

$$S_F(x-y) = \int \frac{\mathrm{d}^4k}{(2\pi)^4} e^{-i(x-y)\cdot k} \frac{i}{\not{k} - m + i\epsilon} = \int \frac{\mathrm{d}^4k}{(2\pi)^4} e^{-i(x-y)\cdot k} \frac{i(\not{k}+m)}{k^2 - m^2 + i\epsilon} = \langle 0|\,\hat{T}\{\hat{\psi}(x)\overline{\hat{\psi}}(y)\}\,|0\rangle\,.$$

As a final note, we have to highlight that also the grassman derivatives are anticommuting variables, since they are defined as the integral over grassman variables, and thus they inherit the anticommutation property of the grassman variables. This means that when we take functional derivatives with respect to the sources η and $\overline{\eta}$, we have to be careful about the order of the derivatives, since they do not commute.

5.2.5 | Derivatives of the Fields

We have already seen that

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\cdots\hat{\phi}(x_n)\}\,|\Omega\rangle = \int \mathrm{D}\phi(x) e^{iS[\phi(x)]}\phi(x_1)\phi(x_2)\cdots\phi(x_n),$$

and we now want to understand what happens when we substitute one of the fields with its derivative, for example

$$\int \mathrm{D}\phi e^{iS[\phi(x)]}(\partial_{t_1}\phi(x_1))\phi(x_2)\cdots\phi(x_n),$$

can be expanded using linearity of the path integral as

$$\dots$$

Thus derivatives inside the path integral can be treated as functional derivatives outside the Green's functions, since we can integrate by parts and then we can use the properties of the functional derivative to compute correlation functions involving derivatives of the fields.

....

5.3 | Quantization of QED

Quantizing gauge theories in path integral formulation is more complicated than quantizing scalar theories, since we have to deal with the gauge symmetry of the theory:

$$A_\mu(x) \to A_\mu(x) + \partial_\mu \alpha(x),$$

which implies that the path integral is not well defined, since we are integrating over an infinite number of gauge equivalent configurations. In order to fix this problem, we have to introduce a gauge fixing term in the lagrangian, which breaks the gauge symmetry and allows us to define a well defined path integral.

The action in a QED theory is given by

$$S = \dots$$

In order to obtain the propagator, we have to invert the operator that appears in the quadratic part of the action, which is given by

$$\hat{O} = -i(\partial^2 g_{\mu\nu} - \partial_\mu \partial_\nu),$$

which then can be used to obtain

$$\hat{O} D_F^{\nu\rho}(x-y) = -i(\partial^2 g_{\mu\nu} - \partial_\mu \partial_\nu) D_F^{\nu\rho}(x-y) = \delta_\mu{}^\rho \delta^{(4)}(x-y).$$

However this equation has no solution, since the operator $\hat{O}$ is not invertible due to the gauge symmetry of the theory: it has a non-trivial null space

$$\partial_\mu \alpha(x) \in \mathrm{Null}(\hat{O}).$$

Equivalently, if the field is of the form

$$\hat{A}^\mu(x) = \partial^\mu \alpha(x),$$

then the action is null $S = 0$ and the path integral diverges, since we are integrating over an infinite number of gauge equivalent configurations. This is a manifestation of the fact that the gauge symmetry is a redundancy in the description of the theory, and thus we have to fix this redundancy in order to define a well defined path integral.

Also in the momentum space the same problem arises, since the operator $\hat{O}$ has a non-trivial null space:

$$D_F^{\mu\nu}(x-y) = \dots$$

where it is easy to notice how p^ν is annihilated by $p^2 g_{\mu\nu} + p_\mu p_\nu$, leading to a non-invertible operator and thus we cannot solve for $\Pi^{\mu\nu}$.

Example (Massive spin 1 case). ...

We have to find a way to define a domain of integration over inequivalent configurations, and thus we have to introduce a gauge fixing term in the lagrangian, which breaks the gauge symmetry and allows us to define a well defined path integral.

[...]

We will use the method by Faddeev and Popov, which consists in inserting a factor of one in the path integral in the form of an integral over the gauge group, and then we can use this factor to fix the gauge and define a well defined path integral (this can be generalized also to non-abelian gauge theories, but we will focus on the abelian case for simplicity).

We have to choose a gauge fixing condition, which is a function $G[A]$ that depends on the gauge field A_μ and that is equal to zero for a specific choice of gauge:

$$G[A_\alpha] = F[A + \frac{1}{e}\partial\alpha] = 0, \quad A_\alpha(x) = A(x) + \frac{1}{e}\partial\alpha(x),$$

for some function $G[A]$ (for example $G[A] = \partial^\mu A_\mu$ for the Lorenz gauge). Then we can use a continuum version of

$$1 = \int \mathrm{d}z_1 \cdots \mathrm{d}z_n \delta(g_1(\mathbf{z})) \cdots \delta(g_n(\mathbf{z})) \det\left(\frac{\partial g_i}{\partial z_j}\right),$$

namely

$$1 = \int \mathcal{D}\alpha \delta(G[A_\alpha]) \det\left(\frac{\delta G[A_\alpha]}{\delta\alpha}\right),$$

where $\delta(G[A_\alpha])$ is the continuum limit of a product of delta functions that impose the gauge fixing condition at each point in space-time, and $\det\left(\frac{\delta G[A_\alpha]}{\delta\alpha}\right)$ is the Faddeev-Popov determinant, which arises from the change of variables in the path integral when we insert the factor of one.

If we now assume that $\frac{\delta G[A_\alpha]}{\delta\alpha}$ is independent of A_μ (true for the Lorenz gauge), then we can pull it out of the path integral and we find

$$\ldots$$

Inserting these results in the path integral, we find

$$\int \mathrm{D}A \exp[iS[A]] = \int \mathrm{D}\alpha \det\left(\frac{\delta G[A_\alpha]}{\delta\alpha}\right) \int \mathrm{D}A \exp[iS[A]]\delta(G[A_\alpha]) = \ldots$$

where we have shifted the integration variable $A_\mu \to A_\mu - \frac{1}{e}\partial_\mu\alpha$ in the last step, and we have used the fact that the action is gauge invariant $S[A] = S[A_\alpha]$. Now we can finally fix the gauge by imposing

$$G[A] = \partial_\mu A^\mu - \omega(x) = 0,$$

for a generic function $\omega(x)$. We still get

$$\frac{\partial}{\partial\alpha} G[A_\alpha] = \det\left(\frac{1}{e}\partial^2\right),$$

so that the path integral becomes

$$\int \mathrm{D}A \exp[iS[A]] = \det\left(\frac{1}{e}\partial^2\right)\left(\int \mathrm{D}\alpha\right) \int \mathrm{D}A \exp[iS[A]]\delta(\partial_\mu A^\mu - \omega(x)),$$

where the first term $\det\left(\frac{1}{e}\partial^2\right)\left(\int \mathrm{D}\alpha\right)$ is a constant and the expression is true for any choice of $\omega(x)$. We can then average over $\omega(x)$ with a gaussian weight: we avrage over a gaussian distribution centered in $\omega = 0$:

$$\begin{cases} f(A) = g(A,\omega) = \sum_\omega g(A,\omega)h(\omega), \\ \sum_\omega h(\omega) = 1, \end{cases} \qquad \begin{cases} f(A) = \int \mathrm{d}\omega\, g(A,\omega)h(\omega), \\ \int \mathrm{d}\omega\, h(\omega) = 1, \end{cases}$$

so that we can write

$$1 = N(\xi) \ldots$$

which after an integration over $\int \mathrm{D}\omega$ will be of the form

$$\int \mathrm{D}A e^{iS[A]} = N_0 \int \mathrm{D}A e^{iS[A]+i\int \mathrm{d}^4x \frac{1}{2\xi}(\partial_\mu A^\mu)^2},$$

[...]

gauge invariant operator...

momentum space...

The propagator depends on the gauge fixing parameter ξ...

In practice one oftenly choose a vealue of ξ:

- $\xi = 1$ is the **Feynman gauge**, which is the most commonly used gauge in perturbative calculations and is the one adopted until now.
- $\xi = 0$ is the **Lorentz gauge**, also called the **Landau gauge**, which is a popular choice for non-perturbative calculations and for lattice QCD.

5.4 | Symmetries in Quantum Field Theory

6 Renormalization and Regularization

[...]

6.1 | Divergences in Quantum Field Theory

To introduce the topic we go back to one of the easiest examples of a quantum field theory, the ϕ^4 theory. The Lagrangian density is given by

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4.$$

We consider one of the simplest processes, the $2 \to 2$ scattering process

$$\phi(p_1)\phi(p_2) \to \phi(p_3)\phi(p_4),$$

which at leading order is given by the tree-level Feynman diagram

$$i\mathcal{M}_{LO} = [\textit{Draw LO}] = -i\lambda.$$

The differential cross section is given by

$$\frac{\mathrm{d}\sigma_{LO}}{\mathrm{d}\Omega}\Big|_{\mathrm{CM}} = \frac{1}{64\pi^2 E_{cm}}\frac{|\mathbf{p}_{out}|}{|\mathbf{p}_{in}|}|\mathcal{M}_{LO}| = \frac{1}{64\pi^2}\frac{\lambda^2}{s}.$$

We now have to compute the amplitude $i\mathcal{M}$ at next-to-leading order (NLO). The Feynman diagrams contributing to the NLO amplitude are given by

$$i\mathcal{M} = i\mathcal{M}_{LO} + i\mathcal{M}_{NLO} + \mathcal{O}(\lambda^3),$$

$$i\mathcal{M}_{NLO} = [\textit{Draw NLO}] = i(\mathcal{M}_1 + \mathcal{M}_2 + \mathcal{M}_3).$$

Let us study the diagrams one by one. Having defined $p = p_1 + p_2$, the first diagram is given by

$$i\mathcal{M}_1 = \frac{(-i\lambda)^2}{2}\int \frac{\mathrm{d}^4 k}{(2\pi)^4}\frac{i}{k^2 - m^2 + i\epsilon}\frac{i}{(k+p)^2 - m^2 + i\epsilon} = (-i\lambda)^2 iV(p^2),$$

since the result hs to be lorentz invariant and can only depend on the Lorentz invariant p^2 (the $\frac{1}{2}$ global factor was due to the symmetry of the diagram).

The second and third diagrams are more or less the same: they can be obtained from the first diagram by replacing p with $p_1 - p_3$ and $p_1 - p_4$ respectively:

$$\mathcal{M}_2 = \mathcal{M}_1\big|_{p \to p_1 - p_3}, \qquad \mathcal{M}_3 = \mathcal{M}_1\big|_{p \to p_1 - p_4}.$$

We can thus write

$$i\mathcal{M}_{NLO} = i(-i\lambda)^2[V(s) + V(t) + V(u)],$$

thus our work is restricted to the computation of the loop integral $V(p^2)$, which is a bit unfortunate since it is divergent. Indeed, if we take the limit $k^\mu \to \infty$ we have

$$k^\mu \to \infty \quad \Rightarrow \quad V(p^2) \sim \int \frac{\mathrm{d}^4 k}{(2\pi)^4}\frac{1}{k^4} \sim \log\Lambda \to \infty,$$

which is the definition of a logarithmic divergence: we are in the presence of an **ultraviolet (UV) divergence**, since the divergence is due to the contribution of high momentum modes in the loop integral.

[...]

Observations

- In quantum mechanics, when computing an amplitude for an event, we need to coonsider all the different intermediate states that can contribute to the process. As $\Delta x \Delta p \sim O(\hbar) \sim \Delta E \Delta t$, ...

- Contrary to the previos observation, we cannot neither expect the thery to be valid at arbitrarily high energies, nor to be able to measure the amplitude with infinite precision (not even QED or the Standard Model). Some new effects...

- Experiments at low energy scales are not sensitive to the details of the UV completion of the theory, but only to a few parameters...

- We are trying to compute $i\mathcal{M}$ and $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}$ as functions of the free parameters of the lagrangian, namely λ and m. [...]

  But how do we ...

[...]

Summary: ...

[...]

Solution. The solution can be achieved as a two step process:

1. **Regularization**: we introduce a regulator in the theory, which is a parameter that makes the loop integrals finite. For example, we can introduce a cutoff Λ in the loop integral, such that we only integrate over momenta k such that $|k| < \Lambda$

$$\int_{-\infty}^{\infty} \mathrm{d}^4 k \to \int_{-\Lambda}^{\Lambda} \mathrm{d}^4 k,$$

   and then we can take the limit $\Lambda \to \infty$ at the end. This way, we have to deal with a finite loop integral depending on a **regulator**, but we have to be careful when taking the limit $\Lambda \to \infty$ at the end, since we can get infinite results.

2. **Renormalization**: we redefine the parameters of the theory (the mass and the coupling constant) in such a way that the physical observables (the amplitude and the cross section) are finite in the limit $\Lambda \to \infty$: rewriting observables as functions of other observables. The result will become finite[1] in the limit $\Lambda \to \infty$ where the regulator is removed.

UV divergence as manifestation of our ignorance of the UV completion of the theory... For example for macroscopic objects...

We should be able to study physics at low energy scales (and high distances) without knowing the details of the UV completion of the theory, but only a few parameters that encode the effects of the UV completion at low energy scales.

Similarly, we should be able to relate...

[1] Under certain condition that we will have to further stress.

6.1.1 | Classical Example: Infinite Wire Potetial

Consider an infinite wire along the z axis with a uniform charge density λ. We want to compute the electric potential $V(r)$ at a distance r from the wire.

Without computing anything, we can note from some dimensional analysis that the potential already has some problems:

$$V(r) = \lambda f(r), \quad [V] = 1, \quad [\lambda] = [L]^{-1} = 1,$$

thus it seems that $f(r)$ should be dimensionless and the potential will result in being constant, which is not what we expect.

By trying to compute the potential we find that the problem is given by a divergence in the integral:

$$V(r) = \int_{-\infty}^{\infty} \mathrm{d}x \frac{\lambda}{4\pi r(x)} = \frac{\lambda}{4\pi} \int_{-\infty}^{\infty} \frac{\mathrm{d}x}{\sqrt{r^2 + x^2}} = \infty,$$

where we should however notice that

$$E_i = -\partial_i V(r) = \frac{\lambda}{4\pi} \int_{-\infty}^{\infty} \mathrm{d}x \frac{r_i}{(r^2 + x^2)^{3/2}} = \frac{\lambda}{2\pi r} \hat{r}_i,$$

is finite, which seems to be a contradiction. From these some observations follows naturally:

- Infinite wires do not exist in nature, [...]
- If $r \ll$ lenght of the wire, de details of what happens at the end of the real finite wire are not relevant, and we can still use the infinite wire approximation to compute the electric field, which is what we are interested in.
- The potential $V = V(r)$ is not a physical observable, since it is not measurable: only differences of potential are measurable, and these are finite: $\Delta V = V(r_2) - V(r_1)$.

Thus we can apply the solution in two steps as we explained before:

- **Regularization**: we introduce a regulator in the theory, which is a parameter that makes the integral finite. For example, we can introduce a cutoff Λ in the integral, such that we only integrate over x such that $|x| < \Lambda$

$$\int_{-\infty}^{\infty} \mathrm{d}x \to \int_{-\Lambda}^{\Lambda} \mathrm{d}x,$$

  and then we can take the limit $\Lambda \to \infty$ at the end:

$$V(r) = \frac{\lambda}{4\pi} \int_{-\Lambda}^{\Lambda} \frac{\mathrm{d}x}{\sqrt{r^2 + x^2}} = \frac{\lambda}{4\pi} \log\left(\frac{\sqrt{r^2 + \Lambda^2} + \Lambda}{\sqrt{r^2 + \Lambda^2} - \Lambda} \right).$$

  The divergence is still there in the limit $\Lambda \to \infty$, but at least we have a finite result depending on the regulator Λ.

- **Renormalization**: we redefine the parameters of the theory (the potential) in such a way that the physical observables (the electric field) are finite in the limit $\Lambda \to \infty$. We can choose an arbitrary distance r_0 (not too far from the wire) in order to renormalize the potential:

$$V_R(r) = V(r) - V(r_0),$$

which is an observable and indeed it is finite in the limit $\Lambda \to \infty$:

$$V_R(r) = -\frac{\lambda}{4\pi} \log\left(\frac{r^2}{r_0^2}\right) + \mathcal{O}(\frac{1}{\Lambda^2}),$$

as $\Lambda \to \infty$. The divergence has thus disappeared in the renormalized potential V_R, which is the physical observable, and we have a finite result for the electric field: ...

6.2 | Regularization Schemes

A regularization scheme is a way to make the loop integrals finite, and there are many different regularization schemes that can be used in quantum field theory. The regularization we just showed is called **cutoff regularization**

$$\int_{-\infty}^{\infty} \mathrm{d}^4 k \to \int_{-\Lambda}^{\Lambda} \mathrm{d}^4 k,$$

which can be used in any theory, but it has the disadvantage of breaking Lorentz invariance and gauge invariance (and other symmetries), making intermediate steps in calculations more complicated (for example we cannot make use of the Ward identity anymore). It is not very convenient and outside textbook examples it is not used in practice.

[...]

Let us present some of the most common regularization schemes used in quantum field theory:

- **Modified interaction** (Pauli-Villars regularization): we can modify the form of the interaction in the high energy regime, since it should not affect the low energy physics. For example, we could modify the Coulomb potential at short distances by introducing:

  $$\frac{\lambda \mathrm{d}x}{4\pi r} \to \frac{\lambda \mathrm{d}x}{4\pi}\left(\frac{1}{r} - \frac{1}{\sqrt{r^2+\Lambda^2}}\right),$$

  which will reduce to the original potential in the limit $\Lambda \to \infty$, but it will be finite for any finite value of Λ. The integral of the potential will now be finite

  $$\dots$$

  [...] Loop integrals can be regularized in a similar way by modifying the propagator at high energies as

  $$\frac{i}{k^2 - m^2 + i\epsilon} \to \frac{i}{k^2 - m^2 + i\epsilon} - \frac{i}{k^2 - \Lambda^2 + i\epsilon},$$

  which makes it finite for any finite value of Λ and reduces to the original propagator in the limit $\Lambda \to \infty$. This is the **Pauli-Villars** regularization scheme.

- **Dimensional regularizarion**. The previous regularization scheme have a clear physical interpretation: ... The method that is almost exclusively used today is called *dimensional regularization* (by Giambiagi, Bollini ...), which do not have a clear physical interpretation, but it is very convenient for calculations and it preserves all the symmetries of the theory.

  Dimensional regularization is based on the idea of performing the loop integrals in d dimensions instead of 4 dimensions, where d is a smaller number than 4, such that the loop integrals are finite. For example, the integral we are using as an example in the ϕ^4 theorywill be convenient to compute in $d = 2, 3$ dimensions. It turns out that we can compute it as an analytic function of the number of spacetime dimensions d:

  $$\int_{-\infty}^{\infty} \mathrm{d}^4 k \frac{1}{k^4} \to \int_{-\infty}^{\infty} \mathrm{d}^d k \frac{1}{k^4} = \dots$$

  this function will have a pole for $d = 4$, which will be removed in smaller dimensions, and we can take the limit $d \to 4$ at the end of the calculation, which will be equivalent to taking the limit $\Lambda \to \infty$ in the cutoff regularization scheme.

  [...]

We can also apply it to our toy-example of the infinite wire potential, by performing the integral in d dimensions instead of 4 dimensions:

$$V(r) = \dots \begin{cases} x \to \mathbf{x} = \begin{pmatrix} x_1 & x_2 & \cdots & x_d \end{pmatrix} \\ \mathrm{d}x \to \mathrm{d}^d x = \mathrm{d}x_1 \mathrm{d}x_2 \cdots \mathrm{d}x_d \\ x^2 \to |\mathbf{x}^2| = \sum_{j=1}^{d} x_j^2 \end{cases}$$

such that we can rewrite the potential as

$$V(r) = \dots$$

where μ is a scale with mass dimension $1 = [\mu]$. [...] [...] The integral is now convergent for $0 < d < 1$, where it is analytic in d. We can thus analytically continue the result in the whole complex plane with the exception for some points in which it will have poles. We can show that ...

$$\dots$$

and we get the usual result for V_R as $d \to 1$.

- **Renormalization without regulators**. In principle, it is not necessary to introduce a regulator (Λ or d) in order to renormalize the theory: we can indeed introduce directly at the integrand level some physical, finite and renormalized quantities. Considering the toy example of the infinite wire potential, we could have introduced directly the renormalized potential at the integrand level as

$$V_R(r) = V(r) - V(r_0) = \frac{\lambda}{4\pi} \int_{-\infty}^{\infty} \mathrm{d}x \left(\frac{1}{\sqrt{r^2 + x^2}} - \frac{1}{\sqrt{r_0^2 + x^2}} \right) = -\frac{\lambda}{4\pi} \log \left(\frac{r^2}{r_0^2} \right),$$

which is finite and does not require any regulator. [...]

6.3 | One Loop Integrals in Dimensional Regularization

In general, the loop integrals we have to compute in quantum field theory are of the form

$$\int \frac{\mathrm{d}^D k}{(2\pi)^D} \frac{1}{D_1^{\alpha_1} D_2^{\alpha_2} \cdots D_n^{\alpha_n}},$$

where we have defined the "denominators of the propagators" as

$$\begin{cases} D_i = l_i^2 - m_i^2 + i\epsilon, \\ l_i^\mu = k^\mu + \sum_{j=1} \beta_{ij} p_j^\mu, \end{cases}$$

where p_j are the external momenta and β_{ij} are some coefficients that depend on the topology of the diagram. We can also promote the metric tensor to D dimensions (1 timelike and $D-1$ spacelike dimensions) as

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -1 \end{pmatrix}, \quad \text{such that} \quad g_\mu{}^\mu = \delta^\mu{}_\mu = D.$$

Since it is not convenient to treat an integral with many denominators, we can use the **Feynman parameters** to rewrite it as an integral with a single denominator:

$$\frac{1}{D_1 \cdots D_n} = \int_0^1 \mathrm{d}x_1 \cdots \int_0^1 \mathrm{d}x_n \, \delta\left(1 - \sum_{j=1}^n x_j\right) \frac{(n-1)!}{(x_1 D_1 + \cdots + x_n D_n)^n},$$

where the delta function is needed to ensure that the sum of the Feynman parameters x_i is equal to 1. By taking derivatives with respect to the propagator denominators D_i we can also get the following more general formula:

$$\left(\frac{\partial}{\partial D_i}\right) \implies \frac{1}{D_i^{\alpha_1} \cdots D_n^{\alpha_n}} = \int_0^1 \mathrm{d}x_1 \cdots \int_0^1 \mathrm{d}x_n \, \delta\left(1 - \sum_{j=1}^n x_j\right) \frac{\Gamma(\alpha_1 + \cdots + \alpha_n)}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_n)} \frac{x_1^{\alpha_1 - 1} \cdots x_n^{\alpha_n - 1}}{(x_1 D_1 + \cdots + x_n D_n)^{\alpha_1 + \cdots + \alpha_n}},$$

with the condition of $\alpha \equiv \sum_{i=1}^n \alpha_i$.

In order to analitically continue the integral to non integer values of the exponents α_i we have to introduce the **Gamma function**, which is defined as

$$\Gamma(z) = \int_0^\infty \mathrm{d}t \, t^{z-1} e^{-t},$$

which is a generalization of the factorial function to complex numbers, since for positive integers n we have $\Gamma(n) = (n-1)!$. The Gamma function has some important properties that we will use in the following:

- It has poles at $z = 0, -1, -2, \ldots$ with residue $\mathrm{Res}(\Gamma(z), z = -n) = \frac{(-1)^n}{n!}$.
- It satisfies the following functional equation: $\Gamma(z+1) = z\Gamma(z)$, which is obtained by performing an integration by parts in the definition of the Gamma function.
- By direct integration we can show that $\Gamma(1) = 1$.

- The last two properties together imply that $\Gamma(n) = (n-1)!$ for positive integers $n \geq 0$.

It converges for $\mathrm{Re}(z) > 0$, but can also be analitically continued to the whole complex plane with the exception of the poles at $z = 0, -1, -2, \ldots$. Thus we are now ready to write the more general formula for the loop integrals in dimensional regularization (and in Feynman parameterization):

$$\frac{1}{D_1^{\alpha_1} \cdots D_n^{\alpha_n}} = \ldots$$

Proof. Let us start from noticing that the following integral after a change of variable $x \to x' = Dx$ can be rewritten in terms of gamma functions as

$$\int_0^\infty \mathrm{d}x\, x^{\alpha-1} e^{-Dx} = D^{-\alpha} \int_0^\infty \mathrm{d}x'\, x'^{\alpha-1} e^{-x'} = \frac{\Gamma(\alpha)}{D^\alpha}.$$

If we apply this formula (referred to as the *Schwinger's trick*) to the denominator of the propagators n times, we obtain:

$$\frac{1}{D_1^{\alpha_1} \cdots D_n^{\alpha_n}} = \ldots$$

But we want to get to a Feynman parametrized form, thus we have to introduce the delta function to ensure that the sum of the Feynman parameters x_i is equal to 1. We can do this by using the following identity:

$$1 = \int_0^\infty \mathrm{d}g\, \delta\left(g - \sum_{j=1}^n x_j\right),$$

and then rescaling the Feynman parameters as $x_i \to g x_i$, in order to get

$$\frac{1}{D_1^{\alpha_1} \cdots D_n^{\alpha_n}} = \ldots$$

But now it is not difficult to recognize that the integral over g is again of the form of the Schwinger's trick, and we can perform it to get the final result:

$$\ldots$$

which leads us to the desired formula. □

Example (D-dimensional solid angle). We will show that the solid angle in D dimensions is given by

$$\Omega_D = \int \mathrm{d}\Omega_D = \frac{2\pi^{D/2}}{\Gamma(D/2)},$$

where $\mathrm{d}\Omega_D$ is the measure of the solid angle in D dimensions. We can start from the Gaussian integral in D dimensions, and we will compute it both in cartesian coordinates and in spherical coordinates:

[...]

6.3.1 | Gamma Functions and Wick Rotation

Studying the properties of the Gamma function we can see that it has poles at $z = 0, -1, -2, \ldots$. This means that if we have a loop integral that is divergent in 4 dimensions, it will have a pole at $D = 4$ in dimensional regularization. In particular, around $z = 0$, we can expand as

$$\Gamma(\epsilon) = \frac{1}{\epsilon} - \gamma_E + \mathcal{O}(\epsilon),$$

as $\epsilon \to 0$. Here, γ_E is the Euler-Mascheroni constant, which is defined as

$$\gamma_E = \lim_{n\to\infty} \left(\sum_{k=1}^{n} \frac{1}{k} - \log n \right) \approx 0.5772,$$

but its numerical value is of no importance for us, since it always cancels out when deriving physical observables. The important thing is that knowing how the gamma function behaves around zero lets us understand how it behaves around all of its poles, since we can always exploit one of its properties to write:

$$\Gamma(z+1) = z\Gamma(z) \quad \Rightarrow \quad \Gamma(-1+\epsilon) = \frac{\Gamma(\epsilon)}{-1+\epsilon},$$

[...]

With feynman parameterization, we can always ...

[...]

Thus let us consider a **tadpole integral** of the following form in dimensional regularization:

$$I = \int \frac{\mathrm{d}^D k}{(2\pi)^D} \frac{1}{(k^2 - m^2 + i\epsilon)^\alpha}, \quad \alpha = 1, 2, 3, \dots$$

[*Draw of complex plane and wick rotation*]

Considering firstly the integral over the time component k^0 of the loop momentum, we can perform a **Wick rotation**: we can rotate the contour of integration in the complex plane of k^0 from the real axis to the imaginary axis counter-clockwise without crossing any poles, by defining

$$k^0 = \pm(\sqrt{\mathbf{k}^2 + m^2} - i\epsilon).$$

With this choice of the contour, the integral does not change, since we are not crossing any poles (we can imagine to close the contour ...); this Wick rotation is equivalent to performing a change of variable

$$\begin{cases} k^0 = ik_E^0, \\ \mathbf{k} = \mathbf{k}_E, \end{cases}$$

and it corresponds to performing the loop integral in Euclidean space (rotation to euclidean coordinates) instead of Minkowski space: indeed we have

$$\begin{aligned} k^2 &\equiv (k^0)^2 - |\mathbf{k}|^2, && \text{Minkowski space} \\ k_E^2 &\equiv (k_E^0)^2 + |\mathbf{k}_E|^2 = -k_E^2, && \text{Euclidean space} \end{aligned}$$

impliying that $k^2 = -k_E^2$. The tadpole integral then

$$I = \dots$$

where we could restore the feynman prescription easily by performing the change of variable $m^2 \to m^2 - i\epsilon$ in the final result. We could now use the Schwinger's trick

$$\frac{1}{D^\alpha} = \dots$$

to rewrite the denominator of the propagator as an integral over a Feynman parameter x:

$$I = \dots$$

Now the integral in the euclidean space is just a gaussian integral, which we can compute by using the formula for the solid angle in D dimensions:

$$\int \mathrm{d}^D k_E \exp\left(-k_E^2 x\right) = \left(\frac{\pi}{x}\right)^{D/2},$$

such that we get

$$\dots$$

using again schwinger's trick with $D \to m^2$ and $\alpha \to \alpha - D/2$ we obtain

$$\dots$$

which putting everything together gives us the final result for the tadpole integral in dimensional regularization:

$$I = \dots$$

[...]

So to compute a one loop integral in dimensional regularization we can follow the following steps:

- $\dots$
- ...
- $\dots$

We can now apply this procedure to compute the one loop correction to the amplitude in the ϕ^4 theory.

6.3.2 | 2-2 Scattering in ϕ^4-Theory at NLO

We first consider the ϕ^4 theory in D dimensions; doing dimensional analysis for our theory:

$$\dots$$

we can write the interaction lagrangian density as

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{4!}\mu^{4-D}\phi^4,$$

where μ is a mass scale, with

$$[\mu] = 1 \implies [\lambda] = 0,$$

where the constraint on the mass dimension of λ is given by the requirement that the interaction lagrangian density has mass dimension D.

We can now compute the amplitude for the $2 \to 2$ scattering process at NLO in dimensional regularization. Firstly, the new Feynman rule for the vertex is given by

$$\text{Draw of vertex} = -i\lambda\mu^{4-D},$$

and this implies

$$\begin{aligned} i\mathcal{M}_{LO} &= vertex = -i\lambda\mu^{4-D}, \\ i\mathcal{M}_{NLO} &= [Draw\ NLO] = i(-i\lambda)^2\mu^{4-D}[V(s) + V(t) + V(u)], \end{aligned}$$

where the second factor of μ is absorbed into the vertex factor:

$$iV(p^2) = \frac{1}{2}\mu^{4-D}\int \frac{\mathrm{d}^D k}{(2\pi)^D}\frac{i}{k^2-m^2}\frac{i}{(k+p)^2-m^2}.$$

Let us underline two observations:

- The integral $V(p^2)$ is still dimensionless with this choice of the vertex factor, as it should be since the amplitude has to be dimensionless.
- The feynman prescription is understood: we have to perform the change $m^2 \to m^2 - i\epsilon$ in the final result if we need to make it explicit.

Now using the Feynman parameterization for $N = 2$ we can rewrite the integral as

$$\frac{1}{D_1 D_2} = \int_0^1 \mathrm{d}x \frac{1}{((1-x)D_1 + xD_2)^2},$$

and

$$iV(p^2) = -\frac{\mu^{4-D}}{2}\int \dots$$

which can be shifted by $k^\mu \to k^\mu - xp^\mu$ in order to obtain

$$iV(p^2) = \dots$$

If we now define $M^2(p^2) = m^2 - (x(1-x)p^2)$, we can recgnize the form of the tadpole integral we computed in the previous section, and we can use the result we obtained for it to get

$$iV(p^2) = \dots$$

where we can notice we have not eliminated the divergence yet, since we have a pole at $D = 4$ in the gamma function; but we have obtained a finite result for the loop integral, which is what we wanted to achieve with regularization. As promised, both $iV(p^2)$ and thus $i\mathcal{M}$ functions are analytic in D with the exception of some poles. At some points we will be interested in expanding the result around $D = 4$; let us define ϵ as

$$D = 4 - 2\epsilon,$$

such that the limit $\epsilon \to 0$ corresponds to the limit $D \to 4$. We can then use the following properties

$$\begin{cases} \Gamma(2-\frac{D}{2}) = \Gamma(\epsilon) = \frac{1}{\epsilon} - \gamma_E + \mathcal{O}(\epsilon), \\ (4\pi)^{-D/2} = \frac{1}{16\pi^2}(1+\epsilon\log(4\pi)) + \mathcal{O}(\epsilon^2), \\ (M^2)^{D/2-2} = (M^2)^{-\epsilon} = 1 - \epsilon\log M^2 + \mathcal{O}(\epsilon^2), \\ \mu^{4-D} = \mu^{2\epsilon} = 1 + \epsilon\log\mu^2 + \mathcal{O}(\epsilon^2), \end{cases}$$

in order to get the following expansion for $iV(p^2)$ around $D = 4$:

$$iV(p^2) = -\frac{i}{32\pi^2}\left(\frac{1}{\epsilon} - \gamma_E + \log(4\pi) - \int_0^1 \mathrm{d}x \log\left(\frac{M^2(p^2)}{\mu^2}\right)\right) + \mathcal{O}(\epsilon).$$

The integral in $\mathrm{d}x$ appearing in the previous expression can be computed analytically, but it is not important for us to get its explicit form, since we are interested in the **regulated amplitude**, which is given by

$$\mu^{D-4}(i\mathcal{M}) = \dots$$

where this step concludes the regularization procedure, since we have obtained a finite result for the amplitude depending on the regulator ϵ, which is what we wanted to achieve with regularization. We can now move to the renormalization procedure, in order to get a finite result for the amplitude in the limit $\epsilon \to 0$.

Renormalization of the Amplitude

Thus we now start from a regulated amplitude, which is a finite function of the regulator ϵ, and we want to get a finite result for the amplitude in the limit $\epsilon \to 0$ by redefining the parameters of the theory (the mass and the coupling constant) in such a way that the physical observables (the amplitude and the cross section) are finite in the limit $\epsilon \to 0$. We can rewrite observables as functions of other observables, and the result will become finite in the limit $\epsilon \to 0$ where the regulator is removed.

Thus we start by replacing the bare parameters m and λ of our lagrangian with physical, observable quantities. Since LO is independent of the mass, which furthermore can be written as

$$m = m_{\text{phys}} + \mathcal{O}(\lambda),$$

the renormalization of m comes into play only at $\mathcal{O}(\lambda^3)$ for the amplitude (but we would need...)

Therefore we have to focus on the renormalization of the coupling constant λ.

Remark. *We should also multiply by ...*

We thus want to understand how to define a λ_{phys} to renormalize the coupling constant λ.

[...]

$$\sqrt{s} = E_{cm} = 2m \implies s = 4m^2,$$

[...]

We define

$$\mu^{D-4}(i\mathcal{M})\Big|_{s=4m^2}^{t=u=0} = -i\lambda_{\text{phys}},$$

as a renormalization condition, which allows us to define the physical coupling constant λ_{phys} as the value of the amplitude at a specific kinematic point (the renormalization point)

$$\lambda_{\text{phys}} = \lambda - \frac{3\lambda^2}{32\pi^2}(\dots) + \mathcal{O}(\lambda^3).$$

This is the **renormalization condition** that we have to impose in order to define the physical coupling constant λ_{phys} in terms of the bare coupling constant λ and the regulator ϵ. By inverting this relation perturbatively, we can express the bare coupling constant λ as a function of the physical coupling constant λ_{phys} and the regulator ϵ:

1. We know $\lambda_{\text{phys}} = \lambda + \alpha\lambda^2 + \mathcal{O}(\lambda^3)$.
2. We want to get an ansatz of the form $\lambda = \lambda_{\text{phys}} + \alpha'\lambda_{\text{phys}}^2 + \mathcal{O}(\lambda_{\text{phys}}^3)$.

By plugging the ansatz into the first relation we get

$$\lambda = \lambda + (\alpha + \alpha')\lambda^2 + \mathcal{O}(\lambda^3) \implies \alpha' = -\alpha,$$

such that we can use this relation to rewrite the amplitude as a function of the physical coupling constant λ_{phys} and the regulator ϵ:

$$\mu^{-2\epsilon}(i\mathcal{M}) = -i\lambda_{\text{phys}} - i\frac{3\lambda_{\text{phys}}^2}{32\pi^2}(\dots) + \mathcal{O}(\lambda_{\text{phys}}^3),$$

which is now a finite result at $\epsilon \to 0$ (equivalently at $D \to 4$), since the divergence has been absorbed into the redefinition of the coupling constant. We have thus obtained a finite result for the amplitude in the limit in which the regulator is removed, by redefining the parameters of the theory (the coupling constant) in such a way that the physical observables (the amplitude) are finite.

We can now use this result to compute the cross section for the $2 \to 2$ scattering process at NLO, for example, or other physical observables that can be derived from the amplitude.

> **Remark.** *We could have chosen a different renormalization condition (i.e. a different definition of the physical coupling constant), and we would have obtained a different result for the amplitude. Let us for example choose the following renormalization condition:*
>
> $$\mu^{D-4}(i\mathcal{M})\Big|_{s=s_0,\,t=t_0,\,u=u_0} = -i\lambda'_{phys},$$
>
> *which will be different from the previous one, since we are evaluating the amplitude at a different kinematic point. This is not a problem, since the physical observables (the cross section) will be the same in both cases, as they should be, since they are independent of the choice of the renormalization condition. But proceding with the calculation*
>
> $$\mu^{-2\epsilon}(i\mathcal{M}) = -i\lambda'_{phys} - i\frac{3(\lambda'_{phys})^2}{32\pi^2}\int_0^1 \mathrm{d}x\,(\dots) + \mathcal{O}((\lambda'_{phys})^3),$$
>
> *which is different from the previous result, since we are evaluating the integral at a different kinematic point. However, the physical observables will be the same in both cases, since if we now rewrite perturbatively the physical coupling constant λ'_{phys} as a function of λ_{phys}:*
>
> $$\lambda'_{phys} = \lambda_{phys} + \alpha''\lambda_{phys}^2 + \mathcal{O}(\lambda_{phys}^3),$$
>
> *which will end up giving us the same result for the amplitude as a function of λ_{phys} and ϵ as before, up to the perturbative order we are considering, and thus the same result for the physical observables.*

Thus the renormalization scheme is not unique, since we can choose different renormalization conditions, but the physical observables will be the same in all the different renormalization schemes, and it all comes down to a choice of how to define the physical parameters of the theory in terms of the bare parameters and the regulator.

6.3.3 | A Note on Analytic Continuation

Let us finally compute the integral in the Feynman parameter x that appears in the expression for the amplitude, which is given by

$$\dots$$

[...]

$$\ldots$$

The analytic continuation to $p^2 = s \geq 4m^2$, but doing so we would get a negative argument for $w - 1$ in the logarithm, which is not defined for real numbers. The logarithm as a complex function has a branch cut on the negative real axis, and we have to specify how to go around it in order to get a well defined result. If $z > 0$ we have

$$\begin{cases} \ldots \\ \ldots \end{cases}$$

What is the correct imaginary part to choose for the logarithm? We can get it by employing the feynman prescription, which is equivalent to performing the change $M^2 \to M^2 - i\epsilon$ or $m^2 \to m^2 - i\epsilon$ or even $p^2 \to p^2 + i\epsilon$ in the final result, and the easiest one is to perform the change $p^2 \to p^2 + i\epsilon$, for small ϵ and check the sign of $i\pi$ (even numerically).

We will find...

[...]

6.4 | Renormalization with Counterterms

[...]

New Notation

[...]

6.4.1 | On Shell Renormalization Conditions

[...]

6.4.2 | Counterterms

[...]

6.4.3 | Renormalization of the Propagator

[...]

6.5 | Renormalization Schemes

[...]

Minimal Subtraction Scheme

[...]

Modified Minimal Subtraction Scheme

[...]

$$\frac{\mathrm{d}}{\mathrm{d}\mu_R}\mathcal{M}(\lambda(\mu_R), \mu_R) = 0.$$

6.5.1 | Scale Dependence and Beta Functions

We can define the beta function for the coupling constant λ with the **running coupling equation**:

$$\mu_R^2 \frac{\mathrm{d}}{\mathrm{d}\mu_R^2}\lambda(\mu_R) = \beta(\lambda).$$

[...]

$$\beta_0 = \frac{3}{32\pi}\dots$$

[...]

Let us try to solve the running coupling equation in perturbation theory. We can write $\lambda(\mu_R)$ as a power series in λ:

$$\beta(\lambda) = \beta_0\lambda^2 + \beta_1\lambda^3 + \dots$$

Let us assume that we know the value of λ at some reference scale μ_0. We can then write:

$$\overline{\lambda} = \lambda(\mu_R^2 = \mu_0^2).$$

We know wnat to understand how λ changes as we change the scale μ_R. We can define the ration at which we change the scale as:

$$\tau = \log\left(\frac{\mu_R^2}{\mu_0^2}\right).$$

We can then writw lambda as a perturbative funtion of $\overline{\lambda}$ as

$$\lambda = \overline{\lambda}\left(1 + c_1(\tau)\overline{\lambda} + c_2(\tau)\overline{\lambda}^2 + \dots\right),$$

with the idea that the coefficients $c_i(\tau)$ will capture the scale dependence of λ. We can then plug this expansion into the running coupling equation along with the expansion of the beta function to find the coefficients $c_i(\tau)$. we will expand the left and right hand sides of the running coupling equation and match the coefficients of powers of $\overline{\lambda}$ to find the functions $c_i(\tau)$.

To start, we need boundary conditions for the functions $c_i(\tau)$ (it is a differential equation). We know that at $\tau = 0$ (which reads $\mu_R = \mu_0$), we have $\lambda = \overline{\lambda}$, which implies that $c_i(0) = 0$ for all i. This gives us the initial conditions we need to solve for the functions $c_i(\tau)$.

The first non trivial order is the order $\overline{\lambda}^2$. At this order, we have:

$$\mathcal{O}(\lambda^2): \quad \dots$$

which means that

$$c_1' = \beta_0, \quad c_1 = \beta_0 \log \tau = \beta_0 \log\left(\frac{\mu_R^2}{\mu_0^2}\right).$$

We can now write lambda (at this order) as a function of $\overline{\lambda}$ and τ as

$$\lambda = \overline{\lambda}\left(1 + \beta_0 \log\left(\frac{\mu_R^2}{\mu_0^2}\right)\overline{\lambda}\right) + \mathcal{O}\left(\overline{\lambda}^3\right).$$

Thus the running coupling equation at this order completely predicts the scale dependence of λ in terms of the beta function coefficient β_0.

[...]

Let us compute the next order in the perturbative expansion. At order $\overline{\lambda}^3$, we have:

$$\mathcal{O}(\lambda^3): \quad c_1'\overline{\lambda}^2 + c_2'\overline{\lambda}^3 = \beta_0\left(\overline{\lambda}^2 + 2c_1\overline{\lambda}^3 + \dots\right) + \beta_1\overline{\lambda}^3 + \dots$$

just by recakking the expansion of λ in terms of $\overline{\lambda}$. We can then match the coefficients of $\overline{\lambda}^3$, using the solution for c_1 we found at the previous order, to find:

$$-c_1 = \beta_0\tau \implies c_2' = 2\beta_0^2\tau + \beta_1,$$

which can now be solved as

$$\begin{aligned} c_2 &= \dots \\ &= \beta_0^2 \log^2\left(\frac{\mu_R^2}{\mu_0^2}\right) + \beta_1 \log\left(\frac{\mu_R^2}{\mu_0^2}\right). \end{aligned}$$

Thus we have found at leading order and the next order the scale dependence of λ in terms of the beta function coefficients β_0 and β_1: if we change the scale μ_R, we can predict how λ changes based on the values of β_0 and β_1. Let us make some further observations:

- The functional form of the scale dependence of λ is completely determined by the running coupling equation. This means that if we know the beta function coefficients, we can predict how λ changes with scale.
- The coefficient β_1 can be fixed by renormalizing at two loops: this is because the beta function at order λ^3 receives contributions from two loop diagrams, so we need to renormalize at two loops to find β_1.
- ...

It is easy to convince ourselves that at higher orders in perturbation theory, the scale dependence of λ will be given by a polynomial in $\log\left(\mu_R^2/\mu_0^2\right)$ of degree equal to the order in perturbation theory.

$$\begin{aligned} c_3' &= \dots \\ c_3 &= \beta_0^3 \log^3\left(\frac{\mu_R^2}{\mu_0^2}\right) + \beta_0\beta_1 \log^2\left(\frac{\mu_R^2}{\mu_0^2}\right) + \beta_2 \log\left(\frac{\mu_R^2}{\mu_0^2}\right), \end{aligned}$$

where now β_2 can be derived from three loop renormalization, β_3 from four loop renormalization, and so on.

This is because at each order in perturbation theory, we will have contributions to the beta function that are proportional to powers of λ, and when we solve the running coupling equation, we will find that the scale dependence of λ is given by a polynomial in $\log\left(\mu_R^2/\mu_0^2\right)$ whose coefficients are determined by the beta function coefficients at that order.

[... highest powers of logarithms at each order in perturbation theory are determined by the lowest order beta function coefficient β_0 ...]

This observation is important: let us imagine to have two very different scales

$$\mu_R^2 \gg \mu_0^2 \quad \text{or} \quad \mu_R^2 \ll \mu_0^2.$$

If this is the case, then the logarithm $\log\left(\mu_R^2/\mu_0^2\right)$ can become of order $\mathcal{O}(1)$. In this scenarion, we can predict how at all perturbative orders the scale dependence of λ will be given by a polynomial in $\log\left(\mu_R^2/\mu_0^2\right)$, having *large logarithms*: the leading contribution is called the **leading logarithm**, LL.

$$\lambda^{k+1} \log^k\left(\frac{\mu_R^2}{\mu_0^2}\right),$$

whoch we can consider of order $\mathcal{O}(\lambda)$ if

$$\lambda \log\left(\frac{\mu_R^2}{\mu_0^2}\right) \sim \mathcal{O}(1).$$

Then the running coupling equation allows us to predict the leading logarithm at all orders in perturbation theory, which is a non trivial result. This is because the leading logarithm at order λ^{k+1} is determined by the lowest order beta function coefficient β_0, which can be computed from one loop renormalization. Thus we can predict the leading logarithm at all orders in perturbation theory just by computing the one loop beta function coefficient β_0.

Similarly, the next to leading logarithm (NLL) have form

$$\lambda^{k+2} \log^k\left(\frac{\mu_R^2}{\mu_0^2}\right),$$

i.e. one higher power of λ than the leading logarithm, but the same power of the logarithm. The NLL at order λ^{k+2} is determined by the beta function coefficients β_0 and β_1, which can be computed from one loop and two loop renormalization, respectively. Thus we can predict the NLL at all orders in perturbation theory just by computing the one loop and two loop beta function coefficients β_0 and β_1 (i.e. renormlization up to two loops).

Let us now focus on the leading logarithm. Since they only depend on the lowest order beta function coefficient β_0, we can solve the running coupling equation at leading logarithm accuracy by only keeping the leading logarithm contribution to the beta function, which is given by $\beta_0\lambda^2$ ($\beta_1 = \beta_2 = \cdots = 0$). We are solving the running coupling equation non perturbatively, using a truncated expansion of the beta function, which is a non trivial result. We have

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\lambda = \beta_0\lambda^2 + \mathcal{O}\left(\lambda^3\right).$$

We neglect the higher order contributions to the beta function, which is a consistent approximation at leading logarithm accuracy. We can then solve this equation by separation of variables:

$$\int_{\overline{\lambda}}^{\lambda} \frac{\mathrm{d}\lambda}{\lambda^2} = \beta_0 \int_0^{\tau} \mathrm{d}\tau,$$

where recall that $\overline{\lambda} = \lambda(\mu_0)$ and $\lambda = \lambda(\mu_R)$. We can then perform the integrals to find

$$-\frac{1}{\lambda(\mu_0^2)} + \frac{1}{\lambda(\mu_R^2)} = \beta_0 \tau = \beta_0 \log\left(\frac{\mu_R^2}{\mu_0^2}\right).$$

Solving for $\lambda(\mu_R^2)$, we find the leading logarithm resummed expression for the running coupling:

$$\lambda(\mu_R^2) = \frac{\lambda(\mu_0^2)}{1 - \beta_0 \lambda(\mu_0^2) \log\left(\frac{\mu_R^2}{\mu_0^2}\right)} = \lambda(\mu_0^2) \sum_{k=0}^{\infty} \left(\beta_0 \lambda(\mu_0^2) \log\left(\frac{\mu_R^2}{\mu_0^2}\right)\right)^k,$$

where we have explicited the fact that the leading logarithm at order λ^{k+1} is given by $\beta_0^k \lambda^{k+1} \log^k(\mu_R^2/\mu_0^2)$. This is a non trivial result: we have found a non perturbative expression for the running coupling at leading logarithm accuracy, which resums the leading logarithms at all orders in perturbation theory. This **resummation** is important when the logarithm becomes large, as it allows us to make reliable predictions even in this regime.

Resummation in quantum field theory is a powerful technique that allows us to improve the convergence of perturbative expansions by summing up certain classes of terms to all orders in perturbation theory. In the case of the running coupling, resummation allows us to capture the leading logarithmic behavior of the coupling at all orders, which is crucial for making accurate predictions when the scale dependence becomes significant.

[...]

One can show that if we had included an additional term in the beta function, such as $\beta_1 \lambda^3$, we would have found connections proportional to...

6.5.2 | Resummation

TODO: change and move back

[...]

Landau Pole

The denominator of the resummed expression for the running coupling can become zero at a certain scale

$$\mu_R = \Lambda,$$

which is called the **Landau pole**. This indicates that the coupling becomes infinite at that scale, which is a sign of a breakdown of the theory.

We can solve this equation for $\mu_0 = \Lambda$ to find the scale at which the Landau pole occurs (the scale at which the theory becomes non perturbative):

$$\lambda(\mu_R^2) = \frac{1}{\beta_0 \log\left(\frac{\Lambda^2}{\mu_R^2}\right)}.$$

it is not difficult to see that

$$\begin{cases} \lambda(\mu_R^2 \to 0) \to 0, \\ \lambda(\mu_R^2 \to \Lambda^2) \to \infty. \end{cases}$$

[*Plot of divergence in the coupling*]

[...]

The **fixed point** is a point where the running of lambda stops, i.e. where the beta function is zero $\beta(\lambda) = 0$. For example, in a ϕ^4 theory, we have a trivial infrared fixed point at $\lambda = 0$, which is an attractive fixed point in the infrared. Other possibilities can arise in more complicated theories, such as non abelian gauge theories, where we can have non trivial fixed points that are not at zero coupling. The existence of fixed points is important for understanding the behavior of the theory at different scales, and can have implications for phenomena such as phase transitions and critical behavior.

[...]

Why would we want to use a scheme such as MS or $\overline{\text{MS}}$ instead of the on shell scheme?

[...]

Consider $\mathcal{M}(\phi\phi \to \phi\phi)$ for $p^2 \gg m^2$:

$$\int \dots$$

in the on shell scheme, we would have

$$i\mathcal{M}_{NLO} = \dots$$

At L-loops we expect to have contributions of the form

$$i\mathcal{M} \sim \lambda\lambda^L \log^L\left(\frac{p^2}{m^2}\right)$$

[*draw L loops*]

the perturbative expansion breaks down when $\lambda \log(s/m^2) \sim \mathcal{O}(1)$.

The on-shell schemes is defined as $s = 4m^2$, but at order $\mathcal{O}(m^2)$ the scheme is ill defined, since we have null scales....?

In $\overline{\text{MS}}$ we have instead

$$\dots$$

where

1. The massless limit is fine
2. choosing $\mu_R^2 \sim \mathcal{O}(s)$ allows us to resum the large logarithms, which is important for making reliable predictions at high energies. It avoids large logarithms in the perturbative expansion, which can spoil the convergence of the series and lead to unreliable predictions.

The running couplin equation in MS or $\overline{\text{MS}}$ is ...

[...]

$$\mu_R \sim \mathcal{O}\left(\text{ scale describing the problem }\right).$$

[...]

In principle observables do not depend on the renormalization scheme, but in practice they do at finite order in perturbation theory. The choice of renormalization scheme can affect the convergence of the perturbative expansion and the size of higher order corrections, so it is important to choose a scheme that is appropriate for the problem at hand.

If we truncate the series in lambda at some order, the choice of renormalization scheme can affect the numerical value of the coupling constant and the predictions for observables. Seeing how much the predictions change when we change the renormalization scheme can give us an estimate of the theoretical uncertainty associated with truncating the perturbative expansion.

The $\overline{\mathrm{MS}}$ scheme can be used to renormalize the mass of the theory, but it is not a physical mass (it is not the pole of the propagator or the eigenvalue of $\hat{P}^2$), since it depends on the renormalization scale μ_R.

[...]

[...]

[...]

6.5.3 | Renormalization without Regulators

As we mentioned, in principle we can perform renormalization without introducing a regulator. This is called **renormalization without regulators**.

As an example, let us consider the 2-2 scattering in ϕ^4 theory.

$$i\mathcal{M}_{NLO} = (-i\lambda)^2 \dots$$

where $\delta\lambda$ is the counterterm for the coupling constant, and the integral is divergent and we had

$$iV(p^2) = \frac{1}{2}\int \frac{\mathrm{d}^4 k}{(2\pi)^4} \frac{i}{(k^2 - m^2)((k+p)^2 - m^2)}.$$

In the on shell scheme, we have defined

$$\delta\lambda = -\lambda^2 \int$$

to cancel the divergence of the integral, so that the NLO contribution to the amplitude is finite. We can then write the NLO amplitude as

$$i\mathcal{M}_{NLO}\big|_{os} = \dots \quad = \dots$$

where we have defined

$$\begin{aligned} i\tilde{V}(p^2) &= iV(p^2) - iV(4m^2) \\ &= \dots \\ &= \dots \end{aligned}$$

which is finite. Thus we have performed renormalization without introducing a regulator, by defining the counterterm in such a way that it cancels the divergence of the integral, and then writing the amplitude in terms of a finite quantity $\tilde{V}(p^2)$. One can compute this finite quantity explicitly with Feynman parameters...

[...]

This strategy becomes quite inconvenient...

6.6 | Integrals in Dimensional Regularization

[...]

$$I = \int \mathrm{d}^D k f(k) \implies \mathrm{d}^D k f(k)\big|_{k^\mu \to \pm\infty} = 0.$$

This has some consequences.

- loop integrals...
- integration by parts...
- scaleless integrals vanish. Let us say that we have $[f(k)] = \Delta_f$ an integer, which means that any dimensional quantity must be set to zer, except for the loop momentum k. Then, we have
$$\int \mathrm{d}^D k f(k) = \lambda^{D+\Delta_f} \int \mathrm{d}^D k f(k),$$
as we scale $k \to \lambda k$. This can only be satisfied if the integral vanishes, since $\lambda^{D+\Delta_f} = 1$ only if $D + \Delta_f = 0$, such that in general it is not the case.

Example.
$$\int \mathrm{d}^D k \frac{1}{(k^2)^\alpha} = 0,$$
for any α, since the integrand is scaleless. Thus from integration by parts we have

$$\dots\dots$$

By analytic continuation...

Example.
$$\int \mathrm{d}^D k \frac{1}{(k^2 - M^2)^\alpha} = 0,$$
but only if $\alpha \leq 0$:

- ($\alpha = 0$): ...
- ($\alpha = -1$): ...

Thus dimensional regularization defines how we treat scaleless integrals, which is a very convenient feature. If the integral diverges, we can consistently ....

6.6.1 | Integrals with Numerators

In most theories, momenta and loop moemnta can appear in the numerator of the integrand, and we need to know how to deal with them.

[...]

Dirac Algebra in D Dimensions

In dimensional regularization, we need to define the Dirac algebra in D dimensions.

We can consistently restrict the fermions to have only two polarizations (plus two for antifermions), ...

[...]

Anticommutation relations are almost the same, but now we are considering a D-dimensional $g^{\mu\nu}$:

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} \times \mathbb{I}_4.$$

where the identity is still 4×4 because we are keeping the number of polarizations fixed. We can also define the trace of two gamma matrices as

$$\dots$$

All trace identities we found are still valid up to the promotion of the metric to D dimensions, and the fact that the trace of the identity is still 4. However we have

$$\gamma_\mu \gamma^\mu = \frac{1}{2} \cdots = g^\mu_\mu \times \mathbb{I}_4 = D \times \mathbb{I}_4,$$

Example (exercise). We already computed it in four dimensions, but we can also compute it in D dimensions:

$$\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma \gamma_\mu = \dots$$

6.7 | Systematics of Renormalization

[...]

- DOes this always work for all ultraviolet divergences?
- What about other theories?

[...]

Definition 6.1 (Superficial Degree of Divergence)**.** Given a Feynman graph G, we define its superficial degree of divergence $D(G)$ as the subtraction of the maximum power of loop momenta in the numerator of the integrand (including $\mathrm{d}^D k_i$), minus the maximum power of loop momenta in the denominator of the integrand.

[...]

Example (Feynman Graphs in $D = 4$). [...]

[... naive expectation ...]

Example (Divergences in Feynman Graphs). [...]

The naive expectation can be wrong in two ways:

1. less divergent...
2. more divergent...

Example (...).

[... Divergences in sub-loops ...]

For any graph G we have

- L
- I
- E
- V

[...]

- More external legs E makes the graph less divergent (more convergent).
- More loops L makes the graph:
  - more convergent for $N < 4$,
  - more divergent for $N > 4$,
  - the same for $N = 4$.

ϕ^N Theory: $N = 4$ case

[...]

ϕ^N Theory: $N < 4$ case

[...]

ϕ^N Theory: $N > 4$ case

[...]

6.7.1 | Higher-Loop Renormalization

We need to address the question of how to deal with divergences in sub-loops. In the previous example we had

[*drawing of graph with subloops*]

where we had $D(G) < 0$ but still a divergence due to the sub-loop. We need to make sure that we can consistently renormalize the sub-loop, and then the whole graph.

The divergence of the sub-loop can however be renormalized by a one-loop counterterm, which is a local operator. This means that the divergence of the sub-loop can be absorbed into a redefinition of the parameters of the theory, and we can consistently renormalize the whole graph:

[*drawing of sum of two graphs = finite*]

[*drawing sum of subloop and counterterm*]

[...]

BPHZ Theorem

In general, disentangling all the divergences of a graph can be very complicated, but it can be shown that it is always possible to do so. Indeed, even sub-loop divergences can be renormalized by counterterms at lower loop-orders; after this procedure, we are left with a graph that is superficially divergent, meaning that it can be renormalized by a counterterm of the same form as the original Lagrangian. This is the content of the BPHZ theorem, which is of paramount importance in quantum field theory:

Theorem 6.1 (BPHZ Theorem)**.** *Any ultraviolet divergence of a graph can be renormalized (removed) by the counterterms corresponding to superficially divergent 1PI green functions.*

(Bogoliubov, Parasiuk, Hepp, Zimmermann)

The proof is rather complicated and can be found in more advanced textbooks on quantum field theory, but the main idea is to use the fact that the divergences of a graph can be disentangled by using the fact that they are local, and then to use the fact that the counterterms are also local to absorb them.

TODO: add reference to proof

Thus the BPHZ theorem guarantees that we can consistently renormalize any quantum field theory, as long as we have a finite number of counterterms corresponding to superficially divergent 1PI green functions. This means we can study the renormalizability of theories using only their superficial divergences, without having to worry about the divergences of sub-loops.

6.7.2 | Systematics of Renormalization for QED

We are going to repeat the analysis of the superficial degree of divergence for QED, and see how it works in practice.

Let us call E_ψ and I_ψ the number of external and internal Dirac (fermionic) lines, and E_γ and I_γ the number of external and internal photon lines.

[*draw of fermion line with two vertex to each of which one photon line is attached*]

We are interested in computing how many times a fermionic or photon line attaches to a vertex. We can distinguish the case of internal and external lines, since the former can be attached to two vertices, while the latter can only be attached to one vertex. We have

$$V = 2I_\gamma + E_\gamma$$
$$2V = 2I_\psi + E_\psi$$

Using these relations plus Euler's formula

$$L = I - V + 1, \quad I = I_\psi + I_\gamma,$$

then we can write

$$I_\gamma = \dots$$
$$I_\psi = \dots$$

Thus as $k \to \infty$ we know that the integrand of the Feynman graph behaves as

$$\frac{1}{\not{k}} \sim \text{fermion line (to be drawn)}$$
$$\frac{1}{k^2} \sim \text{photon line (to be drawn)}$$

so that in the end we can compute the superficial degree of divergence as

$$D(G) = \dots$$

[...] This shows furthermore that for a sufficiently large number of external legs the graph is convergent, while for a small number of external legs the graph is divergent. In particular

$$D(G) \geq 0 \quad \text{for} \quad (E_\gamma + \frac{3}{2}E_\psi) \leq 4,$$

thus the superficially divergent 1PI Green functions are

[*draw of 6 1PI green functions*]

(i) The first one is zero since...

(ii) The third one is zero for Furry's theorem: *Green functions with an odd number of external photon legs vanish...* It implies that also the first one is zero.

(iii) The second graph is renormalized by counterterms from the renormalization constants Z_A, Z_m, Z_ψ and Z_e. This is true also for the fifth and sixth graph, which are renormalized by the same counterterms.

(iv) We are left with the fourth graph, which can be shown to be finite even if it is superficially divergent, due to gauge invariance. This is a non-trivial result, which can be shown by using the Ward-Takahashi identities, which are a consequence of gauge invariance. Let us see this in detail.

Considering the fourth graph, its superficial degree of divergence is $D(G) = 0$, so it is logarithmically divergent. Only largest terms as $|k^\mu| \to \infty$ are independent of m, $p^\mu \ldots [\ldots]$ [*draw*] $[\ldots]$

This must come from a lagrangian term of the form

$$\mathcal{L} \ldots$$

which would not be gauge invariant: this counterterm cannot be generated by a gauge invariant theory, so it must be zero. This is a consequence of the Ward-Takahashi identities, which are a consequence of gauge invariance...

Equivalently, we can use Bose symmetry to show that the counterterm must be of the form... [...] since photons are bosons, the counterterm must be symmetric under the exchange of the photon legs, so it must be of the form... [...] from the ward takanashi identity

$$p_{1\mu} C^{\mu\nu\rho\sigma} = p_{2\mu} C^{\mu\nu\rho\sigma} = \cdots = 0,$$

so the constant in front of the counterterm must be zero, so the counterterm must be zero, so the graph is finite.

6.7.3 | Renormalizability from Dimensional Analysis

It can be shown that the renormalizability of theories is related to ...

Consider a general theory with an arbitrary number of

- scalar fields ϕ_i,
- Dirac fields ψ_i,
- massless spin 1 vector fields A_i,

UV limit for fields...

Furthermore, we know that Feynman rules in vertexes are polynomials in the momenta, so the leading terms... we can neglect all masses and external momenta...

[*draw >o< and integral where we simplify momenta and masses*]

$$\int \frac{\mathrm{d}^4 k}{(k^2 - m^2)((k+p)^2 - m^2)} \sim \int \frac{\mathrm{d}^4 k}{k^4} \sim \int \frac{\mathrm{d}^4 k}{|k|^4},$$

We can notice how, after removing the couplings, the superficial degree of divergence is related to the mass dimension

$$D\left(\int \mathrm{d}^D k_1 \cdots \mathrm{d}^D k_L \frac{1}{k^\alpha}\right) = DL - \alpha = \left[\mathrm{d}^D k_1 \cdots \mathrm{d}^D k_L \frac{1}{k^\alpha}\right],$$

we are looking at an amputed 1PI green function, without external legs.

A generic amputed green funxtion in momentum space has the form

$$\ldots$$

where we had the delta enforcing momentum conservation on the lhs, and the integral of the product of propagators and vertexes on the rhs. The superficial degree of divergence is then given by the mass dimension of the amputated green function $\mathcal{M}(E_\phi, E_\psi, E_\gamma)$, which will be given by

$$[\mathcal{M}(E_\phi, E_\psi, E_\gamma)] = \ldots$$

which is the same as the superficial degree of divergence $D(G)$ of the graph G corresponding to the 1PI green function $\mathcal{M}(E_\phi, E_\psi, E_\gamma)$. This is a general result, which holds for any theory, and it allows us to determine the superficial degree of divergence of a graph by simply looking at the mass dimension of the corresponding green function.

If we now consider an interaction lagrangian of the form

$$\mathcal{L} = \sum_j g_j O_j,$$

as a sum of couplings g_j times operators O_j (depending on the fields $O = O(\phi_i, \psi_i, A_i)$, as products of fields and their derivatives). In $D = 4$ we ....

If we now consider a graph G with N_i vertices of type i, then the graph itself will have a form

$$G \sim \prod_i g_i^{N_i} \int \mathrm{d}^D k_1 \cdots \mathrm{d}^D k_L \frac{1}{k^\alpha},$$

By dimensional analysis, we can write

$$\dots$$

hence, its superficial degree of divergence will be given by

$$D(G) = \dots$$

This shows that the superficial degree of divergence of a graph is related to the mass dimension of the couplings g_i and the operators O_j in the interaction lagrangian. Let us now look at some special cases to see how this works in practice.

- $\Delta_j \geq 0$: ... The maximum degree of divergence is independent of the number of vertices (therefore even independent on the number of loops), so the theory is **renormalizable**. The number of divergent green functions does not depend on the number of loops, so we can renormalize the theory by a finite number of counterterms.

  Since $\Delta_j = [g_j] \geq 0 \iff [O_j] \leq 4$, which means that the operators in the interaction lagrangian must have mass dimension less than or equal to 4, so they can only be products of fields and their derivatives with mass dimension less than or equal to 4. This is a very strong constraint on the form of the interaction lagrangian, and it is one of the reasons why renormalizable theories are so special.

- $\Delta_j > 0$: ... if this is strictly the case for all operators in the interaction lagrangian, then the theory is **super-renormalizable**. The argument is similar, since with more loops we have more negative contributions to the superficial degree of divergence, so the theory becomes more and more convergent as we add more loops, so we only have a finite number of divergent green functions, which can be renormalized by a finite number of counterterms....?

- $\Delta_j < 0$: ... for at least one $\Delta_j < 0$, the theory is **non-renormalizable**. The argument is that the superficial degree of divergence becomes more and more positive as we add more loops "$-N_j \Delta_j$ " ...

  Every green function becomes divergent at sufficiently high loop order, so we have an infinite number of divergent green functions, which can only be renormalized by an infinite number of counterterms, so the theory is **non-renormalizable**.

We have thus found a way to relate the renormalizability of a theory to the mass dimension of the couplings and operators in the interaction lagrangian, which is a very powerful tool to determine the renormalizability of a theory by simply looking at its lagrangian. For a larger collection of theories, the degree of divergence depend of the formula

$$D(G) = 4 - E_\phi - \frac{3}{2}E_\psi - E_\gamma - \sum_j N_j \Delta_j,$$

so we can determine the renormalizability of a theory by looking at the mass dimension of the couplings and operators in the interaction lagrangian, and the number of external legs of the green functions we are interested in.

Example (Massive Spin 1). Why massive spin 1 is different from massless spin 1? The reason is that the propagator of a massive spin 1 field behaves as

$$wiggleline = \frac{i}{k^2 - m^2 + i\epsilon}\left(-g_{\mu\nu} + \frac{k_\mu k^\mu}{m^2}\right),$$

which as $k \to \infty$ behaves as

$$\xrightarrow{|k|\to\infty} \frac{i}{k^2}\frac{k_\mu k^\nu}{m^2} \not\sim \frac{1}{k^\alpha},$$

and the leading term is not independent of the mass m, so it does not behave as a massless spin 1 field: it breaks the assumption we made in the analysis of the superficial degree of divergence, so it can lead to a different behavior of the superficial degree of divergence, and therefore to a different renormalizability of the theory.

Indeed, if we were to consider a theory of QED with a massive photon, we would find that ...

[...]

Remark (Higgs Mechanism). *The Higgs mechanism allows us to give mass to the photon without breaking gauge invariance, so we can have a theory of massive spin 1 fields that is still renormalizable. This is one of the reasons why the Higgs mechanism is so important in particle physics, since it allows us to have a consistent theory of massive spin 1 fields, which are essential for the description of the weak interactions.*

...

Example (Gravity). We will not be concerned with gravity in this course, but it is interesting to note that gravity is a non-renormalizable theory: the gravitational potential is given by

$$\sim G\frac{m_1 m_2}{r},$$

and since the coupling constant of gravity (the Newton constant G) has mass dimension -2, so it has $\Delta_j < 0$, so it is non-renormalizable. This is one of the reasons why we need a theory of quantum gravity, since we cannot have a consistent theory of quantum gravity without a renormalizable theory.

We can obtain a QFT by considering the quantization of the metric field $g_{\mu\nu}$, but this theory is non-renormalizable, so we need to find a way to make it renormalizable, which is one of the main goals of quantum gravity research. This theory would also give us some insights and results, but these would be correct only up to a certain energy scale, after which the non-renormalizability would become a problem.

6.7.4 | Renormalizable Theories

We can now state the condition for a theory to be renormalizable in terms of the UV behaviour of the fields and the mass dimension of the couplings and operators in the interaction lagrangian:

A theory is renormalizable if all its ultraviolet divergences can be renormalized by a finite number of counterterms at all orders.

It follows that...

We have also seen that lagrangians...

However, even in renormalizable theories we may need to add other interaction terms to the lagrangian, which are not present in the original lagrangian, to make it finite.

Example (Yukawa theory). In the Yukawa theory we have an interaction lagrangian of the form

$$\mathcal{L}_{int} = -g\phi\bar{\psi}\psi,$$

which, in $D = 4$ has a coupling g with mass dimension $[g] = 0$...

[*drawing of 1PI green functions a-f*]

(a) the first one

(b) the second one...

Thus in the end we have to add terms to the lagrangian in order to obtain the correct counterterms to renormalize the theory, so the final lagrangian will be of the form

$$\mathcal{L}_{int} = -g\phi\bar{\psi}\psi - \frac{g_3}{3!}\phi^3 - \frac{g_4}{4!}\phi^4,$$

thus ...

Let us look at the mass dimensions of the couplings in the final lagrangian: we have $[g] = 0$, $[g_3] = 1$ and $[g_4] = 0$, so all the couplings have non-negative mass dimension, so the theory is renormalizable. This is a very interesting result, since it shows that loop corrections generate the ϕ^3 and ϕ^4 interactions, even if they are not present in the original lagrangian, but the theory is still renormalizable, since the couplings of these interactions have non-negative mass dimension.

This is an example of a more general rule:

If a theory is renormalizable, it must include all the renormalizable interactions compatible with the field content and the symmetries of the theory.

In particular, we need to add all the interactions such that $[g_j] \geq 0$ or equivalently $[O_j] \leq 4$, which are compatible with the field content and the symmetries of the theory, since these interactions can be generated by loop corrections.

We can thus say that in quantum field theory, the lagrangian of a renormalizable theory is not just a choice, but it is determined by the field content and the symmetries of the theory, since we need to include all the renormalizable interactions compatible with the field content and the symmetries of the theory.

The symmetries of a theory are not one of the properties of the theory, but they are what directly defines the theory, since they determine the form of the lagrangian, and therefore the dynamics. This is one of the reasons why symmetries are so important in quantum field theory.

6.7.5 | Non-Renormalizable Theories

A non renormalizable theory is a theory that has an infinite number of divergent green functions, which can only be renormalized by an infinite number of counterterms, so it is not a consistent theory at all energy scales.

[...]

Non renormalizable theories can still be useful as **effective field theories**, namely as low energy approximations of a more fundamental theory, which is renormalizable. In this case, the non-renormalizable theory is only valid up to a certain energy scale, after which the non-renormalizability becomes a problem, and we need to use the more fundamental theory to describe the physics at higher energies. This approximation is called **UV** completion.

Consider a non renormalizable theory with an interaction lagrangian of the form

$$\mathcal{L}_{int} = \lambda \hat{O}(x), \quad \text{with } [\lambda] < 0,$$

where $\hat{O}(x)$ is an operator with mass dimension greater than 4.

We could imagine to rewrite the coupling λ as

$$\lambda = \frac{g}{M^k}, \quad k = -[\lambda] > 0,$$

where

- g is a dimensionless coupling, which is the coupling of the more fundamental theory, and it is renormalizable,
- M is a mass scale, which is the scale of the more fundamental theory, and it is the scale at which the non-renormalizability becomes a problem.

If we are interested in studying the physics at energy scales $E \ll M$, then we can neglect the non-renormalizable interactions, since they are suppressed by powers of E/M, and we can use the renormalizable theory to describe the physics at low energies...

Non renormalizable theories are thus perfectly valid and predictive below some energy scale, and they can be used to describe the physics at low energies, as long as we are aware of their limitations.

Moreover:

- The low energy limit of a QFT (which may also be renormalizable) is an effective field theory, which is non-renormalizable, since it contains all the interactions compatible with the symmetries of the theory, including non-renormalizable interactions. The original QFT is the UV completion of the effective field theory. An example is the fermi theory of beta decay, which is a non-renormalizable effective field theory (since its interaction lagragian contains a four-fermion interaction with coupling of mass dimension -2); if we compute the same decay in the standard model, we find that the four-fermion interaction is generated by the exchange of a W boson, which is a renormalizable theory, and it is the UV completion of the fermi theory.

- A renormalizable theory is generally not valid at all energy scales. Beyond a certain energy scale, new physics may appear. Below such energy scale, these effects can be described by adding suitable non-renormalizable operators to the lagrangian of the renormalizable theory. This means that in the end even renormalizable theories can be viewed as the first term in a large M expansion of the more foundamental lagrangian.
- The lagrangian of a non-renormalizable theory is determined by the field content and the symmetries of the theory, since we need to include all the interactions compatible with the field content and the symmetries of the theory. While generally there is an infinite number of interactions compatible with it, for low energy processes we can neglect all the interactions above a certain energy scale.

7 QED at the Loop Level

[...]

Exercises

Sheet 1

Sheet 2

6. Unpolarized scattering of fermions in the Yukawa theory

Compute the unpolarized (i.e. averaged over initial and summed over final polarization states) squared tree-level scattering amplitude $|\mathcal{M}|^2$ unpolarized for elastic scattering of two identical massless Dirac fermions (e.g. $e^-e^- \to e^-e^-$ for $m_e \to 0$) in the Yukawa theory, as a functions of the Mandelstam invariants and the mass M of the scalar.

For the Yukawa theory, the interaction Lagrangian is given by

$$\mathcal{L}_{\text{int}} = -g\phi\bar{\psi}\psi,$$

and the vertices are represented by a factor of $-ig$. The tree-level scattering amplitude for the process

$$e^-(p_1)e^-(p_2) \to e^-(p_3)e^-(p_4),$$

can be computed using the Feynman rules for the Yukawa theory, which involve the exchange of a scalar particle in the t-channel and u-channel:

$$i\mathcal{M} = \text{draw t-chan} + \text{u-chan},$$

where using Feynman rules, we have

$$i\mathcal{M} = (-ig)^2 \left[\bar{u}_3 u_1 \frac{i}{t - M^2} \bar{u}_4 u_2 - \bar{u}_4 u_1 \frac{i}{u - M^2} \bar{u}_3 u_2\right] = \mathcal{M}_1 + \mathcal{M}_2,$$

with $t = (p_1 - p_3)^2$ and $u = (p_1 - p_4)^2$, and the minus sign arises due to the fact that the u-channel is obtained by exchanging the two identical fermions in the final state w.r.t. the t-channel, which introduces a minus sign (see extra rules for fermions).

[...]

$$|\mathcal{M}_{\text{unpol.}}|^2 = \dots$$

[...]

$$\sum_s \ldots = \ldots$$
$$\sum_s \ldots = \ldots$$
$$\sum_s \ldots = \ldots$$
$$\sum_s \ldots = \ldots$$

and the final, more difficult, is given by

$$\sum_s \ldots = \ldots$$
$$= \ldots$$

If we now use the limit for $m_e \to 0$ we can compte:

$$\ldots$$

and now, using the properties of the Mandelstam invariants, we can write the final result as a function of s and t, since $s + t + u = 0$. Thus we have

$$|\mathcal{M}_{\text{unpol.}}|^2 = \ldots$$

[...]

$$|\mathcal{M}_{\text{unpol.}}|^2 = \ldots$$

[...]

Remark. *When the total mass approaches zero, i.e. $M \to 0$, we get a constant matrix element for the unpolarized scattering amplitude, which is given by $|\mathcal{M}_{unpol.}|^2 \to 3g^4$. This is a consequence of the fact that the propagator of the scalar particle becomes massless, so that it behaves as $1/t$ or $1/u$, and since t and u are negative, we get a constant contribution to the scattering amplitude.*

7. Decay of a scalar into a fermion-antifermion pair

In the Yukawa theory, compute the total decay rate Γ of a scalar into a fermion antifermion pair, as a function of the masses M of the scalar and m of the Dirac fermion. Which condition must M and m satisfy for this decay to be possible?

The vertex of this process is given by

$$i\mathcal{M} = \text{draw vertex} = -(ig)\bar{u}_1 v_2,$$

[...]

$$|\mathcal{M}_{\text{unpol.}}|^2 = \ldots$$

Now, computing the square of the momenta of the final state particles, we have

$$M^2 = P^2 = (p_1 + p_2)^2 = 2m^2 + 2p_1 \cdot p_2, \implies 2p_1 \cdot p_2 = M^2 - 2m^2,$$

which let us write the unpolarized squared amplitude as

$$|\mathcal{M}_{\text{unpol.}}|^2 = 2g^2(M^2 - 4m^2).$$

Now using the formula for the differential decay rate of a particle of mass M into two particles of mass m we have

$$\mathrm{d}\Gamma = \frac{1}{2M}|\mathcal{M}_{\text{unpol.}}|^2 \mathrm{d}\Pi_2,$$

where $\mathrm{d}\Pi_2$ is the two-particle phase space measure, which can be computed keeping in ming that the decay is computed in the center of mass frame of the particle, so that it is given by the expression we explicitly computed in the lectures:

TODO: add reference to the lectures

$$\mathrm{d}\Pi_2 = \mathrm{d}\Omega \frac{1}{16\pi^2}\frac{|\vec{p}_f|}{E_{c.m.}}, \qquad \begin{cases} E_{c.m.} = M, \\ |\vec{p}_f| = |p_1| = \sqrt{E_1^2 - m^2}, \\ P = \begin{pmatrix} M \\ \vec{0} \end{pmatrix}, \quad p_1 = \begin{pmatrix} E_1 \\ \vec{p}_f \end{pmatrix}. \end{cases}$$

We thus have to compute expressions for E_1 and $|\vec{p}_f|$ as a function of M and m. We can do this by using the fact that the momenta of the final state particles are on-shell, so that we have

$$\dots$$

and

$$\dots$$

With these expressions, we can write the total differential decay rate as

$$\frac{\mathrm{d}\Gamma}{\mathrm{d}\Omega} = \dots$$

and since the integral over the solid angle is given by $\int \mathrm{d}\Omega = 4\pi$, we can write the total decay rate as

$$\Gamma = \int \frac{\mathrm{d}\Gamma}{\mathrm{d}\Omega}\mathrm{d}\Omega = \dots$$

[...]

Remark. *The condition for the decay to be possible is given by the fact that the total energy of the final state particles must be greater than the mass of the initial state particle, so that we must have $M > 2m$. When we are in the limit for $M \to 2m$, the decay rate vanishes $\Gamma \to 0$: this is because the phase space volume for the decay vanishes, since the final state particles are produced at rest, so that the decay is kinematically forbidden.*

8. Compton scattering in QED

In QED, compute the unpolarized squared tree-level scattering amplitude for elastic scattering of a massless electron and a photon, as a function of the Mandelstam invariants. Use this result to compute the differential cross section $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}$ for this process in the centre of mass.

[...]

$$\begin{aligned} i\mathcal{M} &= \text{draw sum of 2 diagrams} \\ &= (-ie)^2 \left[\bar{u}_3 \gamma^\nu \frac{\not{p}_1 + \not{p}_2}{s} \gamma^\mu u_1 + \dots \right] = i\mathcal{M}_1 + i\mathcal{M}_2, \end{aligned}$$

[...]

$$|\mathcal{M}_{\text{unpol.}}|^2 = \dots$$

where now we have to compute the A, B and C terms. Starting from C we have

$$\begin{aligned} C &= \frac{1}{4} \sum_s \dots = \text{product of diagrams} \\ &= \frac{e^4}{2su} \dots \\ &\times (\epsilon_2^*)_\mu (\epsilon_4^*)_\nu (\epsilon_2)_\rho (\epsilon_4)_\sigma \end{aligned}$$

where the last term is given by $(\epsilon_2^*)_\mu (\epsilon_4^*)_\nu (\epsilon_2)_\rho (\epsilon_4)_\sigma = g_{\mu\rho} g_{\nu\sigma}$, enforcing

$$C = \dots$$

If we now use the result computed in exercise 5, in which we demonstrated that

$$\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma \gamma_\mu = -2\gamma^\sigma \gamma^\rho \gamma^\nu,$$

it easy to see that this implies

$$\implies \dots$$

Thus we can further simplify the expression for C as

$$C = -\frac{e^4}{2su} \operatorname{Tr}[\dots]$$

We need to show that $\gamma^\mu \gamma^\nu \gamma^\rho \gamma_\mu = 4 g_{\nu\rho} \mathbb{I}$.

Proof. We can start by defining ... □

Thus now we have all the ingredients to compute the final value for the C-term, which is given by

$$\begin{aligned} C &= \dots \\ &= \dots \\ &= -\frac{4e^4}{su}(s + t + u)(-t) = 0, \end{aligned}$$

since we are in the massless limit, so that $s + t + u = 0$. Thus we have that the interference term between the two diagrams vanishes.

We can now move on to the computation of the A-term, which is given by

$$\begin{aligned} A &= \frac{1}{4} \sum_s |\mathcal{M}_1|^2 = \frac{1}{4} \text{ product of diagrams} \\ &= \frac{e^4}{4s^2} \operatorname{Tr}[\dots] \end{aligned}$$

where now we need to compute the following product of gamma matrices:

$$\gamma^\mu \gamma^\nu \gamma_\mu = \dots$$

which will enforce

$$\begin{aligned} A &= \frac{4e^4}{4s^2}\,\mathrm{Tr}[\dots] \\ &= \frac{4e^4}{4s^2}\,\mathrm{Tr}\Big[\not{p}_2\not{p}_1\not{p}_3\not{p}_2\Big] \\ &= ... \\ &= ... = -2e^4\frac{u}{s}, \end{aligned}$$

where we used ...

The B-term can be computed via crossing symmetry, since it is given by the same expression as A but ...

[...]

Thus we can compute the unpolarized squared amplitude as

$$|\mathcal{M}_{\text{unpol.}}|^2 = \dots$$

Again, we are in the center of mass frame

$$E_1 = E_2 = E_3 = E_4 = \frac{E_{cm}}{2} = E,$$

so that we can write the momenta four-vectors as

$$p_1 = \begin{pmatrix} E \\ 0 \\ 0 \\ E \end{pmatrix}, \quad p_2 = \begin{pmatrix} E \\ 0 \\ 0 \\ -E \end{pmatrix}, \quad p_3 = \begin{pmatrix} E \\ 0 \\ E\sin\theta \\ E\cos\theta \end{pmatrix}, \quad p_4 = \begin{pmatrix} E \\ 0 \\ -E\sin\theta \\ -E\cos\theta \end{pmatrix},$$

and thus the Mandelstam invariants read

$$\begin{aligned} u &= (p_4 - p_1)^2 = -2p_1\cdot p_4 = -2(E^2 + E^2\cos\theta) = -2E^2(1+\cos\theta), \\ s &= E_{cm}^2 = 4E^2, \\ \frac{u}{s} &= -\frac{1+\cos\theta}{2}. \end{aligned}$$

So that we can write the unpolarized squared amplitude as a function of the scattering angle θ as

$$|\mathcal{M}_{\text{unpol.}}|^2 = 2e^4\left(\frac{1+\cos\theta}{2} + \frac{2}{1+\cos\theta}\right).$$

Finally, we can compute the differential cross section for this process in the center of mass frame as

$$\begin{aligned} \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} &= \dots \\ &= \dots \end{aligned}$$

9. Ward identity in sQED

In sQED, explicitly show that the Ward identity is valid for an amplitude of Compton scattering (i.e. the same we did in the lecture but replacing the fermion with a scalar). Note that the amplitude is a sum of three diagrams.

$$\text{Draw of vertex} = \dots$$

$$\text{Draw of vertex} = \dots$$

$$\mathrm{d}\gamma = \phi\gamma = \text{Draw of generic diagram} = \dots$$

We want to show that $(p_2)_\mu i\mathcal{M}^{\mu\nu} = 0$ at tree level, where p_2 is the momentum of the outgoing photon. We can do this by explicitly computing the amplitude for each diagram and then summing them up, so that we have

$$\begin{aligned}(p_2)_\mu i\mathcal{M}^{\mu\nu} &= (p_2)_\mu(\text{sum of diagrams}) \\ &= \dots \\ &+ \dots\end{aligned}$$

where now we need to compute the following terms:

$$p_1 \cdot p_2 = \frac{s - m^2}{2}, \quad p_2 \cdot p_4 = -\frac{t}{2}, \quad p_2 \cdot p_3 = -\frac{u - m^2}{2},$$

and with these expressions we can compute

$$\begin{aligned}(p_2)_\mu[2p_1^\mu + p_2^\mu] &= s - m^2 + 0 = s - m^2, \\ (p_2)_\mu[p_1^\mu - p_4^\mu + p_3^\mu] &= \frac{1}{2}(s - m^2 + t - u + m^2) = -(u - m^2),\end{aligned}$$

where for the last equality we used the fact that $s + t + u = 2m^2$. Thus we have

$$\begin{aligned}(p_2)_\mu i\mathcal{M}^{\mu\nu} &= e^2\left[-i(p_1^\nu + p_2^\nu + p_3^\nu) + i(2p_1^\nu - p_4^\nu) + 2ip_2^\nu\right] \\ &= ie^2(p_1^\nu + p_2^\nu - p_3^\nu - p_4^\nu) = 0,\end{aligned}$$

where the last equality follows from the conservation of momentum at the vertex, which implies that $p_1 + p_2 = p_3 + p_4$.

Remark. *All of the three diagrams are necessary to satisfy the Ward identity and thus the cancellation to happen: if we consider only one or two of them, we would get a non-zero contribution to $(p_2)_\mu i\mathcal{M}^{\mu\nu}$, which would violate the Ward identity. The Ward identity is a consequence of gauge invariance, which is a fundamental symmetry of the theory, and thus must be satisfied by all physical amplitudes.*

10. Scaling Green functions and scattering amplitudes

Show that, under a transformation of the fields of the form

$$\phi(x) \to c\phi(x),$$

where c is a constant, then:

(a) *The Greeen functions of the theory scale as*

$$\dots$$

(b) *The amputated Green functions of the theory scale as*

$$\dots$$

(c) *Scattering matrix elements are invariant under such transformation.*

Hint: *Use the LSZ formula and recall that* $\sqrt{Z} \equiv \langle\Omega| \, \phi(0) \, |\mathbf{p}\rangle \big|_{\mathbf{p}=\mathbf{0}}$.

Defining

$$G_N(x_1, \ldots, x_N) = \langle\Omega| \, \hat{T}\{\phi(x_1) \cdots \phi(x_N)\} \, |\Omega\rangle \, ,$$

which is the Green function of the theory, we can compute how it scales under the transformation of the fields. We have:

(a) Since...

(b) For the amputated Green functions, we have...

(c) For the scattering matrix elements, we recall the LSZ reduction formula as...

Appendices

A | Notation and Conventions

Conventions on Units and Natural Units

We are going to use natural units throughout these notes, where the speed of light c and the reduced Planck constant $\hbar$ are set to 1:

$$c = 1, \quad \hbar = 1.$$

In natural units, all physical quantities are expressed in terms of a single unit of energy (or mass) and measured in Electronvolts (eV) and multiples thereof, since the dimensions of length and time are related to energy through the relations $E = mc^2$ and $E = \hbar\omega$. This simplifies many equations in quantum field theory and allows us to focus on the essential physics without being distracted by factors of c and $\hbar$. In these units, we have:

$$\begin{aligned} [E] &= [M] = [L]^{-1} = [T]^{-1}, \\ [\hbar] &= 1, \\ [c] &= 1. \end{aligned}$$

We report some useful conversion factors between natural units and SI units:

$$\begin{aligned} 1\,\mathrm{GeV} &\approx 1.602 \times 10^{-10}\,\mathrm{J}, \\ 1\,\mathrm{GeV} &\approx 1.783 \times 10^{-27}\,\mathrm{kg}, \\ 1\,\mathrm{GeV}^{-1} &\approx 0.1975\,\mathrm{fm}, \\ 1\,\mathrm{GeV}^{-1} &\approx 6.582 \times 10^{-25}\,\mathrm{s}, \end{aligned}$$

and also

$$\begin{aligned} \hbar c &\approx 197.3\,\mathrm{MeV} \cdot \mathrm{fm}, \\ 1\,\mathrm{fm} &\approx 10^{-15}\,\mathrm{m}, \end{aligned}$$

where the femtometer (fm) is a common unit of length in nuclear and particle physics, often called a Fermi: it is approximately the length of a proton or neutron radius:

$$r_p \sim 0.8\,\mathrm{fm} = \frac{0.8}{197.3}\,\mathrm{MeV}^{-1}.$$

Conventions on Electromagnetism

In order to avoid confusion, we will use the **Heaviside-Lorentz** units for electromagnetism throughout these notes: in these units, the electric charge e is dimensionless and appears explicitly in the interaction terms of the Lagrangian, making it easier to identify the strength of electromagnetic interaction, while Maxwell equations take a more symmetric form.

Explicitly, in Heaviside-Lorentz units, we have the following relations:

- $\epsilon_0 = \mu_0 = 1$, where ϵ_0 is the *vacuum permittivity* and μ_0 is the *vacuum permeability*: this means that the electric and magnetic fields have the same units and that the speed of light c is equal to 1 (these units are consistent with the natural units we are using).
- The **electric charge** e is dimensionless and connected to the fine-structure constant α by
$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} = \frac{e^2}{4\pi} \approx \frac{1}{137}.$$
- The **electric potential** Φ is defined as
$$\Phi = \frac{Q}{4\pi R},$$
where Q is the charge and R is the distance from the charge: this means that the electric potential has the same units as the electric field E (which is consistent with the fact that in Heaviside-Lorentz units, $\epsilon_0 = 1$).
- The **Maxwell equations** take the following form:
$$\begin{aligned} \nabla \cdot \mathbf{E} &= \rho, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\ \nabla \times \mathbf{B} &= \mathbf{J} + \frac{\partial \mathbf{E}}{\partial t}, \end{aligned}$$
where $\mathbf{E}$ is the electric field, $\mathbf{B}$ is the magnetic field, ρ is the charge density, and $\mathbf{J}$ is the current density: this symmetric form of Maxwell equations is one of the reasons why Heaviside-Lorentz units are often preferred in theoretical physics, especially in quantum field theory. Basically the Heaviside-Lorentz units are a rationalized version of the Gaussian units, where the factors of 4π are absorbed into the definitions of the fields and charges, leading to simpler equations and making it easier to identify the physical constants that govern the interactions.

Conventions on the Metric and Spacetime

We will use the **mostly minus** convention for the metric tensor $g_{\mu\nu}$ in Minkowski spacetime, which is defined as
$$g_{\mu\nu} = g^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$
This means that the time component of the metric is positive, while the spatial components are negative; using Einstein summation convention, we have the following relations among the components of the metric and v^μ, w^μ four vectors:
$$\begin{aligned} v \cdot w &= g_{\mu\nu} v^\mu w^\nu = v^0 w^0 - \mathbf{v} \cdot \mathbf{w}, \\ v^2 &= v \cdot v = g_{\mu\nu} v^\mu v^\nu = (v^0)^2 - \mathbf{v}^2, \\ v_\mu = g_{\mu\nu} v^\nu &\implies v_0 = v^0, \quad \mathbf{v} = v^i = -v_i. \end{aligned}$$

We will use greek letters μ, ν, ρ, σ to denote spacetime indices that run from 0 to 3, while Latin letters i, j, k will be used to denote spatial indices that run from 1 to 3:

$$\mu, \nu, \rho, \sigma \in \{0,\ 1,\ 2,\ 3\} = \{t,\ x,\ y,\ z\},$$
$$i, j, k \in \{1,\ 2,\ 3\} = \{x,\ y,\ z\}.$$

Fourier Transforms

We will use the following conventions for Fourier transforms of a function $f(x)$ in four-dimensional spacetime:

$$f(x) = \int \frac{\mathrm{d}^4 p}{(2\pi)^4} e^{-ip\cdot x} \tilde{f}(p),$$
$$\tilde{f}(p) = \int \mathrm{d}^4 x e^{ip\cdot x} f(x).$$

We highlight that the Fourier transform of a function $f(x)$ is denoted by $\tilde{f}(p)$, and that the integration measure includes a factor of $(2\pi)^{-4}$ in the inverse Fourier transform, which is a common convention in quantum field theory: in the space of coordinates, we have the measure $\mathrm{d}^4 x$, while in the momentum space, we have the measure $\frac{\mathrm{d}^4 p}{(2\pi)^4}$. This convention ensures that the Fourier transform and its inverse are consistent and that the normalization of the fields and operators in quantum field theory is correct.

Feynman Diagrams

In these notes, we will use Feynman diagrams as a graphical representation of the interactions between particles and fields in quantum field theory. We will use the following conventions for Feynman diagrams:

TODO: Check and change in the end

- Time will be represented on the horizontal axis, going from left to right, which is a common convention in quantum field theory; this means that the left side of the diagram will represent the initial state of the particles, while the right side will represent the final state after the interaction has occurred.

- Solid lines will represent fermions (spin-1/2 particles), while dashed lines will represent bosons (spin-0 or spin-1 particles).

- Arrows on the lines will indicate the direction of particle flow, with arrows pointing towards the interaction vertex for incoming particles and away from the vertex for outgoing particles.

- Wavy lines will represent gauge bosons (spin-1 particles that mediate interactions), such as photons, W and Z bosons, and gluons.

- The vertices in the diagrams will represent interaction points where particles interact according to the rules of the theory, with each vertex corresponding to a specific term in the Lagrangian of the quantum field theory.

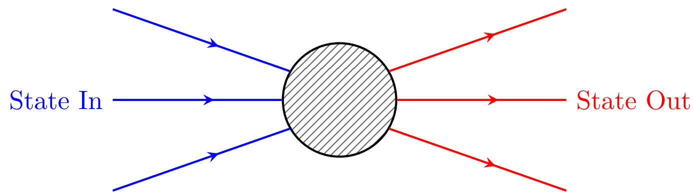


Figure 1: Generic Feynman diagram representing an interaction between particles, where the blue lines represent incoming particles (initial state) and the red lines represent outgoing particles (final state). The black box represents the interaction vertex, which corresponds to a specific term in the Lagrangian of the quantum field theory.

B | Computations and further developments

Differential Cross Section

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Decay Rate

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